Computers in Biology and Medicine 198 (2025) 111166

Contents lists available at ScienceDirect

Computers in Biology and Medicine

journal homepage: www.elsevier.com/locate/compbiomed

Use of machine learning for predicting stress episodes based on wearable sensor data: A systematic review

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ARTICLE INFO

Keywords:
Stress prediction
Wearables
Machine learning
Sensors
Sensor data

ABSTRACT

Objective:

This study consists of a systematic literature review that aims to explore the potential of integrating wearable sensor data and machine learning (ML) techniques for predicting stress episodes. It aims to identify prevalent sensors, key physiological features, and the effectiveness of ML methods in real-world stress monitoring and prediction.

Methods:

This systematic review follows the PRISMA methodology, analyzing literature from January 2010 to June 2025. Data sources included IEEE Xplore, Elsevier, Springer, Multidisciplinary Digital Publishing Institute (MDPI), and paper repositories such as PubMed Central and the Association of Computing Machinery (ACM). The inclusion criteria encompassed studies that employed wearable devices for ML stress prediction, focusing on physiological data such as heart rate variability (HRV), skin conductance, and sleep patterns. Articles were screened for originality, clinical relevance, and methodological rigor.

Results:

Key findings highlighted the use of diverse wearable sensors, including electrodermal activity (EDA), photoplethysmography (PPG), and accelerometers. Commonly extracted features included HRV metrics, skin conductance levels, and respiratory patterns. ML models, such as Random Forest (RF), Support Vector Machines (SVM), and deep neural networks (DNN), have demonstrated high predictive accuracy (*e.g.*, up to 99%). Despite promising results, challenges such as small sample sizes, variability in data quality, and the need for standardized protocols were noted.

Conclusion:

Wearable sensors combined with ML algorithms provide scalable, real-time stress monitoring solutions, emphasizing proactive healthcare management. However, advancing this field requires addressing limitations through interdisciplinary collaboration and focusing on the accessibility and usability of technologies.

Significance:

This study highlights the transformative role of wearable technologies in predicting stress, with implications for personalized health interventions, mental health support, and enhanced healthcare efficiency.

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https://doi.org/10.1016/j.compbiomed.2025.111166
Received 18 December 2024; Received in revised form 3 August 2025; Accepted 27 September 2025
Available online 7 October 2025
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1. Introduction

The rapid adoption of wearable devices in healthcare has transformed the monitoring of physiological and behavioral data, enabling significant changes in clinical treatment and personal health management [1,2]. These technologies offer continuous, real-time insights into individual health conditions, facilitating early diagnosis and intervention for potential health issues before they become critical [3–5]. The increasing demand for efficient and proactive prediction and management solutions is particularly evident in the context of stress, which has become a widespread health concern that impacts both mental and physical well-being [6,7].

The purpose of this review is to investigate the use of wearable devices and machine learning (ML) techniques in stress prediction. By monitoring stress-related indicators such as heart rate variability (HRV), physical activity levels, body temperature, and sleep patterns, wearable technology offers a unique solution for real-time stress monitoring [7–9]. These devices enable individuals and healthcare professionals to identify early signs of stress, allowing for proactive interventions that can significantly improve quality of life and reduce healthcare costs associated with chronic stress [10].

Accurately predicting and managing stress is challenging due to the intricate interactions between physiological and environmental factors [11]. Traditional methods of stress evaluation are subjective and often fail to capture the dynamic, real-time nature of stress responses [12,13]. While these methods have value, they cannot provide objective, real-time insights into stress levels. Objective stress measurement methods, such as cortisol level measurement, offer a more accurate and quantifiable approach. Cortisol, often referred to as the "stress hormone", is released in response to stress and can be measured through blood, saliva, or sweat samples. Advances in wearable technology now allow for continuous, non-invasive cortisol measurement in sweat, providing real-time stress monitoring that complements traditional subjective methods [14–16]. These advances underscore the need for innovative solutions that provide ongoing, unbiased, and personalized insights into the dynamics of stress.

One potential solution is the combination of wearable sensor data and ML algorithms [17–19]. With the help of sophisticated sensors, wearable technology allows for the collection of detailed behavioral and physiological data in natural environments. When combined with machine learning techniques, these data can be analyzed to identify subtle patterns and correlations, paving the way for accurate stress predictions.

Devices like smartwatches and fitness trackers provide rich datasets, including HRV, body temperature, physical activity, and sleep patterns, which are essential for stress analysis [13,20,21]. ML enables pattern recognition and predictive modeling, transforming raw sensor data into actionable insights for stress prediction [22]. Proactive stress management through personalized alerts, targeted interventions, and customized stress reduction strategies tailored to individual needs is a significant benefit of these technologies [23].

By combining wearable sensor data with ML algorithms, this review explores how the integration of these technologies enhances the ability to provide objective, scalable, and continuous stress monitoring [18]. ML-driven models, applied to the complex datasets gathered by wearable devices, can uncover subtle patterns and correlations, offering actionable insights and personalized stress forecasts. This innovative approach is crucial due to the complexity of accurately predicting stress given the interaction of physiological and environmental factors [11].

To provide an overview of the typical stress detection workflow, a schematic of a general wearable stress sensing system is presented in Fig. 1. This system commonly involves the acquisition of physiological signals from wearable sensors such as electrodermal activity (EDA), heart rate variability (HRV), and skin temperature. The raw data undergoes preprocessing to remove noise and artifacts, followed by feature extraction or transformation. These features are then input into machine learning models that classify stress states or predict stress levels. In more advanced systems, the output may be used for real-time feedback, diagnosis, or long-term stress monitoring and intervention planning.

This systematic review aims to provide a comprehensive understanding of the current practices, benefits, and limitations of integrating wearable sensor data with machine learning for stress monitoring. It seeks to inform healthcare professionals about the potential applications of this technology, explore its role in proactive stress management, and identify areas for future research. By addressing these challenges, this study aims to advance the integration of wearable technologies in stress prediction, ultimately benefiting both individuals and healthcare systems.

The main contributions of this systematic review are:

- **Significance of Wearable Sensors in Stress Prediction:** Wearable devices have revolutionized the monitoring of physiological and behavioral data, enabling continuous assessment of stress-related biomarkers such as heart rate variability, physical activity levels, and sleep patterns. These devices provide critical insights into the physiological underpinnings of stress, contributing to the development of personalized stress management strategies.
- **Geographic and Demographic Representation:** The studies included in this review demonstrate a diverse participant pool spanning various demographics and geographic regions. While a significant portion of the research focuses on younger populations, increasing emphasis is being placed on older age groups and individuals in high-stress occupations, showcasing the broad applicability of wearable-assisted stress monitoring.
- **Parameters for Measuring Stress and Associated Features:** Core physiological parameters such as heart rate variability, skin conductance, respiratory patterns, and sleep quality were frequently extracted from wearable devices. These metrics are pivotal for understanding stress dynamics and designing targeted interventions, particularly in populations exposed to chronic stress or mental health challenges.
- **Innovative Approaches in Data Collection and Analysis:** Studies have utilized cutting-edge technologies, including advanced electrodermal activity sensors, photoplethysmography, and accelerometers. Machine learning techniques, including deep neural networks and ensemble models, have been leveraged to analyze sensor data, enhancing the precision of stress detection and prediction in real-world settings.
- **Limitations and Recommendations for Future Research:** The review highlights several challenges, including small sample sizes, variability in data quality, and limited longitudinal studies. Future research should focus on larger, more diverse cohorts, standardizing data collection protocols, and improving the affordability and accessibility of wearable technologies. Additionally, interdisciplinary collaboration and patient-centered research will be essential for refining wearable-assisted stress prediction systems and expanding their utility across diverse populations.

2. Methodology

This systematic review aims to investigate the application of machine learning in predicting stress episodes using wearable sensor data.

With advancements in technology and the growing acceptance of artificial intelligence (AI) and machine learning (ML), interest has risen in how these technologies can enhance the accuracy and efficacy of stress prediction models. This research adopts a systematic review approach, following the "Preferred Reporting Items for Systematic Reviews and Meta-Analyses" (PRISMA) standards [24], to examine literature published between January 2010 and June 2025. The focus is on utilizing wearable devices and sensor data to monitor physiological metrics, including heart rate variability, physical activity levels, body temperature, and sleep patterns, to predict episodes of stress.

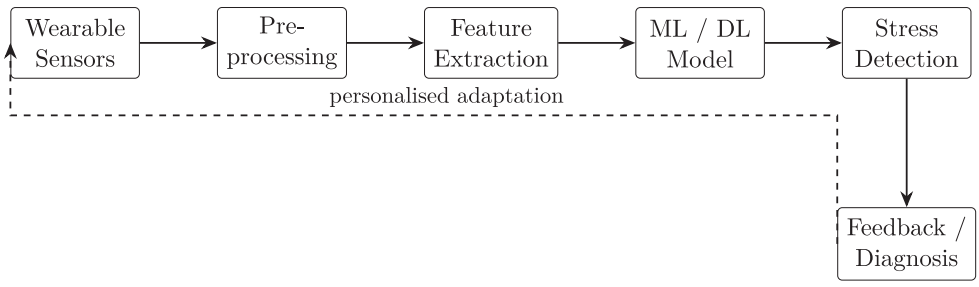


Fig. 1. General architecture of a wearable stress sensing, detection, and feedback system.

2.1. Research questions

In this section, we present three fundamental research questions that guide our systematic review aimed at exploring the core aspects of this evolving field. These questions are designed to examine the critical components of stress prediction using wearable sensor data, focusing on:

RQ1. What are the most commonly used sensors to detect stress episodes?
RQ2. What features are extracted most frequently from the data captured by wearable devices for the identification of stress episodes?
RQ3. How effective are the methods applied in these studies?

This research aims to provide an in-depth analysis of the current practices, benefits, and potential applications of wearable technology in stress prediction. It aims to inform healthcare practitioners and identify new research areas, ultimately benefiting patients and providers by integrating wearable sensor technology into stress management.

2.2. Search strategy

To address these research questions, a comprehensive search strategy was developed in accordance with best practices in systematic reviews. The searches were conducted on multiple electronic databases, including IEEE Xplore, Elsevier, Springer, and the Multidisciplinary Digital Publishing Institute (MDPI), as well as online repositories such as PubMed Central and the Association for Computing Machinery (ACM), to ensure a broad range of relevant studies was captured. A combination of search terms related to stress detection, wearable sensors, machine learning, and physiological data was used, including terms like "stress detection with sensors", "stress detection with wearables", and "stress detection". To further broaden the scope, stemming was applied to account for variations in terminology, and phrase permutations were utilized to capture different word orderings. The search targeted studies published between January 2010 and June 2025 to reflect the progression of wearable technologies in stress prediction. All search results were imported into Mendeley, a reference management software, for further screening and analysis.

The NLP toolkit's [25] input parameters include keywords for identifying relevant articles and conditions for selected articles. In this research, the search keywords were "stress prediction with wearables", "stress prediction with mobile devices", "stress prediction with sensors", and "stress prediction with physiological data". The search was conducted four times across digital libraries, with duplicates removed based on unique identifiers. The inclusion criteria were used to evaluate the remaining papers. The search was conducted on June 20, 2025.

2.3. Inclusion criteria

The inclusion criteria of this systematic review are the following:

- **Relevance to Stress Prediction:** Studies must directly investigate stress prediction or related physiological monitoring using wearable devices. It includes research that focuses on assessing, evaluating, or analyzing stress indicators or their applications in mental health.
- **Use of Wearable Devices, Sensors, or Technology:** Studies must involve the use of wearable devices, sensors, or other technologies for physiological data collection relevant to stress prediction. This criterion encompasses research that leverages mobile devices, wearable sensors, and other technological solutions to enhance stress monitoring and management.
- **Healthcare and Mental Health Relevance:** Studies should have a healthcare or mental health focus, covering topics such as stress monitoring, mental health interventions, emotional health, or stress-related conditions in clinical or preventative settings.
- **Clear Objectives:** Studies must present their objectives clearly, specifying their approach to stress prediction using wearable sensor data and machine learning techniques.
- **Publication Year Range:** Articles published between January 2013 and June 2025 were selected to reflect recent advancements in wearable technology and AI for stress prediction.
- **Study Types:** The systematic review included various research categories, such as randomized trials, clinical trials, comparative studies, and other experimental designs focusing on stress prediction. Review articles and surveys were also included for comprehensive background information.
- **Target Beneficiaries and Users:** The review covers studies targeting different beneficiary groups, such as individuals in high-stress environments, healthcare providers, psychologists, and caregivers. This broad inclusion aims to provide insights into wearable-assisted stress prediction across diverse contexts.
- **Focus on Physiological and Behavioral Indicators:** Studies must address key indicators related to stress, including both physiological (e.g., HRV, body temperature, sleep patterns) and behavioral indicators (*e.g.*, physical activity, emotional state, and cognitive load). It also extends to research focused on early detection, prevention, and intervention for stress-related conditions. The emphasis is on understanding the full spectrum of stress indicators for comprehensive stress management.
- **Technological Considerations:** Studies should address challenges related to accessibility, data privacy, user acceptability, adoption, usability, reliability, safety, security, and trust within the context of wearable-assisted stress prediction.
- **Paper Language:** Only studies published in English were considered.

2.4. Exclusion criteria

The exclusion criteria of this systematic review are the following:

- **Exclusion of Review Papers:** Review papers, including systematic reviews, scoping reviews, and survey articles, were excluded to maintain a focus on original, primary research studies rather than summaries of existing literature.

- **Exclusion of Theoretical and Position Papers:** Position papers that present theoretical perspectives or communications without original data were excluded to ensure that only studies with empirical findings were included.
- **Exclusion of Commentary Articles:** Commentary articles, such as editorials and opinion pieces lacking original research findings, were not considered in the review to prioritize methodologically rigorous studies with new data on wearable-assisted stress prediction.

2.5. Extraction of study characteristics

The extraction focused on key aspects: year of Publication, Location, Population Studied, Purpose of the Study, Sensors or Equipment Used, Methods Employed, Features Analyzed, and Main Results. However, the diversity in extracted information led to the grouping of specific data into broader categories for better clarity.

All data were obtained from the full texts of the included studies. Abstracts were only used to evaluate possible eligibility during the first screening stage. Abstracts by themselves did not yield any findings or conclusions. In cases where essential methodological details were missing or unclear in the full text, those studies were flagged and either further verified or excluded to maintain rigor and data reliability.

Table 1 summarizes the basic characteristics of each study: **Year of Publication**, **Population**, **Purpose of Study**, **Sensor Used**, **Diseases or health conditions**, and **Main Results**. This table establishes relationships between population, study purpose, and targeted health conditions. Also, Table 2 focuses on the technological aspects, detailing: **Stress Assessment Methods**, **Supervised Learning Methods**, **Data Processing/Feature Extraction/Unsupervised Methods**, **Features Used**, and **Time Window Length/Resolution**. This table connects the technological setup with the analytical methods and study outcomes. This analysis intends to map the existing frameworks, aligning sensors, methods, and analytical features with real-world applicability in stress episode prediction.

3. Results

The PRISMA 2020 methodology [24] was employed to identify, screen systematically, and include studies for analysis. Fig. 2 illustrates the selection process, supported by the NLP toolset used for automation during screening [25]. During the identification phase, a total of 27,705 records were retrieved from six databases: 8205 from NCBI:PubMed, 13,073 from Springer, 365 from ACM, 548 from IEEE Xplore, 2498 from MDPI, and 3016 from Elsevier. No additional sources were searched. Before the screening, 6416 records were removed: 3542 due to duplication, 1666 by automation tools, and 1208 for other reasons. It resulted in 21,705 records proceeding to the screening phase. During screening, records were excluded for reasons including irrelevance (n = 5587), insufficient data (n = 3629), non-English language (n = 1318), wrong population (n = 1073), theoretical or position papers (n = 387), unclear methodology (n = 799), incorrect study design (n = 2500), insufficient outcome measures (n = 2200), data collection issues (n = 2700), incomplete reporting (n = 1000), and focus on non-human subjects (n = 769). Following this rigorous process, 66 studies were ultimately included in the review, forming the basis for both the qualitative synthesis and the quantitative meta-analysis.

The included studies span from January 2013 to June 2025, with a notable surge in publications after 2020. As summarized in Tables 1 and 2, approximately 34% of the studies were published in 2024, indicating a sharp rise in interest in wearable sensors and machine learning for stress detection. Another 30% were published in 2023, followed by 16% in 2022 and 9% in 2025. The remaining 29% of studies were distributed across the years 2013 to 2021, reflecting the field's steady evolution from its early exploration to a period of rapid expansion.

In terms of disease or population focus, only 8% of the included studies investigated stress-related conditions in specialized clinical cohorts, namely, individuals with intellectual disabilities, metabolic syndrome, mild cognitive impairment, rheumatic disease/chronic pain, or cardiac arrhythmia (Table 1). A further 8% centered on psychological or physical stressors inherent to high-stress occupations such as industrial shift work, firefighting, and construction, where the primary aim was operational safety and fatigue management. Mental-health–specific applications, covering major depressive disorder, anxiety disorders, obsessive-compulsive disorder, substance-use relapse prediction, and residual depressive symptoms, were examined in 14% of the papers, underscoring the expanding use of wearables for continuous psychiatric monitoring beyond short-term laboratory stress induction.

Regarding sensor types and equipment, heart rate monitoring (HRM) sensors, such as electrocardiography (ECG) and heart rate variability (HRV) sensors, were the most prevalent, used in approximately 79% of the reviewed studies. Electrodermal activity (EDA) sensors were also widely adopted, appearing in 70% of the studies. Accelerometers were used in 58% of cases, often for motion tracking or context-aware stress analysis. Skin temperature sensors were reported in 35% of studies. Additionally, 4% of the studies featured emerging hybrid biosensing devices, including wearable electrochemical and localized surface plasmon resonance (LSPR) sensors for cortisol detection in sweat.

From a methodological perspective, statistical methods were employed in 98% of the studies, providing the analytical foundation for data interpretation. Machine learning techniques, particularly Random Forest (RF) and Support Vector Machines (SVM), were applied in 80% of the studies, reflecting their widespread adoption in physiological signal classification. Ensemble models were featured in 26% of the studies, highlighting a growing trend toward model robustness through algorithmic integration. Deep learning methods were employed in 14% of the studies, primarily in contexts that require complex pattern recognition from multimodal or high-frequency data streams.

Feature extraction remains a cornerstone of predictive accuracy in stress detection research. Physiological features, such as heart rate variability (HRV) indices, electrodermal activity (EDA) parameters, and skin temperature, were utilized in approximately 88% of the studies, underscoring their foundational role. Behavioral features, including physical movement patterns and activity tracking via accelerometers, were incorporated in 45% of the studies. Additionally, contextual and environmental data, such as ambient temperature, humidity, and location, were considered in 18% of the studies. These trends reflect a growing shift toward holistic stress modeling that integrates both intrinsic physiological signals and extrinsic environmental and behavioral factors.

The primary sensor or technology for each study organizes the following sections focused on the study, such as Electrodermal Activity (EDA) and related biometrics (Section 3.1), Heart Monitoring and Related Technologies (Section 3.2), Specialty and Hybrid Sensors (Section 3.3) and Other Sensors (Section 3.4).

3.1. Electrodermal activity (EDA) and related biometrics

Ciccarelli et al. [26] investigates the use of wearable technology for stress detection in controlled laboratories, such as the Zephyr BioHarness 3 and Empatica E4. The study conducted external validation of stress detection models across different lab-based datasets, emphasizing the importance of validating wearable stress detection devices across various settings, as their inherent variability may compromise the generalizability of stress induction techniques. Wearable technology has the potential to facilitate proactive stress management in industrial and occupational settings, notwithstanding these obstacles.

In [28], the study utilizes physiological biomarkers, including ECG, EDA, and respiration, to validate wearable technologies for stress monitoring. It classifies stress levels in controlled lab settings and real-world

Table 1
Study analysis.

Paper	Year	Population	Purpose of study	Sensor used	Diseases/Conditions	Main results
Ciccarelli et al. [26]	2025	Industrial workers	Investigate the use of wearable devices for stress detection in controlled laboratory settings	Empatica E4 wristband (Electrodermal Activity, Heart Rate Variability); Zephyr BioHarness 3	Work-related stress in industrial settings	High accuracy in classifying stress levels using machine learning algorithms with HRV and EDA signals
Qiao et al. [27]	2025	5 subjects	Develop a wearable electrochemical biosensor for cortisol detection in sweat	PANI Hydrogel-based electrochemical sensor	Stress	The sensor demonstrated excellent antifouling performance and accurate cortisol detection with a detection limit of 33 pg/mL.
Darwish et al. [28]	2025	WESAD dataset [29]	Develop a framework for multimodal stress detection using wearable devices, transitioning from controlled lab settings to real-life scenarios.	Electrodermal Activity (EDA) sensors; Electrocardiography (ECG) sensors, Respiration	Stress	Achieved maximum accuracies of 99.78% and 99.61% for binary and five-class classification, respectively, using ECG, EDA, and respiration as reliable biomarkers for stress.
Singh et al. [30]	2025	VerBIO [31], UTD [32], WESAD [29], SWELL-KW [33], MSD Nurses [34], DriveDB [35]	To develop a deep learning framework for early stress detection using electrodermal activity (EDA) signals.	Electrodermal Activity (EDA) sensors	Stress, Anxiety	Achieved 95% accuracy for early stress detection with EDA signals. The deep learning approach outperforms the multimodal fusion of signals.
Tanwar et al. [36]	2025	Proprietary dataset	To develop a stress recognition system using autoencoders with wearable data.	Chest and wrist-worn wearable sensors	Stress	Achieved 96% accuracy for binary stress classification and 90% accuracy for multilevel classification (baseline, stress, fun).
Schreiber et al. [37]	2025	VerBIO [31], UTD [32], WESAD [29], SWELL-KW [33], MSD Nurses [34], DriveDB [35]	To integrate multimodal vital signs for stress detection and evaluation using wearable devices.	Heart rate, skin conductivity, temperature, accelerometer	Stress, Physical Stress	Achieved 79% accuracy for tri-class classification and 89% for binary classification (baseline vs. stress). Multimodal data showed advantages over single-sensor data.
Mai and Chung [38]	2024	15 participants (stress-inducing tasks)	To develop a wearable EEG-based stress detection system with embedded tiny neural networks for on-chip processing	Behind-the-ear EEG sensor	Mental Stress	The system achieved 95.32% accuracy in stress detection using EEG signals with on-chip processing and a user-friendly smartphone app for monitoring
Andrić et al. [39]	2024	Public datasets with controlled lab stress induction	To predict anticipatory stress using wrist-worn biomarker signals for early detection.	Blood volume pulse, skin conductance, skin temperature, accelerometer	Stress, Mental Health Issues	Random forest classifier showed promising results for anticipatory stress detection. Identified anticipatory physiological changes before full stress onset.
Hossain et al. [40]	2024	Healthy participants during cycling exercises	To develop and evaluate a wearable sensor suite for monitoring electromyogram (EMG) and cortisol levels	EMG, sweat cortisol, pH, temperature sensors	Stress, Muscle Tension, Hormonal Response	The system detected stress levels by combining real-time EMG signals and sweat cortisol measurements during exercises, achieving insightful results on muscle tension and hormonal response

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Table 1 (*continued*).

Paper	Year	Population	Purpose of study	Sensor used	Diseases/Conditions	Main results
Habib et al. [41]	2024	People with Major Depressive Disorder (MDD)	To create a deep learning model for detecting MDD from single-channel ECG signals	ECG sensor	Major Depressive Disorder (MDD)	MDDBranchNet improved MDD detection accuracy by 7% by using ECG-derived signals. It achieved 76% accuracy with a leave-subject-out strategy
Fontes et al. [42]	2024	Not reported	Proposal of a deep learning-based system for stress detection using remote photoplethysmography (rPPG) from facial videos	rPPG (camera-based)	Stress, Mental Health	The system achieved 95.83% accuracy for stress detection using facial video-based rPPG with hybrid deep learning models
Gungormus et al. [43]	2024	67 elderly adults	Develop semi-automatic mHealth system combining physiological and psychological data to monitor chronic pain	Empatica E4 wristband (Electrodermal Activity, Heart Rate Variability); Smartphone microservices platform	Rheumatic disease; Chronic pain	HRV and skin conductance predict multiple pain parameters; psychological stress relates to pain; wearable and clinical data integration enables early pain detection
de Vries et al. [44]	2024	71 individuals	Develop AI-based stress detection system for individuals with intellectual disabilities	Empatica E4 wristband (Electrodermal Activity, Heart Rate Variability); Movesense sensor (Heart Rate Variability); SentiSock wearable sensor	Intellectual Disabilities	Accuracy: 73%
Al-Alim et al. [45]	2024	240 participants from WESAD [29] and SWELL [46] datasets	Develop ML approach for real-time stress detection using wearable physiological data	Empatica E4 wristband (EDA, accelerometer, gyroscope, PPG, BVP); Respiration sensors	Stress	Random Forest, KNN, SVC, Decision Tree, XGBoost; SMOTE used for balancing; Accuracy up to 98.98% in three-level classification
Pickert et al. [47]	2024	12 healthy adults	Develop a wearable accelerometer-based system for ballistocardiography (BCG) signal analysis and heartbeat detection without ECG reference	Chest-mounted tri-axial accelerometer	Not Reported	Personalized heartbeat detection accuracy: 97.6% at rest, 71.9% during physical stress; enables ECG-independent cardiac monitoring
Nasirzadeh et al. [16]	2024	26 construction workers	Assess accuracy and comfort of off-the-shelf wearable temperature sensors for continuous monitoring in construction sites	VivaLNK (axillary temperature); Cosinuss (tympanic temperature); Equivital (axillary temperature, ECG, EDA)	Physical fatigue; Heat stress; Infectious diseases	All sensors showed bias <1 °C vs. clinical thermometers; Equivital rated most comfortable and practical for extended wear
Rescio et al. [48]	2024	20 participants	Develop a deep learning-based platform for worker stress detection using minimally intrusive wearable and ambient sensors	Electrocardiography (ECG); Photoplethysmography (PPG); Electrodermal Activity (EDA); RGB Camera; Thermal Infrared (IR) Imaging	Stress	1D-CNN achieved 95.38% accuracy for 3-class stress detection
Nuamah [49]	2024	35 participants	Investigate effects of recurrent task-induced acute stress on task performance, heart rate variability, and pupil response	Empatica E4 wristband (EDA, HRV); Tobii Pro Glasses 2 (pupil tracking)	Stress	Not Reported
Béquet et al. [50]	2024	85 participants	Investigate the efficacy of tactile respiratory guidance for stress regulation during autonomous vehicle takeovers	Custom wristband with 3 vibration motors for respiratory guidance	Stress regulation in driving context	Not Reported
Tanwar et al. [51]	2024	WESAD dataset [29]	Propose hybrid attention-based CNN-LSTM model for real-time multimodal stress detection	Empatica E4 wristband (EDA, HRV, skin temperature)	Stress detection	Accuracy: 92.7%; F1-score: 90%
Toshnazarov et al. [52]	2024	44 participants	Evaluate feasibility of stress detection using commodity smartwatches in real-life	Samsung Galaxy Watch 5 (PPG, accelerometer); Android smartphone (contextual data)	Stress	Lab condition (F1-score) 84%; Real-life (F1-score) 71%

(*continued on next page*)

Table 1 (*continued*).

Paper	Year	Population	Purpose of study	Sensor used	Diseases/Conditions	Main results
Carreiro et al. [53]	2024	30 participants	Assess accuracy and engagement of a digital health tool for stress and craving detection in substance use disorder recovery	Garmin Vivosmart 4 (PPG, HRV, accelerometer); ECG sensor	Stress; Craving	Stress detection accuracy: 71.6%; Craving detection accuracy: 71%
Kader et al. [54]	2024	20 healthy adults	Develop a one-channel wearable system combining EEG and GSR signals for mental stress detection using machine learning classifiers	Electroencephalography (EEG); Galvanic Skin Response (GSR)	Stress	Accuracy: 70.3%; EEG+GSR fusion accuracy: 84.6%
Volpes et al. [55]	2024	6 healthy adults	Design and validate a ring-shaped wearable device for synchronous acquisition of ECG, PPG, GSR, and motion signals on the finger	Photoplethysmography (PPG); Electrocardiography (ECG); Galvanic Skin Response (GSR); 3-axis accelerometer	Not reported	Demonstrated capability to detect physiological changes between rest and stress conditions; high-quality synchronous signals at 1 kHz sampling rate; validated PRV as surrogate for HRV in stationary states; motion artifacts impact signal quality during activity
Islam and Washington [56]	2023	WESAD dataset [29]	Develop personalized stress prediction model using self-supervised pre-training on biosignals	Electrodermal Activity (EDA) sensors; Electrocardiography (ECG) sensors	Not reported	Demonstrated personalized SSL models outperform supervised baselines with fewer labels, enabling efficient stress prediction with minimal annotation effort
Dahal et al. [57]	2023	WESAD dataset [29]; SWELL dataset [46]	Develop global stress detection framework combining HRV features and Random Forest model	Blood Volume Pulse (BVP); Electrocardiography (ECG); Electromyography (EMG); Electrodermal Activity (EDA); Heart Rate Variability (HRV); Photoplethysmography (PPG); Respiration; Temperature; Accelerometer	Stress	Accuracy: 99%
Parousidou et al. [58]	2023	WESAD dataset [29]; SWELL dataset [46]; ADARP dataset [59]; LifeSnaps dataset [60]	Improve stress detection using personalized machine learning techniques	Heart rate (HR) sensor; Electrodermal Activity (EDA) sensor	Not reported	Personalized models: up to 99% F1-score in lab, 66% in-the-wild
Whiston et al. [61]	2023	22 individuals remitted from depression	Explore longitudinal interactions between physiological stress and residual symptoms in remitted depression using wearable sensors	Empatica E4 wristband (Electrodermal Activity, Heart Rate, Accelerometer)	Residual symptoms of depression	Bidirectional relationships found between skin conductance responses and depressive symptoms; potential for relapse prediction
Nan et al. [62]	2023	3 subjects	Develop wearable localized surface plasmon resonance (LSPR) biosensor for sensitive cortisol detection in sweat	Wearable LSPR biosensor with gold nanoparticle (AuNP)-functionalized PDMS substrate	Not reported	Detection limit: 0.1 nM cortisol; Non-invasive, flexible, selective biosensor for cortisol monitoring in human sweat
Zheng et al. [63]	2023	Proprietary dataset	Create flexible stress sensor mimicking spider web structure using laser-induced graphene on PDMS substrate	Laser-induced graphene (LIG) flexible strain sensor on PDMS substrate	Stress	Not reported
Mitro et al. [64]	2023	15 volunteers	Monitor passengers' stress during evacuations	Photoplethysmography (PPG) sensor; Pulse oximeter; Heart rate sensor	Stress	Accuracy: 76%

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Table 1 (*continued*).

Paper	Year	Population	Purpose of study	Sensor used	Diseases/Conditions	Main results
Campanella et al. [65]	2023	29 participants	Evaluate stress detection models using Empatica E4 signals	Empatica E4 wristband (Electrodermal Activity, Heart Rate, Accelerometer, Temperature)	Stress	Accuracy: 76.5% (Random Forest), 74.5% (SVM), 72.8% (Logistic Regression)
Costantini et al. [66]	2023	14 participants	Evaluate Empatica E4 wristband accuracy for HRV and EDA measurement in stressful driving scenarios	Empatica E4 wristband (PPG for HRV; skin electrodes for EDA; accelerometer; temperature sensor)	Not reported	Not reported
Bocharov et al. [67]	2023	5 older adults	Assess physiological cost during combat activities using wearable ECG device	Miniature wearable ECG patch device (Solvaig LTD)	Not reported	Not reported
Stržinar et al. [68]	2023	WESAD dataset [29]	Detect stress using frequency spectrum analysis of wrist-measured electrodermal activity (EDA)	Empatica E4 wristband (EDA)	Stress detection	Binary classification accuracy: 84.87%, F1-score: 82.13%
Vos et al. [69]	2023	Combined datasets (SWELL [46], NEURO [70], WESAD [29], UBFC-Phys) totaling 99 subjects	Develop ensemble machine learning model trained on the synthesized dataset to improve stress prediction generalization	Electrodermal Activity (EDA) sensors; Electrocardiography (ECG) sensors; Heart Rate Variability (HRV) sensors; Accelerometer	Not reported	Accuracy: 85%
Gaitán-Padilla et al. [71]	2023	5 volunteers	Evaluate low-cost optical fiber sensor for stress classification	Polymeric optical fiber sensor worn at chest	Stress	Accuracy: 88.4%; Precision: 91.3%
Almadhor et al. [72]	2023	WESAD dataset [29]	Develop federated learning system for stress detection	Accelerometer; Blood Volume Pulse (BVP) sensor; Skin temperature sensor; Electrodermal Activity (EDA) sensor	Stress	Accuracy: 86.82%
Almadhor et al. [73]	2023	WESAD dataset [29]	Stress detection via machine learning using chest Electrodermal Activity (EDA)	Chest-worn sensors measuring Electrodermal Activity (EDA), Electromyography (EMG), Temperature, Accelerometer, Electrocardiography (ECG), Respiration (RESP)	Not reported	Accuracy: 99.7%, F1-score: 99.8%
Zamani et al. [74]	2023	400 subjects	Develop deep learning model for predicting sleep quality using wearable data	Smartwatches; Smart rings; Smart clothing wristbands; GPS-enabled shoes	N/D	Accuracy: 97.54%; Precision: 95.37%; Recall: 94.91%; F-Score: 94.82%
Zhou et al. [75]	2023	APD [76], WESAD [29], CASE [77] datasets	Explore generalizability of stress and anxiety physiological features across datasets	ECG (HRV, frequency-domain metrics), Electrodermal Activity (EDA) sensors	Stress, Anxiety	Accuracy (CASE): 68.3%, AUC (CASE): 76.1%; Accuracy (WESAD): 56.9%, AUC (APD): 56.1%
Momeni et al. [78]	2022	Proprietary dataset	Develop cost-aware feature selection method for multimodal stress monitoring	Electrodermal Activity (EDA); Photoplethysmography (PPG); Accelerometer; Electrocardiography (ECG)	Not reported	Accuracy: 90.98%
Rominger and Schwerdtfeger [79]	2022	38 firefighters	Develop algorithms to detect stress transitions using HRV-based triggers	Mobile ECG device (ECG, HRV); Accelerometer	Stress	Not reported
Ritsert et al. [80]	2022	19 participants	Analyze heart rate variability (HRV) and breathing rate variability (BRV) to detect anxiety	Heart Rate Variability (HRV) sensors; Breathing rate sensors	Anxiety	Not reported

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Table 1 (*continued*).

Paper	Year	Population	Purpose of study	Sensor used	Diseases/Conditions	Main results
Chalabianloo et al. [81]	2022	32 healthy participants	Evaluate wearable devices for stress detection	Photoplethysmography (PPG) sensors; Electrodermal Activity (EDA) sensors; Accelerometer; Electrocardiography (ECG) sensors; Electroencephalography (EEG) sensors; Barometer	Not reported	Accuracy: 88%
Cummings et al. [82]	2022	99 pregnant women	Assess the feasibility of ECG biosensors and EMA for stress detection during pregnancy	Electrocardiography (ECG) biosensor	Pregnant	ECG adherence rate: 52%
Ehrhart et al. [83]	2022	35 subjects	Generate synthetic time series data to improve stress detection	Galvanic Skin Response (GSR) sensor; Skin Temperature (ST) sensor	Not reported	LSTM accuracy: 88%
Zhang et al. [84]	2022	86 participants	Explore relationship between urban features and physiological stress responses	Electrodermal Activity (EDA) sensors; Skin temperature sensor; Wearable camera; GPS device	Stress	Accuracy: 84%
Zhong et al. [85]	2022	63 subjects	Detect psychological stress in VR high-altitude simulation using ECG	Smart ECG T-shirt with 3-lead ECG sensors	Stress	Accuracy: 78%
Thierfelder et al. [86]	2022	5 adolescents	Monitor stress and compulsive behavior in children with OCD during telepsychotherapy	ECG chest belt; Inertial sensors (hand movements); Eye tracker	Obsessive-Compulsive Disorder (OCD)	Not reported
Iqbal et al. [87]	2021	WESAD dataset [29]	Sensitivity and specificity analysis for stress detection	Electrocardiography (ECG) sensors; Electrodermal Activity (EDA) sensors; Electromyography (EMG) sensors; Respiration sensor; Temperature sensor	Not reported	Logistic regression: 96% accuracy
Hag et al. [88]	2021	DEAP dataset [89]	Optimize EEG-based stress recognition using hybrid feature selection	Electroencephalography (EEG) wearable device	Not reported	Accuracy: 93.88%
Hossain et al. [90]	2021	MIT-BIH Arrhythmia database [91]	Electrocardiography (ECG) denoising technique for QRS detection	Electrocardiography (ECG) sensors	Arrhythmia	Detection rate: 92.96%
Aristizabal et al. [92]	2021	18 healthy adults	Feasibility of wearable devices for stress detection	Electrodermal Activity (EDA) sensors; Photoplethysmography (PPG) sensors; Accelerometer; Optical thermometer	N/D	Accuracy: 96% (combined data), 88% (wearable data alone)
Föll et al. [93]	2021	WESAD dataset [29]	Open-source toolkit for standardizing wearable data processing and feature extraction for machine learning	Electrocardiography (ECG) sensors; Heart rate variability (HRV) sensors; Electrodermal activity (EDA) sensors; Accelerometer	Not reported	Accuracy: HRV (F1 = 57.59%), EDA (F1 = 51.96%), Accelerometer (F1 = 56.55%) using LDA
Dzieżyc et al. [94]	2020	WESAD dataset [29]	Evaluate deep learning models for affect recognition	Electrocardiography (ECG) sensors; Electroencephalography (EEG) sensors; Electromyography (EMG) sensors; Blood Volume Pulse (BVP) sensors; Electrodermal Activity (EDA) sensors	Not reported	F1-score: 73%; AUC: 91%

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Table 1 (*continued*).

Paper	Year	Population	Purpose of study	Sensor used	Diseases/Conditions	Main results
Bolliger et al. [95]	2020	55 academic staff	Investigate day-to-day occupational stress using combined ecological momentary assessment and physiological wearable data	Empatica E4 wristband (EDA, HR, skin temperature, accelerometer); Android smartphone app (sensor and usage data)	Occupational stress	Multimodal data enable rich modeling of day-to-day stress in real-world work environments
Delmastro et al. [96]	2020	Frail older adults	Evaluate cognitive and physical training impact on stress in MCI older adults using wearable physiological sensors	Electrocardiography (ECG) sensors; Electrodermal Activity (EDA) sensors; Heart Rate (HR); Heart Rate Variability (HRV) sensors	Mild cognitive impairment	Random Forest and AdaBoost: 85.3% accuracy; adaptive stress response observed; cognitive performance improved after physical exercise
Akbulut et al. [97]	2020	30 patients	Stress recognition in metabolic syndrome patients	Electrocardiography (ECG); Galvanic Skin Response (GSR); body temperature; Oxygen Saturation (SpO2) sensors	Metabolic syndrome	Accuracy: 92%
Can et al. [98]	2020	32 participants	Enhance person-independent stress detection	Electrocardiography (ECG) sensors; Heart Rate Variability (HRV) sensors; Electrodermal Activity (EDA) sensors; Accelerometer	Stress	Accuracy: HRV: 94.44%; EDA: 92.89%
Zubair and Yoon [99]	2020	14 graduate students	Real-time stress detection using ultra-short Photoplethysmography (PPG) signals	Photoplethysmography (PPG) sensors; Heart Rate Variability (HRV) sensors	Stress	Accuracy: 94.33%
Mozgovoy [100]	2019	18 participants	Improve stress recognition using motion intensity	Galvanic Skin Response (GSR); Heart Rate (HR); Accelerometer	Stress	Not reported
Rodrigues et al. [101]	2018	21 firefighters	Investigate stress impact on cognitive performance using wearable ECG and HRV	Vital Jacket® wearable ECG with accelerometer	Stress	Acute stress caused changes in HRV parameters (AVNN, SDNN, LF/HF) and impaired cognitive performance after TSST
Betti et al. [102]	2018	15 participants	Evaluate wearable system for stress detection	Zephyr BioHarness (ECG, HRV); Shimmer Sensor (EDA); MindWave Mobile EEG headset (EEG)	Stress	Accuracy: 86%; sensitivity: 84%; specificity: 90%
Xu et al. [103]	2015	39 healthy adults	Develop cluster-based methodology for inter-subject differences in stress responses	Electrocardiography (ECG) sensors; Electrodermal Activity (EDA) sensors; Galvanic Skin Response (GSR) sensors; Heart Rate Variability (HRV) sensors	Not reported	Accuracy: 85.2%
Seoane et al. [104]	2013	Spanish Armed Forces personnel	Real-time stress assessment using wearable sensorized garments	Textile electrodes; Galvanic Skin Response (GSR) sensors; temperature sensors; Electrocardiography (ECG) amplifier	Stress	Not reported

situations with up to 99.78% accuracy using machine learning algorithms. According to the results, wearable technology can improve early stress detection and management by bridging the gap between experimental and practical applications. While the study achieved high accuracy on internal validation (99.78%), external validation using real-world data showed significantly lower performance (accuracy: 0.73, F1-score: 0.58), highlighting the challenges in translating lab-trained models to real-world environments. However, limitations include the fact that stress reactions vary depending on the environment. In addition, the study examines wearable technology that supports these physiological modalities and provides valuable guidance on how to utilize it for real-time stress monitoring.

The study [30] looks for early indicators of stress using electrodermal activity (EDA) sensors in wearable technology. It created a model based on deep learning that is capable of real-time stress level classification. Healthy adults participated in the study, which was carried out in the United States. Using a convolutional neural network, the model achieved high accuracy in predicting stress levels. According to the study, EDA signals processed by a deep learning model may be a valuable tool for real-time stress monitoring in daily situations. The small sample size and controlled setting of the study, however, may limit the applicability of the results to broader populations.

Using wearable sensors to integrate multiple physiological signals, the VitaStress sought to increase the accuracy of stress detection [37].

Table 2
Outcomes study analysis.

Paper	Stress assessment methods	Supervised learning methods	Data processing/Feature extraction/Unsupervised methods	Features used	Time window length/Resolution
Ciccarelli et al. [26]	Heart Rate Variability (HRV), Electrodermal Activity (EDA)	Support Vector Machine (SVM)	Signal processing with Welch's method for HR spectral analysis, bandpass filtering, and EDA decomposition using Continuous Decomposition Analysis (CDA)	HRV, EDA (tonic and phasic components), RR intervals, PPG	Data resampled to ensure consistency; no fixed time window, but baseline and task-specific phases analyzed
Qiao et al. [27]	Cortisol levels in sweat (bio-marker for stress)	No specific supervised learning methods mentioned	Signal processing using differential pulse voltammetry (DPV), electrochemical impedance spectroscopy (EIS), and X-ray photoelectron spectroscopy (XPS)	Cortisol concentration, sensor impedance	Real-time monitoring of cortisol levels, no specific time window length given, continuous measurement during testing phases
Darwish et al. [28]	ECG, EDA, Respiration signals collected using wearable devices	Random Forest (RF), XGBoost (XGB), K-Nearest Neighbors (KNN)	Signal preprocessing (filtering, segmentation), Feature extraction from time and frequency domains (PSD, Skewness, Kurtosis, etc.), Hyperparameter optimization	HR, HRV, EDA, RESP, Frequency domain features (FFT)	300 s, 390 s window lengths with 50% overlap
Singh et al. [30]	Electrodermal Activity (EDA) signals from wrist-worn devices	Deep learning models: FNN, CNN, Wide and Deep	Advanced preprocessing techniques; feature extraction from EDA signals; multimodal signal fusion (EDA, ECG, PPG)	EDA signal features; ECG and PPG signals (when fused)	Continuous monitoring; no specific time window length provided
Tanwar et al. [36]	Stress recognition using wearable data	Autoencoder-based neural network architecture	Feature extraction via autoencoder; classification into stress, non-stress, and amusement states	Features extracted from wearable sensor data	Continuous monitoring; no specific time window length provided
Schreiber et al. [37]	Multimodal vital signs for stress detection	Machine learning classifiers	Data collection from multiple sensors; classification into baseline, physical stress, and overall stress states	Heart rate, RR intervals, respiration, skin conductance, temperature, accelerometer data	Continuous monitoring; no specific time window length provided
Mai and Chung [38]	EEG signals captured from behind-the-ear (BTE) electrode, including temporal resolution for stress-related brain activity analysis	Convolutional Neural Networks (CNN)	Signal-to-spectrogram conversion (FFT), filtering, noise removal, real-time on-chip processing	EEG spectrogram images, frequency bands (Delta, Theta, Alpha, Beta, Gamma)	Continuous EEG data collected in 1-s segments (128 data points per segment)
Andrić et al. [39]	Biomarker signals from wrist-worn devices for stress anticipation	Random Forest classifier	Data mining methods for feature extraction; classification into no-stress, stress anticipation, stress, and post-stress states	Blood volume pulse, skin conductance, skin temperature, acceleration data	Continuous monitoring; no specific time window length provided
Hossain et al. [40]	Sweat cortisol levels and EMG signals measured during cycling exercise	Not reported	Signal processing of EMG using power spectral density, data correction for temperature and pH variations in sweat	EMG signals, sweat cortisol, pH, and temperature data	Continuous measurement during exercise with focus on interval measurements (e.g., every 5 min during cycling)
Habib et al. [41]	ECG signals, R-R intervals, Horizontal Visibility Graph (HVG)	Convolutional Neural Networks (CNN)	Signal resampling, ECG segmentation, Normalization, Derived signal features (R-R signal, HVG)	ECG signal, R-R signal, Degree distribution time series of HVG	20 s, 30 s, 60 s segments with overlapping
Fontes et al. [42]	Remote photo-plethysmography (rPPG) from facial videos	Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), 1D Convolutional Neural Network (1D-CNN)	Hyperparameter optimization, data augmentation, feature extraction from rPPG signals, deep learning-based models	rPPG signal features, physiological signals extracted from facial video frames	Continuous, video frames processed for real-time stress analysis (no fixed time window length)

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Table 2 (*continued*).

Paper	Stress assessment methods	Supervised learning methods	Data processing/Feature extraction/Unsupervised methods	Features used	Time window length/Resolution
Gungormus et al. [43]	Perceived Stress Scale, Stress Vulnerability Scale, Heart Rate Variability, Skin Conductance (mean values)	Multiple Linear Regression	Heart Rate Variability computed from Blood Volume Pulse at 1 Hz; Skin Conductance sampled at 4 Hz; Data aggregated by activity phase (e.g., sitting, walking, shopping)	Heart Rate Variability, Skin Conductance, Perceived Stress, Stress Vulnerability	HRV: 1-s intervals; SC: 0.25-s intervals; Aggregated by activity phases during the shopping task
de Vries et al. [44]	Stress labels from caregiver observations and video annotations (baseline, stress, recovery phases)	Random Forest (general/personal-ized), Decision Tree, AdaBoost, k-Nearest Neighbors	Sliding 60 s windows with 50% overlap; Z-score normalization; Principal Component Analysis	Electrodermal Activity, Heart Rate, Skin Temperature, Accelerometer metrics (e.g., mean, slope)	60 s window with 30 s step (50% overlap)
Al-Alim et al. [45]	Self-reported stress labels collected via Ecological Momentary Assessment and physiological markers aligned with free-living daily life	K-Nearest Neighbors, Support Vector Machines, Decision Tree, Random Forest, XGBoost	Sliding windows (length unspecified); SMOTE applied for class balancing; features z-normalized	Electrocardiogram, Skin Temperature, Skin Conductance (from SWEET dataset)	Not explicitly stated; aligned with user-reported stress events in free-living settings
Pickert et al. [47]	No explicit stress assessment; focused on cardiac signal quality under rest and physical stress conditions	None (rule-based detection)	Preprocessing: bandpass filter (20–35 Hz), absolute acceleration, lowpass filter (5 Hz), total acceleration calculated from 3 axes; Rule-based heartbeat detection adapted from Pan–Tompkins algorithm	Seismocardiography from tri-axial accelerometers; derived systolic and diastolic peaks	Real-time sections of 1.5 s with 50 ms overlap; 2 s initial learning phase
Nasirzadeh et al. [16]	Continuous body temperature monitoring to infer heat-related stress risks in construction workers	None – evaluation of wearable sensor accuracy and comfort	Bland–Altman analysis; Mean Absolute Error and Root Mean Square Error computed; comfort feedback via survey; data segmented into 30 min intervals	Skin temperature (via wearable devices), reference core temperature, ambient temperature, self-reported comfort levels	Continuous logging at device-dependent resolution (e.g., ≈1min for typical wearables), analyzed in 30 min blocks
Rescio et al. [48]	Stress levels labeled based on physiological response data collected during real work shifts with minimally intrusive wearable and ambient sensors	1D Convolutional Neural Network (outperforms LSTM, GRU)	Signal denoising and segmentation; deep learning-based automatic feature extraction from raw time-series; compared architectures (1D-CNN, LSTM, GRU)	Electrodermal Activity, Heart Rate (ECG/PPG), Skin Temperature, Breathing rate	Real-time processing; window length as per model input (e.g., ≈60 s segments); system achieves stress vs. no-stress classification at 95% accuracy
Nuamah [49]	Acute stress induced by repeated difficult arithmetic tasks; stress quantified via subjective Visual Analogue Scale, vagally mediated heart rate variability (vmHRV), and task-evoked pupillary response (TEPR)	None (statistical modeling)	RMSSD, pNN50, and HF-HRV extracted from ECG; task-locked event averaging for pupil responses; mixed-effects models and repeated-measures correlation analyses	Vagally mediated HRV (RMSSD, pNN50, HF-HRV), pupil dilation/task-evoked pupillary response, subjective stress	Data segmented around repeated task blocks; physiological and pupillary metrics computed per block (exact epoch durations aligned with task trials)
Béquet et al. [50]	Slow-paced breathing guidance provided via wrist-worn tactile wearable; stress inferred through physiological response during task blocks (driving+NDRT)	Logistic Regression used to model individual differences in ease-of-use and physiological regulation	Signal cleaning and segmentation aligned with intervention blocks; statistical comparison of breathing rate, RMSSD, and SWRR between guidance conditions; logistic regression for individual moderators	Respiratory rate, Heart Rate Variability (RMSSD), Steering Wheel Reversal Rate, Subjective distraction and synchronization difficulty ratings	Segmented by condition blocks (Stress/Non-stress with/without respiratory guidance); block durations defined by NDRT intervals (≈minutes each)

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Table 2 (*continued*).

Paper	Stress assessment methods	Supervised learning methods	Data processing/Feature extraction/Unsupervised methods	Features used	Time window length/Resolution
Tanwar et al. [51]	Labeled using WESAD dataset (baseline, stress, amusement)	Attention-based hybrid 1D-CNN + BiLSTM model, compared with SVM, Random Forest, K-Nearest Neighbors	Signals normalized and segmented into 5-s windows with 2-s shift; attention mechanism applied for temporal feature weighting	Electrodermal Activity, ECG, Body Temperature, Respiration, Accelerometer data	5-s windows, 2-s step as per WESAD preprocessing standards
Toshnazarov et al. [52]	Self-reported stress via ecological momentary assessment in real-life and lab settings; smartwatch physiological data	XGBoost, LightGBM, Logistic Regression, AdaBoost, Gradient Boosting, MLP, Random Forest, SVM (lab layer); second-layer field model fuses physiology and context	Sliding windows (1 min), motion and looseness artifact filtering, z-score normalization, contextual feature fusion, two-layer structure (lab and field)	Heart Rate, Heart Rate Variability (IBI), accelerometer motion, contextual smartphone data (location, activity), past stress predictions	1 min physiological-contextual windows in lab and wild; F-1 scores up to 0.84 (lab) and 0.71 (field)
Carreiro et al. [53]	Self-reported stress and craving events collected via ecological momentary assessment over 30 days	Supervised machine learning (e.g., XGBoost, LightGBM, Logistic Regression) to develop group-level digital biomarkers	Continuous physiological monitoring; sliding-window feature extraction; retrospective model training on labeled events; performance evaluated using accuracy and F1 metrics	Electrodermal Activity, Heart Rate, Physical Movement (wearable wrist sensor)	Continuous data collected; matched with EMA reports at ≈15 min intervals; event-based windows extracted (≈20-40min around reports)
Kader et al. [54]	Acute mental stress-induced via Stroop color-word test; labels derived from task condition (stress vs. control) without self-report scales	Naïve Bayes, K-Nearest Neighbors, Support Vector Machine (linear), NB+GSR modeling	EEG and GSR signal preprocessing (artifact removal, threshold filtering); EEG epoch segmentation (≈1 s); band power (delta, theta, alpha, beta); GSR statistical features; feature fusion	Single-channel EEG (Fp1/Fp2 or F7/F8) band power; Galvanic Skin Response (GSR) statistical features (AC-based)	EEG epochs of ≈1 s; GSR aligned to EEG epochs; classification across 86 epochs per condition
Volpes et al. [55]	Compared rest vs. induced orthostatic/physical stress (standing, walking) to evaluate device capability in sensing stress-related changes	None (hardware validation only; statistical analysis)	Filtering and artifact removal; computation of time/frequency/information-domain indices; comparison between postures	Electrocardiogram, Photoplethysmography, Galvanic Skin Response, Motion; derived HR, HRV, Pulse Arrival Time, SpO2, skin conductance metrics	Continuous acquisition; analyzed over posture segments (≈minutes each)
Islam and Washington [56]	Ecological momentary assessment (EMA) stress labels aligned with WESAD dataset segments	1D Convolutional Neural Network pre trained via self-supervised learning, then fine-tuned with supervised classifier (XGBoost/Logistic Regression comparison)	Self-supervised contrastive pretraining on unlabeled EDA segments; z-score normalization; fine-tuning on labeled data per individual	Electrodermal Activity (EDA) from wrist wearables (WESAD)	10-s pretraining windows; stress-classification on 60-s sliding windows with overlap; fine-tuned with <30% labeled data
Dahal et al. [57]	Stress labeled via controlled stress and baseline tasks from WESAD and SWELL datasets	Random Forest (global model), compared with Decision Tree, SVM, Naïve Bayes	HRV features extracted from ECG; minimum redundancy–maximum relevance (mRMR) used to select top 8 features; z-score normalization	Heart Rate Variability metrics	5 min ECG segments aligned with stress tasks for HRV computation
Parousidou et al. [58]	Physiological + Subjective (Perceived Stress)	Machine Learning and Deep Learning	Various personalization methods: user-based, group-based (single- and multi-attribute), fuzzy clustering; K-means and Fuzzy C-means used for grouping	Heart Rate Variability, Acceleration, Sleep, Activity, Demographics (varied by dataset)	Lab (WESAD, SWELL-KW), In-the-wild (ADARP, LifeSnaps); 2 h–4mo durations, up to 71 participants

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Table 2 (*continued*).

Paper	Stress assessment methods	Supervised learning methods	Data processing/Feature extraction/Unsupervised methods	Features used	Time window length/Resolution
Whiston et al. [61]	Physiological stress indexed via skin conductance response (SCR) during daily life; used as a proxy for residual symptom fluctuations in remitted depression	None (network modeling and temporal correlation analysis)	Skin conductance data preprocessed and aggregated; symptom–stress interactions examined via network analysis and longitudinal mixed-effects models	Skin Conductance Response (SCR), wearable-based physiological signals, residual depressive symptoms scores	Longitudinal daily monitoring over several weeks; daily-level aggregation aligned with symptom assessments
Nan et al. [62]	Cortisol concentration in sweat measured by wearable LSPR biosensor; no induced stress tasks or ML modeling	None (biosensor validation only)	Spectral analysis of LSPR peak shifts; calibration using standard cortisol solutions; signal stability under mechanical deformation	Sweat cortisol concentration; peak wavelength shift; biosensor spectral response	Real-time continuous measurement; response time in minutes, high-frequency spectral sampling
Zheng et al. [63]	Piezoresistive strain-based stress sensing using laser-induced graphene/PDMS sensor with spider-web–inspired structure	None (hardware and material validation only)	Spider-web structural fabrication; strain-resistance calibration (gauge factor GF = 36.8); sensor characterization via SEM, Raman spectroscopy, FTIR	Resistance changes under tensile strain (0%–35%); fast response (≈260 ms)	Real-time tests on cyclic strain (10%–30%, 3000 cycles); monitors finger, fist, and arm movement
Mitro et al. [64]	Stress detected via ultra-short-term pulse rate variability during emergency and lab scenarios; labels based on stress conditions	Lightweight machine learning pipeline	Ultra-short (30 s) PPG segments; preprocessing includes moving average subtraction and filtering; time and frequency domain features extracted onboard	Pulse Rate Variability (from PPG), Oxygen Saturation, Pulse Rate	30-s sliding windows with real-time counts executed on microcontroller
Campanella et al. [65]	Stress vs. non-stress induced via tailored protocol (cognitive and emotional tasks), labeled based on task periods	Random Forest, Support Vector Machine (RBF kernel), Logistic Regression; Random Forest performed best	Preprocessing of PPG and EDA; feature selection via chi-squared and Pearson correlation; 27 statistical features extracted	Photoplethysmo-gram (PPG), Electrodermal Activity (EDA)	Task-aligned segments (length unspecified, matched to E4 sampling rates and protocol tasks)
Costantini et al. [66]	Stress and arousal elicited using video clips, scream task, and simulated driving risk levels; subjective appraisal not directly assessed	None (validation study comparing wearable vs. laboratory-grade devices)	Semi-automatic reconstruction of inter-beat intervals and EDA signals; Bland–Altman and Spearman correlation analyses for signal agreement	Heart rate variability (from PPG), heart rate, electrodermal activity (EDA)	Six discrete conditions (baseline, video, scream, low/mid/high-risk driving); analyzed in segments matching experimental phases (≈minutes each)
Bocharov et al. [67]	Continuous ECG monitoring under physical and cognitive load; physiological cost measured via HRV changes	None (descriptive physiological metrics)	ECG signal processed to extract time-domain HRV (SDNN, RMSSD); physiological cost index calculated and statistically compared across activity phases	Heart Rate Variability (SDNN, RMSSD), Heart Rate, Physiological Cost Index	Analyzed in 5 min segments spanning pre-task, during-task, and post-task phases in real-world field contexts
Stržinar et al. [68]	Controlled tasks (baseline, stress, amusement) labeled via WESAD dataset	SVM (linear and RBF), k-Nearest Neighbors, Decision Tree, Logistic Regression	Wrist EDA filtered and normalized; frequency-domain features via FFT in multiple bands; time-to-detection optimized with 60-s windows and 30-s step	EDA spectral features: mean, standard deviation, peak frequency, centroid, slope from low-/high bands	60-s sliding windows, 30-s overlap; used for rapid stress detection
Vos et al. [69]	Stress labels derived from task-induced segments across WESAD, SWELL, NEURO, UBFC-Phys	Ensemble model combining Gradient Boosting and Neural Network; compared to Random Forest, AdaBoost, XGBoost, Gaussian NB	Z-score normalization; sliding windows (25 s); correlation and gain-ratio feature selection; synthesized dataset with 200 subjects created via random sampling	HR, HRV, EDA, BVP, Accelerometer (features aggregated across multiple public datasets)	25–60 s windows depending on source dataset protocols

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Table 2 (*continued*).

Paper	Stress assessment methods	Supervised learning methods	Data processing/Feature extraction/Unsupervised methods	Features used	Time window length/Resolution
Gaitán-Padilla et al. [71]	Stress-induced using Stroop-like word color tests and mental algebra tasks; labeled as stress vs. non-stress	Bagged Decision Tree Classifier (BDTC)	Butterworth filtering of respiration and pulse signals; MATLAB peak detection; 22 features including statistical and inter-peak metrics	HRV, pulse signal peaks, respiration rate, inter-peak intervals, statistical moments	1 min segments for each task phase; total 50 samples (35 train/15 test)
Almadhor et al. [72]	Labels from WESAD dataset (baseline, stress, amusement, meditation, transient) via controlled experimental conditions	Federated Deep Neural Network (two-client setup)	Min–max normalization and SMOTE on client-side; local model training with DNN; model weights aggregated centrally; evaluation by accuracy, precision, recall, F1, AUROC	Wrist electrodermal activity (EDA) processed through SMOTE-normalization pipeline	30–60 s windows per WESAD protocol; client-side splits ≈50% of data each
Almadhor et al. [73]	Chest-based EDA signals labeled from WESAD dataset for stress detection	Stacking ensemble (e.g., RF, SVM, KNN); feature selection using correlation filter	Z-score normalization; feature filtering and stacking-based model pipeline; temporal segmentation	Electrodermal Activity (EDA) features from chest belt (time-domain stats)	Aligned with WESAD window protocol (30–60 s sliding windows)
Zamani et al. [74]	Sleep quality assessed via wearable-collected activity data used as an indirect proxy for stress susceptibility	Deep Belief Network (DBN) with hyperparameters tuned using Enhanced Seagull Optimization; compared against CNN, MLP, Logistic Regression	Data preprocessing with normalization; deep feature learning via DBN; hyperparameter tuning (ESGO); comparison to traditional ML pipelines	Wearable-derived activity and sleep features collected during awake hours (e.g., movement, heart rate)	Aggregated per sleep episode; exact window lengths derived from nightly sleep cycles (≈hours)
Zhou et al. [75]	Anxiety induced using cognitive tasks; labels confirmed via State-Trait Anxiety Inventory (STAI) questionnaire	Support Vector Machine (SVM), k-Nearest Neighbors, Logistic Regression, Decision Tree; SVM	Signal preprocessing; feature extraction including time-domain (mean, SD) and frequency-domain (band powers); feature selection via ANOVA; Leave-One-Subject-Out cross-validation	Electroencephalo-gram (EEG; 8 bands), Electrocardiogram (ECG/HRV), Galvanic Skin Response (GSR), Respiration	1 min sliding windows used for individualized prediction training and evaluation
Momeni et al. [78]	Acute stress-induced under controlled lab tasks; labels aligned with stress vs. non-stress conditions	Decision Tree, k-Nearest Neighbors, Random Forest, XGBoost; feature subsets selected via CAFS optimization	Cost-aware feature selection trading prediction accuracy and energy; self-aware model switching; z-score normalization	Heart rate, skin conductance level, respiratory and pulse wave features	Window lengths adapted to lab protocols (e.g., ≈60 s); energy-scalable models selected dynamically
Rominger and Schwerdtfeger [79]	Stress increases labeled based on emergency vs. routine operations among firefighters over 24 h field recordings	None (rule-based trigger algorithms)	Static and dynamic AddHRVr algorithms applied to RMSSD; binary triggers based on window thresholds (e.g., 7 out of 10 segments); multilevel modeling and bootstrap power simulations to select optimal settings	Root mean square of successive differences (RMSSD) from ECG-derived HRV	1 min streaming windows with dynamic thresholding; algorithm settings tested across combinations (e.g., 10, 20 min silent periods)
Ritsert et al. [80]	Anxiety induced via alternating video clips (anxiety-inducing vs. non-anxiety-inducing)	None (statistical analysis: ANOVA, Wilcoxon tests)	Butterworth bandpass filtering; peak-to-peak interval extraction; Kubios HRV	ECG: MHR, SD, SDNN, RMSSD; RSP: MBR, SD, RMSSD, SDNN; ECG frequency domain: LF, HF, LF/HF	60 s (non-transitional phase); 15 s (transitional phase)
Chalabianloo et al. [81]	Stress states labeled across 4 classes (baseline, stress, recovery, cycling) during controlled tasks	Random Forest, Support Vector Machine, k-NN; models explained with SHAP	Z-score normalization, feature extraction per wearable; multiclass classification with XAI via SHAP	ECG-based HRV metrics, EDA, accelerometry, temperature	60-s windows aligned to task phases; performance aggregated across 32 participants

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Table 2 (*continued*).

Paper	Stress assessment methods	Supervised learning methods	Data processing/Feature extraction/Unsupervised methods	Features used	Time window length/Resolution
Cummings et al. [82]	Passive ECG biosensing combined with EMA emotion reporting during a 4-week intervention in pregnant women; feasibility and acceptability rated via engagement metrics	None – focus on feasibility/scalability (no ML for stress detection)	Continuous chest-strap ECG synchronized with EMA prompts; protocol adherence, data quality, and usage analytics; qualitative feedback assessed	ECG-derived heart rate and HRV metrics; EMA self-reported emotion scores	Continuous (hours daily) over 4 weeks; EMA prompts ≈daily aligned with key intervention moments
Ehrhart et al. [83]	Labels derived from WESAD dataset (baseline, amusement, stress); used for training real and synthetic data classifiers	CNN-LSTM stress classifiers trained on real and GAN-generated data; conditional GAN used to generate labeled synthetic signals	Normalization and temporal windowing applied to raw WESAD signals; conditional GAN trained per class label; evaluation with classifier accuracy and signal fidelity	ECG, EDA, respiration, and accelerometer data (from Empatica/RespiBAN)	1-s sliding windows (256 Hz) for GAN training; evaluated on 60-s aggregated segments
Zhang et al. [84]	Physiological stress responses measured via wrist-worn Empatica E4; responses tied to different urban space types (green, blue, transit, etc.)	None (random-effects and geographically weighted regression models)	Synchronized GPS, wearable camera images, and EDA/skin temperature; stress response change scores computed; spatiotemporal regression analysis	GSR change scores, skin temperature, GPS coordinates, urban feature tags (water, greenery, traffic, seating) derived via image recognition	10-s physiological sampling; aggregated per urban segment/space types
Zhong et al. [85]	Psychological stress classified from ECG data recorded via wearable sensors during designed stress-inducing tasks	Gated Recurrent Unit (GRU) model; compared to CNN, SVM, and LSTM; GRU	ECG preprocessed and segmented into fixed-length sequences; no handcrafted features; end-to-end learning with sequence modeling	Raw ECG signals; temporal features implicitly learned by GRU	10-s ECG segments input into GRU for classification
Thierfelder et al. [86]	Stress and compulsive behavior detected in children with OCD using multimodal sensor data during exposure tasks	Random Forest classifiers used to differentiate between stress, rest, and compulsive actions	Signal segmentation and annotation; statistical and dynamic feature extraction from multimodal streams; manual labeling by clinicians	EDA (phasic/tonic), heart rate, acceleration (mean, SD, slope, peaks), temperature	Windowed features; resolution not explicitly stated, likely 1–5 s based on sensor data stream processing
Iqbal et al. [87]	Controlled stress induced via WESAD tasks (baseline vs. stress); compared sensitivity/specificity of stress biomarkers	Logistic Regression used for classification and sensitivity analysis; no ML classifiers beyond statistical modeling	PPG, ECG, EDA, and EMG signals filtered and segmented; statistical analysis, regression, t-tests, leave-one-out cross-validation	Heart Rate, Respiratory Rate, HRV (ECG/PPG), EDA (tonic/phasic), EMG	5–10 s epochs (dataset-dependent); classification and analysis performed on variable window lengths
Hag et al. [88]	EEG recorded during Stroop and arithmetic tasks to induce mental stress; labels based on experimental conditions	SVM, k-NN, Decision Tree, Random Forest, Naïve Bayes classifiers evaluated	Hybrid feature selection combining Particle Swarm Optimization (PSO) and Grey Wolf Optimizer (GWO); z-score normalization; dimensionality reduction applied	EEG-derived features: PSD (alpha, beta, theta), statistical metrics (mean, SD), Hjorth parameters, entropy	4-s EEG segments from task intervals; 14-channel EEG system
Hossain et al. [90]	Validated ECG denoising under real-life noise	None (evaluation via signal quality metrics)	Variable Frequency Complex Demodulation (VFCDM) sub-band decomposition; adaptive masking to preserve QRS; dynamic reconstruction; baseline drift removal; adaptive mean smoothing	Cleaned ECG waveforms with improved QRS detection characteristics	Applied continuously to full-length ECG (>5 min); no fixed windowing; evaluated on MIT-BIH and armband data
Aristizabal et al. [92]	Acute stress-induced via Trier Social Stress Test (TSST); labels based on physiological signals and self-reported anxiety/stress	Convolutional neural network (deep learning); tested with physiological-only vs physiological + self-report	Wearable signals (EDA, HR, skin temp) and EMA data fused; z-scoring; CNN processing; performance evaluated via accuracy, precision, recall	EDA, heart rate, skin temperature, self-reported anxiety/stress (via EMA)	Continuous signals aggregated per TSST phase (≈minutes), CNN input windows aligned; daily-hours collection over 4-week campaign

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Table 2 (*continued*).

Paper	Stress assessment methods	Supervised learning methods	Data processing/Feature extraction/Unsupervised methods	Features used	Time window length/Resolution
Föll et al. [93]	Provides a robust toolkit to preprocess and feature-engineer wearable data for stress and health applications	None (toolkit paper, ML models shown in evaluation using WESAD)	Signal cleaning (artifact detection via particle filters, etc.), normalization, sliding-window processing, standardized feature generation	>100 features including HRV, EDA, accelerometry, temperature derived via statistical and spectral algorithms	Configurable sliding windows (used 60 s in evaluation)
Dzieżyc et al. [94]	Stress and affect labeled via experimental protocols including Trier Social Stress Test and affective videos across four datasets	End-to-end deep learning using multimodal CNN, ResNet, FCN, LSTM; Bayesian hyperparameter optimization; compared to traditional SVM/RF baselines	Minimal preprocessing (e.g., normalization, downsampling); raw signals directly input to deep networks; No feature engineering	Raw EDA, ECG, EMG, BVP, EOG, Respiration, Temperature, Acceleration from wearable datasets	60-s windows (aligned across datasets)
Bolliger et al. [95]	Day-to-day occupational stress measured via EMA (Android app), wearables (Empatica E4), and smartphone sensors over 15 working days	Planned ML-based acute stress prediction (protocol only; no models reported yet)	Multilevel statistical modeling and ML pipeline development planned; data fusion of EMA, PPG, EDA, temperature, accelerometry, and phone logs	Heart rate, EDA, skin temperature, accelerometer data, smartphone call/SMS/app logs, EMA stress ratings	Continuous sensor sampling (e.g., 64 Hz EDA); EMA 3–5 prompts/day; aggregated per work period over 15 days
Delmastro et al. [96]	Stress during cognitive and physical training in MCI frail older adults measured via wearable ECG and EDA sensors, and evaluated over two randomized sessions	ML-based binary stress detection using HR, HRV, and EDA; Decision Support System proposed for online stress level inference	Evaluated ML classifiers (RF, AdaBoost, SVM, etc.), physiological signal processing, and statistical analysis; DSS architecture integrates wearable data and clinician feedback	Heart rate, HRV, EDA (via ECG chest strap and GSR sensors); SCWT performance; clinical scales (MMSE, FAB, etc.)	HR sampled at 1 Hz, ECG at 250 Hz, EDA at 51.2 Hz; 1-min sliding window analysis; SCWT pre/post-intervention; dataset includes 342 instances per condition
Akbulut et al. [97]	Psychosocial stress assessed via wearable data synchronized with EMA in post-MI patients; no formal stressor task	Statistical analysis only (correlation, regression); no ML model	Wearable data (PPG, EDA, skin temperature) preprocessed (Artifact removal, filtering); EMA self-reports time-aligned; correlation/regression analyses performed	ECG-derived HR, PPG-derived HRV markers, EDA, skin temperature, EMA scores	Continuous sensor recording; aggregate data aligned with EMA entries (multiple times daily)
Can et al. [98]	Daily stress assessed via self-reports in the wild; Smartwatch physiological and motion data used for prediction	Random Forest classifier for binary stress classification; enhanced via clustering and smoothing	Personalized clustering of stress patterns; decision-level smoothing using majority voting; feature standardization applied	HR, HRV, EDA, acceleration, skin temperature; 11 selected features based on importance	2 min non-overlapping sliding windows from continuous data
Zubair and Yoon [99]	Mental arithmetic and Stroop tasks to elicit four stress levels; labeling via experimental conditions	Linear Discriminant Analysis (LDA), k-Nearest Neighbor (kNN), Support Vector Machine (SVM)	HRV feature extraction from ultra-short pulse rate (PR) signals; signal normalization and statistical testing for feature selection	Time-domain and nonlinear HRV features (e.g., RMSSD, SDNN, Poincaré, entropy) from PR signal	Ultra-short segments: 30 s, 60 s, and 120 s; best performance with 60 s windows
Mozgovoy [100]	Self-reported stress levels via questionnaires and physiological biosignal logging in workplace environment	Decision Tree, Random Forest, k-NN, Gradient Boosting	Feature extraction from EDA, HR, TEMP, and motion; time and frequency domain features; motion intensity used as contextual feature	EDA statistics, HR variability, temperature deviation, motion intensity (accelerometer)	5 min windows for sensor data analysis; aggregation per workday intervals
Rodrigues et al. [101]	Acute stress-induced using Trier Social Stress Test and 2-Choice Reaction Time task in firefighters; stress measured via HRV and self-report	No ML classification used; statistical analysis of physiological and performance changes	ECG data collected via wearable platform; HRV features (AVNN, SDNN, LF/HF) computed; paired t-tests applied for comparisons	ECG-derived HRV metrics (AVNN, SDNN, LF/HF), reaction time performance	Session-based data segments (≈minutes per task); pre- and post-stress comparisons

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Table 2 (*continued*).

Paper	Stress assessment methods	Supervised learning methods	Data processing/Feature extraction/Unsupervised methods	Features used	Time window length/Resolution
Betti et al. [102]	Stress-induced via Maastricht Acute Stress Test; stress labels validated with salivary cortisol	Support Vector Machine (SVM) trained on wearables-derived physiological features	ECG, EDA, and EEG signals preprocessed; 15 physiological features extracted; correlation analysis with cortisol (R^2=0.714)	HRV metrics, EDA, EEG features selected	Time-segmented around stress test phases (minutes); continuous recording during lab session
Xu et al. [103]	Acute stress-induced via task–rest cycles, labels derived using State-Trait Anxiety Inventory (STAI); validated within-subjects	General Regression Neural Network applied per cluster; compared to global regression baseline	k-means clustering used to group individuals with similar stress responses; cluster-specific GRNN models trained; physiological features extracted	Physiological features: HRV, skin conductance (EDA), respiration from wearable sensors	Windows aligned to task/rest cycles (e.g., 30–60 s segments); clustered model evaluated per-subject group
Seoane et al. [104]	Ambulatory stress evaluation in defense-related settings using ECG and respiration via sensorized garments; no specific psychological stressor tasks mentioned	No ML classification; physiological analysis of autonomic responses	ECG and respiration signals collected using textile electrodes (tetrodes); signal quality validated in lab and field; ANS responses analyzed statistically	ECG (R–R intervals), respiration rate, respiration amplitude	Continuous signal acquisition during activities; time resolution determined by sensor sampling (ECG: 250 Hz, Respiration: 25 Hz)

Participants in the study included both healthy individuals and those experiencing significant stress. In contrast to single-sensor methods, the study found that integrating multiple signals resulted in more accurate stress identification. A more thorough and accurate method for measuring stress in both clinical and daily settings is through multimodal monitoring. The study's key features include multiple vital signs, ongoing monitoring, and machine-learning techniques for real-time stress detection. The intricacy of the study, however, might make it challenging to apply in less regulated real-world situations.

Gungormus et al. [43] developed a semi-automatic mobile health (mHealth) system using wearable devices to identify pain-related parameters in elderly individuals with suspected rheumatic disease. The study involved 67 participants aged 65–90 years and utilized tools such as the McGill Pain Questionnaire and pressure algometry to collect data on heart rate variability and skin conductance during daily activities. Results showed that physiological and psychological stress indicators significantly correlated with pain dimensions, accounting for variations in pain experiences. The study highlighted the benefits of mHealth systems for pain assessment and early detection of chronic conditions. However, limitations included a small sample size and reliance on self-reported pain measures. Future research should expand the population size and integrate additional biomarkers.

de Vries et al. [44] developed a wearable sensor-based stress detection system for individuals with intellectual disabilities at Mentech's headquarters in Eindhoven, Netherlands. The study included a cohort of 71 individuals with intellectual disabilities, who were categorized into four distinct groups based on their age range of 18 to 58 years. The system developed utilizes electrodermal activity and heart rate variability data to detect stress, as well as machine learning models, including both personalized and generalized models, to differentiate stress levels. The study found that personalized models outperformed generalized models in accurately detecting stress, emphasizing the need for tailored approaches. The system offers several potential benefits, including enabling caregivers to identify and address stress before it escalates into challenging behaviors, thereby improving the quality of life and reducing caregiver burnout. However, limitations include a small sample size and subjective stress labeling.

In [45], the researchers evaluated the application of wearable sensors for stress level monitoring within free-living environments, utilizing data from 240 subjects to assess machine learning models for early stress detection. One thousand two volunteers aged from 29.6 to 49.4 participated in the study. The research tested models like Random Forest, K-Nearest Neighbors, Support Vector Classification, and XGBoost. The Random Forest model showed the best results, with an accuracy of 98.29% for binary classification. The study found that wearable sensors combined with ML models can accurately detect stress, especially in diverse, uncontrolled settings. However, limitations include variability in data quality due to environmental factors and subject compliance in free-living settings. The benefits of wearable technology include real-time health monitoring and early intervention for stress-related conditions, improving the quality of life.

Rescio et al. [48] developed a deep learning-based platform in Italy for detecting workers' stress using minimally intrusive multisensory devices. The platform utilizes wearable and ambient sensors, including ECG and RGB cameras, to collect physiological and visual data. The study involved 20 volunteers aged 24–38 who were subjected to stress-inducing tasks. The platform achieved a 95.38% accuracy in distinguishing two stress levels using a one-dimensional convolutional neural network. Key features extracted included heart rate variability, electrodermal activity, and blink detection. The platform's benefits include real-time monitoring capabilities, minimal sensor placement requirements, and adaptability for various workplace settings. However, limitations include the need for controlled environments and potential discrepancies due to varied work tasks and motion artifacts.

In [49], the authors assessed a study at Oklahoma State University that examined the effects of recurrent task-induced acute stress on 32 students (25.2 ± 2.3 years, 17 females) using wearable sensors. The study found that stress negatively affects task performance, heart rate variability, and task-evoked pupillary response, indicating physiological and cognitive reactivity to stress. The results showed that task performance decreased, vmHRV measures dropped, and TEPR increased under stress conditions, indicating that repeated stress exposure could lead to habituation and performance degradation over time. The study offers a multidimensional view of stress responses using wearable technology; however, its limitations include a reliance on physiological data without additional validation methods.

Tanwar et al. [51] assessed a study focused on developing an attention-based hybrid deep learning model for wearable-based stress recognition using wearable devices like Empatica E4 and RespiBAN. The model integrates multiple physiological signals and deep learning techniques, combining CNN and LSTM networks with an attention mechanism. The WESAD [29] dataset is utilized to evaluate the model's

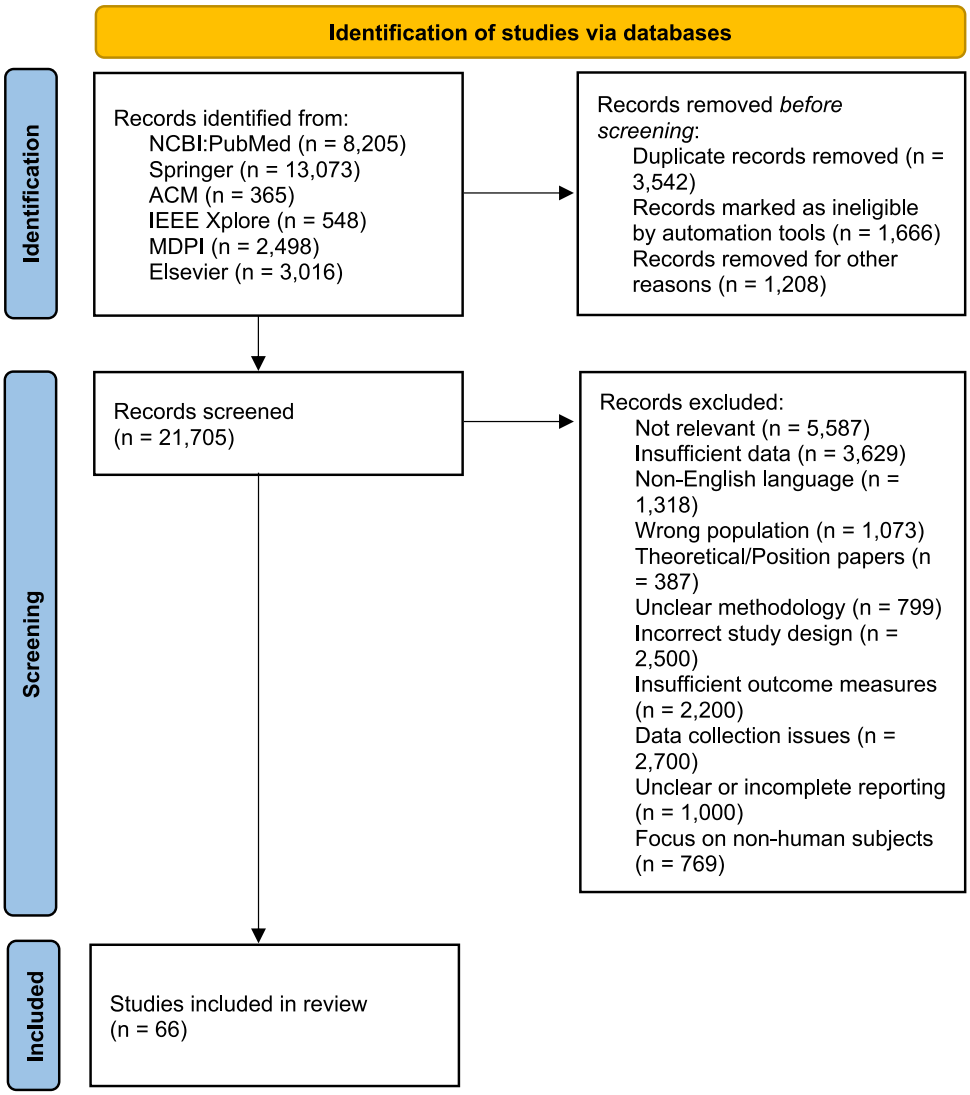


Fig. 2. Diagram showing the process used for selecting the relevant research.

potential for real-world applications. The model achieves a high classification accuracy of 92.7%, surpassing existing CNN and CNN-LSTM models. Key features include efficient feature extraction and dynamic data prioritization through the attention layer. The model is beneficial for healthcare and affective computing applications.

Kader et al. [54] developed a one-channel wearable system for monitoring mental stress on 20 human subjects using electroencephalogram (EEG) and galvanic skin response (GSR) data. The system, which utilized sensors in the prefrontal and ventrolateral frontal regions, was able to distinguish between two levels of mental stress with 70.3% accuracy using EEG alone and 84.6% accuracy when combined. The study highlighted the benefits of a single-channel system, including reduced complexity and improved usability in real-world settings, particularly in workplaces. However, limitations included sensitivity to environmental factors and noise interference in EEG data.

In [55], the authors developed a wearable biomedical device for physiological monitoring using electrocardiographic, photoplethysmographic, and galvanic skin response signals. The device, designed to detect physiological changes during daily activities, was tested on healthy subjects at the University of Palermo. The device collected biosignals from the fingers, providing indices such as heart rate variability, blood oxygen levels, and respiratory rate. The device showed promise in differentiating between rest and stress phases based on HRV and GSR indices. However, limitations were noted during motion-based activities, suggesting the need for artifact correction algorithms. Further validation is needed to optimize performance and confirm reliability.

In [56], the study utilizes wearable devices to predict stress levels by analyzing biosignal data from electrodermal activity (EDA). The biosignals were collected from 15 healthy adults (mean age, 27.5; standard deviation (SD), 2.4). The model, developed using self-supervised learning (SSL), accurately predicts stress using fewer labels, making mobile health interventions more feasible. The methodology involves pre-training a 1D convolutional neural network with raw biosignal data, followed by fine-tuning for stress prediction. SSL models perform better with fewer data points, achieving stability and precision compared to traditional methods. The study concludes that personalized SSL models are effective for individualized stress prediction, enhancing mobile sensing capabilities, and reducing labeling efforts.

In a study that aims to develop a global stress detection framework using heart rate variability (HRV) features and a Random Forest model, Dahal et al. [57] used wearable wristband devices to record HRV data from WESAD [29] and SWELL [105] datasets. The research shows that a reduced set of eight HRV features enhances classification accuracy to over 99%. The model's high accuracy and adaptability across individuals are achieved when validated on combined datasets.

However, limitations include the potential for overfitting, a lack of generalization, and dataset constraints.

Parousidou et al. [58] developed a personalized machine learning model for stress detection using wearable sensors. The goal is to identify stress levels accurately by addressing individual differences. The study employs a continuum of machine learning and deep learning models, exploring user- and group-based personalizations based on attributes such as physical activity and sleep patterns. They achieved personalized models with an F1-score of up to 99% (user) and 66% (group) in-the-wild accuracy. Results show that personalized models outperform generic models, particularly in real-world conditions. However, challenges such as dataset size, environmental variability, and subjective stress measurements in uncontrolled settings remain for scalability and generalizability in real-world applications.

In [61], the study examines the relationship between residual symptoms and physiological stress in individuals with major depressive disorder (MDD) and found that skin conductance responses (SCRs) and residual symptoms, such as sleep and appetite issues, concentration difficulties, and mood disturbances, interact over time. The study utilized electrodermal activity (EDA) as a measure of physiological stress, and 22 participants aged 18 years or older self-reported symptoms for three weeks. Key findings showed that SCRs were central in predicting energy loss, with depressed mood and concentration issues being crucial nodes in symptom networks. The study also highlighted the potential of wearable technology in monitoring physiological stress in everyday life and suggested its potential for advancing personalized treatments in mental health care.

A study utilizing the Empatica E4 bracelet and machine learning techniques, Campanella et al. [65] developed a stress detection method. The bracelet's sensors collected physiological data from 29 subjects while they performed stress-inducing tasks, including cognitive and social stressors. The data was processed using algorithms like Random Forest, SVM, and Logistic Regression. The Random Forest model achieved the highest accuracy at 76.5%. The study concluded that the Empatica E4 bracelet is a viable tool for stress detection, with the Random Forest model showing the most stable accuracy. However, limitations include small sample sizes, laboratory settings, and sensor noise. Future work should focus on increasing sample diversity, refining protocols, and testing in real-life environments.

Costantini et al. [66] validated the Empatica E4 wristband's accuracy in monitoring heart rate variability and electrodermal activity in stressful driving conditions. The study involved 14 participants aged 25 to 41 years who were exposed to six scenarios and compared the wristband's performance to that of a gold standard system. Results showed high reliability for HRV time-domain parameters in low-motion scenarios but lower agreement for EDA measurements due to motion artifacts. The study highlights the benefits of wearable sensors in stress detection but suggests further development and algorithmic enhancements to improve signal quality and reduce artifacts, especially in dynamic driving contexts.

The study [68] explores stress detection using wrist-measured electrodermal activity (EDA) using frequency spectrum analysis. The research uses data from the WESAD [29] dataset, which includes 15 subjects performing real-world tasks. The aim is to improve stress detection accuracy using wrist-worn EDA sensors. The method utilizes sliding windows to segment EDA signals and extract features, including mean, standard deviation, and frequency-based metrics. The results demonstrate a competitive accuracy of 84.87% and an improved time-to-detection compared to other models. The method has low computational cost and high responsiveness, making it suitable for real-time wearable applications. However, the method's generalizability may be affected by the dataset's controlled environment.

In [69], the study used wearable sensors, specifically the Empatica E4 device, to collect physiological biomarkers from various datasets to predict stress levels using machine learning models. The goal was to develop a generalized model for stress prediction, overcoming the limitations of small sample sizes in individual studies. The research involved combining small datasets to create larger synthesized datasets, implementing feature engineering, and applying ensemble techniques. The Leave-One-Subject-Out (LOSO) validation method tested the model's generalization capabilities on unseen datasets. Results showed that models trained on small datasets had limited generalizability, while larger synthesized models showed improved predictive accuracy. However, challenges such as class imbalance and dataset variability necessitated the use of advanced sampling and balancing techniques.

Almadhor et al. [72] assessed a study in which they used wrist-based electrodermal activity (EDA) monitoring for stress detection using a federated learning approach to ensure privacy. The study utilizes data from the Wearable Stress and WESAD [29] Affect Detection dataset, which includes transient, baseline, stress, amusement, and meditation states. Sensors such as accelerometers, blood volume pulse detectors, and electrodermal activity detectors are incorporated into wrist-worn devices like the Empatica E4 smartwatch. The FL-based deep neural network model achieved an accuracy of 86.82%, demonstrating improved performance over traditional models while protecting patient data. The study suggests further exploration of wearable combinations and algorithms to enhance the model's performance and scalability.

In [73], the study uses RespiBAN professional chest-worn devices to detect stress through physiological signals from the Wearable Stress and WESAD [29] dataset. A machine learning stacking model is used to improve stress detection methods. The model has a high accuracy rate of 99.7% in classifying stress. The study suggests that integrating chest-based monitoring into everyday wearable technology could enhance early stress detection and improve healthcare diagnostics. However, the model's generalizability is limited by the dataset's specificity and requires further validation on diverse datasets. The approach shows promise in reducing healthcare costs and enhancing diagnostic precision, but real-world applicability remains a challenge.

Momeni et al. [78] developed a Cost-Aware Feature Selection (CAFS) methodology for multimodal stress monitoring using wearable devices. The method utilizes sensors such as ECG, EDA, PPG, and respiration monitors to capture physiological data. The study involved 60 male participants (20.43±2.17 years) who were exposed to a virtual reality environment to induce acute stress. The authors implemented the CAFS method to balance prediction accuracy and energy consumption, thereby extending the battery life of wearable devices. The method reduced energy consumption by up to 94.37% while maintaining an accuracy of 90.98% on unseen data. The CAFS method utilizes machine learning techniques to train models with varying feature sets, thereby producing energy-scalable stress monitoring models. The study highlights the potential of CAFS for long-term, low-power stress monitoring on wearable platforms.

In the study [81], seven wearable devices were tested with different biosensors (ECG, PPG, EDA) during four sessions on 32 participants aged 28.4 ± 5.98 years. The study used machine learning algorithms to assess stress detection performance. Results showed that ECG-based wearables provided higher accuracy than PPG-based devices, but combining multiple biosignals significantly improved performance. The study concluded that ECG sensors with complementary biosignals were most effective for stress detection. Wearable sensors offer portability and real-time stress monitoring capabilities, but limitations include sensor noise, accuracy differences, and the impact of physical movements on data quality. The research emphasizes the need for careful selection of wearable technology, balancing user comfort, sensor quality, and specific application requirements for effective stress monitoring.

Ehrhart et al. [83] used wearable physiological sensors, specifically the Empatica E4 wristband, to measure Galvanic Skin Response and Skin Temperature from 35 participants aged between 18 and 55 years. The aim was to detect stress-related events using these physiological signals, analyzed using a Conditional Generative Adversarial Network (cGAN) model. The study found that the cGAN model effectively generated synthetic time series data, thereby improving the performance of the stress detection classifier. However, challenges included ensuring

dataset quality and stability during GAN training. Future improvements may involve optimizing hyperparameters and expanding the dataset to validate the model's effectiveness.

In the study [84], wearable sensors were used to study the relationship between urban features and physiological stress responses. The 86 Participants navigated six public open spaces, including water areas, transit zones, green areas, commercial spaces, motor traffic areas, and mixed office and residential areas in Copenhagen, Denmark. The study utilized Microsoft Cognitive Services for image analysis and employed statistical models to investigate how these urban features impacted stress responses. Results showed that urban flow, water bodies, greenery, and seating had a significant influence on stress responses. Green and blue spaces could reduce stress, but their effects depended on the specific urban context. The study concluded that the context and design of buffering spaces are crucial for mitigating stressors like traffic and crowding. However, limitations included sample biases and sensor connectivity issues.

The STRAW project investigated occupational stress among academics using ecological momentary assessments (EMA), physiological data, and smartphone sensor data [95]. The study involved healthy academics working at two institutions in Belgium and Slovenia. Participants used the Empatica E4 wristband to monitor physiological responses and installed a custom app on their Android smartphones to collect sensor and usage data. The aim was to identify correlations between work environment risk factors and stress outcomes. The project employed multilevel statistical modeling and machine learning to develop predictive models for detecting acute stress in real-world settings. Key findings demonstrated the feasibility of integrating wearable technology and smartphone data to track stress in real-time.

A study in Italy used wearable sensors and machine learning to monitor stress and cognitive function in frail older adults with Mild Cognitive Impairment. Delmastro et al. [96] used devices Zephyr BioHarness3 and Shimmer3 GSR+ to collect heart rate, HRV, and EDA data. A mobile health app was used to process the data. The study involved nine frail elderly individuals aged 70–85 who were monitored during exercise sessions. Results showed improved cognitive performance post-exercise, with shorter completion times and fewer errors. Physiological data indicated a mitigated stress response during exercise sessions, suggesting an adaptive response to physical activity. Machine learning algorithms, such as Random Forest and AdaBoost, achieved an accuracy of 85.3% for stress detection. The study concluded that wearable technology combined with machine learning offers an effective way to personalize cognitive training and monitor stress.

Xu et al. [103] developed a personalized stress evaluation method using physiological signals like heart rate variability, galvanic skin response, and electroencephalography. Data were collected from 44 male participants, aged 28.6 ± 7.2 years, during physical and cognitive stress tasks. The study employed k-means clustering and general regression neural networks to categorize subjects based on their physiological patterns. The results showed that clustering improved stress evaluation accuracy, with two clusters being the optimal choice. The cluster-based model accounts for individual differences in stress responses, making it suitable for developing intelligent tools for stress management. However, limitations include the small sample size and challenges in consistently controlling individual stress levels, suggesting the need for further refinement to enhance broader applicability.

The authors of [75] explored the generalizability of physiological markers of anxiety and stress using data from ECG and EDA sensors. The datasets included Anxiety Phases Dataset (APD) [76], Wearable Stress and Affect Detection (WESAD) [29], and Continuously Annotated Signals of Emotion (CASE) [77]. The research used cross-corpus evaluation and leave-one-corpus-out analysis with machine learning classifiers. Results showed that models trained on stress data performed well on previously unseen arousal datasets, suggesting they might be detecting generalized arousal. The study concluded that physiological signals from ECG and EDA can effectively recognize emotional arousal states across datasets, offering the potential for developing more generalized stress detection models.

Seoane et al. [104] developed sensorized garments to assess autonomic nervous system responses under stress, specifically for Spanish Armed Forces personnel during training exercises. The equipment includes a chest strap system, a sensorized glove, and an upper-arm strap with sensors for monitoring heart rate, respiration, galvanic skin response (GSR), and skin temperature. The ATREC project aims to enhance troop safety and readiness by identifying stress indicators using wearable technology. The garments and sensors provide reliable, multiparametric data, including real-time data recording and wireless communication capabilities. The study highlights the garments' readiness for further stress-inducing tasks. Still, it acknowledges potential motion artifacts and the need for further refinement.

Akbulut et al. [97] aimed to assess psychosocial stress in patients with metabolic syndrome (MES) using a wearable sensor-based system. The study involved 30 participants from Istanbul Cerrahpasa Medical Faculty, half of whom had MES or cardiovascular diseases, and the other half were healthy controls. The wearable system was tested in a hospital environment, achieving an average stress detection accuracy of 92% for MES patients and 89% for others. The study found that MES patients had significantly higher stress levels compared to healthy individuals, with stress responses accurately classified using the system. The wearable device effectively measures stress, providing a valuable tool for early intervention, but further validation with larger datasets is needed to strengthen the findings and improve the system's generalizability.

In [98], the authors utilized smartwatches and wearable devices to enhance ambulatory stress detection by employing a hybrid model that incorporates personal stress-level clustering and decision-level smoothing. The study involved 32 participants, primarily primary school teachers aged 25 to 40, and utilized self-reports and wearable sensors to assess stress levels. The results showed the system improved stress detection accuracy using a hybrid clustering approach and decision-level smoothing techniques. The study concluded that person-specific models increase detection accuracy, while hybrid models improve generalization without extensive individual data collection. Benefits include real-life applicability and non-intrusiveness, but limitations include reduced accuracy and challenges related to battery life and device movement artifacts.

Mozgovoy [100] aimed to enhance stress detection in the workplace using wearable biosensors. Eighteen employees wore biometric devices for two months, recording physiological data to monitor their stress levels remotely. The study used unsupervised learning techniques to classify stress-related conditions into emotional overarousal, arousal, and relaxation. The results showed that incorporating Motion Activity (MA) improved stress pattern recognition accuracy, suggesting its usefulness in remote stress monitoring systems. The study highlights the potential benefits of integrating non-physiological measures for better interpreting physiological stress indicators. Still, it acknowledges limitations, such as distinguishing work-related stress from external factors and issues with generalizability.

The study of [102] evaluated a wearable system of physiological sensors for monitoring stress in working environments on 15 healthy participants, comprising 8 males and 7 females, with a mean age of 40.8 ±9.5 years. The system, which included ECG, EDA, and EEG devices, measured heart rate variability, electrodermal activity, and brain activity. The study involved 15 healthy participants and used the Maastricht Acute Stress Test (MAST) to induce stress. The system identified stress with an accuracy of 86%, showing a significant correlation between physiological features and cortisol levels. The study highlighted features such as heart rate variability analysis, skin conductance measures, and EEG signal interpretation. The results support the feasibility of real-time stress monitoring for workplace interventions and stress management. However, challenges such as optimizing sensor accuracy and minimizing data variability remain.

The authors of [87] utilized the WESAD dataset [29] to analyze the biophysiological responses to stress. The results showed that respiratory rate and heart rate were the most effective features for distinguishing between stressed and unstressed states. The study concluded that these parameters can effectively detect stress, providing continuous, non-invasive monitoring. However, limitations include the small sample size and potential confounding factors, such as physical activity, suggesting the need for larger studies to validate the findings and incorporate context-based data for real-life applications.

Conducted at the Well Living Lab in a controlled environment, the authors of [92] aimed to evaluate the accuracy of stress detection models using wearable sensors and self-reported measures. The study involved 18 healthy adults (11 Females and 7 Males) aged 22 to 31 years who underwent the Trier Social Stress Test and provided data on physiological responses, cortisol levels, and self-reported stress and anxiety levels. The results showed that combining wearable sensor data with self-reports significantly improved stress detection accuracy, with a deep learning model achieving 96% accuracy. However, limitations such as small sample sizes and controlled lab conditions may not fully replicate real-life stress scenarios.

FLIRT is a Python-based toolkit designed for generating features from wearable sensor data, addressing issues such as noise, artifacts, and missing data. In [93], it was used to standardize processing and feature extraction from the WESAD dataset [29], which includes physiological data from 15 participants undergoing stress-inducing tasks. FLIRT offers advanced algorithms, such as Kalman filters and sliding window methods, enabling the extraction of feature vectors over 100 dimensions, which are suitable for machine learning models. Results show FLIRT improves data preprocessing and feature generation, enhancing performance in stress detection tasks. It bridges gaps in current wearable data processing tools by providing standardized methodologies, increased reproducibility, and ease of use across multiple sensor types.

The study [94] examines the application of end-to-end deep learning (DL) architectures for affect recognition utilizing physiological sensor data. It utilizes wearable sensors, including ECG, EDA, and EEG, across four datasets to identify emotional and stress states. The research aims to eliminate the need for manual feature engineering. The results show that DL models perform better in stress detection tasks than emotion recognition, with CNN-based architectures outperforming LSTM-based models. The study employs Bayesian optimization for hyperparameter tuning and evaluates the multimodal integration of signals. Results suggest that physiological responses to stress are more easily detectable than those to mild emotional stimuli. The study concludes that DL architectures can recognize stress.

3.2. Heart monitoring and related technologies

Mai and Chung [38] presented an on-chip mental stress detection system utilizing neural networks and a wearable EEG device. To classify stress, the system records EEG signals from a single channel located behind the ear and transforms them into spectrogram images. In cross-validation tests, the system performs well, achieving accuracies of up to 95.32%. It enables portable, real-time stress level monitoring without the need for external processing.

The authors of [39] performed a study to predict the onset of stress using biomarker signals recorded by a wrist-worn device. Participants in the study had varied degrees of ongoing stress. The aim was to create a predictive model that could detect stress before it becomes apparent, allowing for early intervention. Predictive algorithms and the wrist-worn device reliably predict stress events, offering insightful information for proactive stress management. The study highlights the potential of wearable technology for predictive, real-time health monitoring. The study's drawback, however, is the potential decline in accuracy in non-laboratory settings.

Major Depressive Disorder (MDD) can be identified using electrocardiogram (ECG) signals and a deep learning model called MDDBranchNet [41]. The model detects MDD from ECG data with an accuracy of up to 95.32% by combining several deep learning branches. It is a promising tool for early diagnosis of MDD because it is non-invasive, lightweight, and designed for real-time detection. The study notes limitations, including the need for validation of a larger dataset and the possibility of ECG signal variability. Still, it also highlights the potential of ECG as a reliable biomarker for detecting MDD.

Zhong et al. [85] developed a psychological stress detection system on 80 participants using wearable ECG devices in smart shirts. The study employed a virtual reality high-altitude scenario to induce stress, capturing ECG signals at various stress levels. A gated recurrent unit (GRU) neural network model was developed to classify stress states. The proposed GRU model achieved the highest classification accuracy (78%), demonstrating its potential for real-time stress monitoring, especially in high-risk professions. However, limitations included distinguishing mild stress levels and modest overall accuracy. The study's findings show promise for wearable-based, non-invasive stress monitoring, offering real-time intervention and safety improvements in high-stress environments.

The authors of [90] have developed a method for ECG signal denoising using Variable Frequency Complex Demodulation (VFCDM). The method utilizes sensors such as wearable armband ECG devices and data from the MIT-BIH Arrhythmia and Noise Stress Test Databases. The goal is to enhance ECG signal clarity by eliminating noise, such as baseline wander and muscle artifacts, which are essential for accurate QRS complex detection. The VFCDM technique decomposes ECG signals into frequency sub-bands, allowing targeted noise removal while preserving essential signal features. The method is expected to enable long-term monitoring of cardiac conditions, such as atrial fibrillation, but has limitations in terms of computational complexity and sensitivity to electrode noise.

A study involving 14 healthy graduate students aged 25–30 years developed a multilevel mental stress detection system using ultra-short pulse rate variability (PRV) from a low-cost photoplethysmogram (PPG) sensor. The researchers in [99] employed statistical analysis, feature extraction, and machine learning classifiers, including Support Vector Machine (SVM) and Quadratic Discriminant Analysis (QDA), for stress classification. The SVM classifier achieved a high accuracy rate of 94.33% for detecting five distinct stress levels using the proposed features. The system's features, including Point Transition Measures (PTM) and traditional HRV metrics, demonstrated significant sensitivity to varying stress levels. However, the study faced limitations, including its dependency on specific stressors and the need for a controlled environment.

The study [52] conducted in South Korea demonstrates the potential of off-the-shelf smartwatches for stress detection in real-life settings. The research uses a two-layer machine learning model to integrate physiological data and contextual data from smartphone users. The study found stress detection accuracy of 0.84 in laboratory settings and 71% in real-life scenarios. The method utilizes photoplethysmography sensors for heart rate monitoring, thereby addressing data inaccuracies caused by motion and varying wearing conditions. The study suggests that smartwatches offer a practical and accessible alternative to specialized devices, providing high temporal coverage and user context.

Carreiro et al. [53] performed a study in the US that evaluated the RAE Health digital tool, which uses wearable sensors and machine learning to detect stress and craving in substance use disorder (SUD) treatment. Thirty participants wore a Garmin Vivosmart 4 wrist device and used the RAE app for 30 days. The tool's accuracy reached 71% and 69%, respectively, but false positives were observed. The study concluded that the tool, when integrated with a commercially available sensor, demonstrated promising accuracy comparable to that of research-grade devices. However, challenges included higher attrition rates in lower socioeconomic populations and connectivity issues with

Android devices. The study emphasized the need for device-agnostic digital health solutions and further research to explore the impact of these tools on clinical outcomes and optimal intervention dosing.

Mitro et al. [64] developed an AI-enabled smart wristband for real-time monitoring of vital signs and stress detection, specifically for passenger ship evacuations. In the study, 15 volunteers were subjected to cognitive stressors. The wristband uses a photoplethysmography sensor to monitor pulse rate and blood oxygen saturation, while a machine-learning model identifies stress through pulse rate variability. The device achieved 91% accuracy on an unseen dataset and 76% accuracy in external validation. However, limitations include reliance on a single biometric signal and reduced accuracy across different datasets and conditions.

Bocharov et al. [67] performed a study on the physiological costs of Ukrainian defense force officers during combat using miniature wearable ECG devices. The study involved five senior officers, aged 25 to 43 years, who served in combat zones during the ongoing conflict with Russia. The researchers utilized cloud-based technology to analyze data from ECG patches, assessing parameters such as heart rate variability, rhythm disorders, and psycho-emotional indices. The results showed the monitoring system could detect stressors with high sensitivity, providing an integrated view of soldiers' cardiovascular and emotional state. The technology supports decision-making in military operations, optimizing task assignments based on physiological readiness. However, limitations included a small sample size and the need for further validation across diverse scenarios.

The authors of [79] aimed to develop and validate algorithms for detecting stress in firefighters using heart rate variability (HRV) monitoring. The study involved 38 male firefighters aged 26 to 40 years and was conducted over 24 h. The goal was to simulate and validate both static and dynamic HRV decrease-trigger algorithms (AddHRVr) that account for both metabolic and psychosocial stressors. The results showed that a dynamic algorithm showed greater sensitivity in detecting stress transitions compared to static models. This approach offers a robust method for real-time stress monitoring, enabling psychosocial assessments in high-risk professions such as firefighting. However, limitations included the need for further validation in different populations and stressors.

Ritsert et al. [80] addressed a study at Simon Fraser University in Canada that used a portable device to detect anxiety using heart rate variability (HRV) and breathing rate variability (BRV) signals. The device, a BIOPAC MP45, monitored physiological changes in 19 participants aged 18 to 59 years. The results showed significant HRV/BRV differences between anxiety and non-anxiety states, with a decrease in variability measures during anxiety episodes. The study concluded that HRV and BRV are effective biomarkers for detecting anxiety, providing an objective and continuous assessment of physiological responses. However, limitations include a small sample size and reliance on specific stimuli. Future applications could involve enhancing machine learning algorithms for automated anxiety detection.

In [82], the authors explored the use of passive ECG bio-sensing and ecological momentary assessment (EMA) technologies in a prenatal well-being intervention. Ninety-nine pregnant women participated in the research to reduce prenatal stress and improve infant neurodevelopmental outcomes. The study found compliance rates of 67% for EMAs and 52% for biosensors, with a higher baseline of negative affect predicting lower adherence. The study also highlighted the need to tailor interventions based on sociodemographic factors, as higher stress and lower socioeconomic status correlated with lower adherence. The study concluded that mHealth interventions need optimization to minimize participant burden and improve uptake, particularly among disadvantaged groups.

Betti et al. [102] performed a study using a wearable biomonitoring platform, finding that stress significantly altered autonomic responses and impaired cognitive performance in firefighters. The platform, which utilized biomedical sensors to monitor physiological parameters and psychological assessments, was tested on 21 Portuguese firefighters aged 21 to 59 years. The results showed significant changes in heart rate variability and cognitive performance, suggesting that stress negatively impacts cognitive functions. The study's findings could inform the development of interventions for stress management in high-risk occupations, such as firefighting. However, limitations included a small sample size, an artificial stress environment, and a lack of a control group.

The authors of [86] used multimodal sensors in telemedical settings and identified stress and compulsive behaviors in children with obsessive-compulsive disorder (OCD). The study involved five adolescents aged 13–17, monitored using ECG chest belts, inertial motion sensors, and eye-tracking devices. The goal was to provide real-time, personalized data on stress levels during exposure and response prevention therapy sessions in natural, home-based environments. The results showed that multimodal sensors effectively detect biomarkers of stress and anxiety, offering insights for real-time therapeutic applications outside traditional clinical settings. The study highlights the effectiveness of wearable sensors in capturing stress-related physiological and behavioral changes in children with OCD.

3.3. Specialty and hybrid sensors

Qiao et al. [27] developed a wearable electrochemical biosensor that can identify cortisol in perspiration to provide a non-invasive way of monitoring stress. Stability and accuracy are guaranteed by the sensor's use of a conducting polyaniline hydrogel with antifouling qualities. The design of the biosensor ensures dependable performance by reducing interference from environmental factors and skin impurities. Although the biosensor holds great promise for tracking stress in real-time, its effectiveness needs to be confirmed under various real-world scenarios before it can be widely adopted.

In [36], the authors developed a deep learning model based on autoencoders to identify stress in real-time from wearable data, including skin temperature and heart rate. The goal of the Spanish study was to develop an effective stress recognition system for commonplace settings. The model demonstrated how wearable data and deep learning can be utilized to recognize stress in the real world, successfully identifying stress patterns with higher accuracy than conventional techniques. The system can be adjusted to various environmental conditions due to its non-intrusive nature and ongoing real-time monitoring. It is susceptible to individual differences in stress reactions, though, and could result in data overload.

The authors of [42] presented a real-time stress detection method that integrates deep learning techniques with remote photoplethysmography (rPPG). The technique analyzes facial video signals using models such as CNN, GRU, and LSTM. With an accuracy of up to 95.83% on the UBFC-Phys dataset, the results demonstrate a notable improvement in stress detection. For ongoing stress monitoring, this non-invasive system provides a practical and less invasive approach. To ensure robustness, additional validation is necessary, as accuracy may vary in uncontrolled settings. The study highlights rPPG's potential as a more practical and effective real-time monitoring solution.

A wearable biosensor has been developed by [62] for detecting cortisol levels in human sweat using gold nanoparticles on a poly (dimethylsiloxane) substrate. The sensor, developed in Gwangju, South Korea, uses a specific aptamer for precise and selective capture of cortisol, a biomarker for stress. The sensor demonstrated sensitivity across various concentrations and showed stability under various mechanical deformations. The flexible LSPR biosensor offers low cost, non-invasiveness, and quick response times. Still, potential interference from other biomolecules and further testing in diverse populations are needed to validate its efficacy.

In [63], the authors developed a wearable stress sensor using a spider web-inspired structure and laser-induced graphene technology. The sensor aims to improve sensitivity, flexibility, and durability for applications in human motion detection, medical monitoring, and human–computer interaction. The sensor was tested in Guangzhou, China, and

showed high sensitivity, a response time of 260 ms, and durability after 3000 cycles. The sensor's design allows integration into wearable devices for monitoring physiological signals, offering high sensitivity, low production costs, and potential applications in health and robotics. However, improvements in graphene durability and response consistency are needed for broader practical applications.

The authors of [71] aimed to develop a low-cost optical fiber physiological sensor for stress classification, focusing on heart rate variability and respiratory signals. The sensor, made of polymer optical fiber and LEDs, is wearable and suitable for real-time monitoring. The study involved five college students (four males and one female) aged 22.6 ± 0.55 years who underwent relaxation and stress-induction tests. The Bagged Decision Tree Classifier algorithm achieved an 88% accuracy in stress detection. The sensor offers a non-intrusive, adaptable, and cost-effective solution for stress monitoring. However, limitations include a small sample size and the need for more realistic stress induction methods.

Zamani et al. [74] explored the use of wearable-assisted smart health monitoring systems to predict sleep quality using deep learning models. The research uses physiological data from smartwatches to assess sleep efficiency. The Deep Belief Network (DBN) model is trained in controlled environments to improve sleep quality prediction. The methodology uses the DLM-WESHMS approach, which incorporates an Auto-Encoder Algorithm for hyperparameter tuning. The model demonstrates improved accuracy and precision in predicting sleep quality, providing non-invasive, continuous health monitoring. However, limitations include reliance on wearable accuracy and the need for extensive data for model training. The study suggests the potential for integrating these models into e-health solutions for personalized health monitoring.

3.4. Other sensors

A wearable sensor system called StressFit measures stress by measuring sweat cortisol levels and electromyogram (EMG) signals [40]. Real-time monitoring of muscle tension and hormonal reactions is made possible by the system's integration of EMG sensors and electrodes. Wearable stress detection benefits significantly from this multifaceted approach to stress assessment. The system incorporates temperature, pH, cortisol, and EMG sensors to provide continuous, real-time feedback on a person's stress level. To verify its dependability and usability, more testing in practical settings and optimization are required.

Pickert et al. [47] is a study developed at the Peter L. Reichertz Institute for Medical Informatics in Germany, which utilized a wearable accelerometer-based sensor system for non-invasive cardiac monitoring. The study involved 12 heart-healthy adults. The system employs a rule-based algorithm to detect heartbeats independently of an ECG reference, allowing for the autonomous derivation of health parameters. The system accurately detected 97.6% of heartbeats at rest and 71.9% under physical stress. However, the algorithm's sensitivity decreases during physical activity due to movement artifacts. The wearable device has the potential for real-time cardiac monitoring in both everyday and clinical settings. However, the absence of a second sensor and reliance on a single sensor position may limit the accuracy of SCG signal interpretation. Improvements in sensor setup and algorithmic adjustments are recommended.

The authors of [16] conducted a study in Australia that tested three wearable sensors for monitoring body temperature in 26 construction workers (24 males and 2 females) across two construction sites, aiming to address health and safety risks such as heat stress, fatigue, and infectious diseases. The study found high accuracy with a bias of less than one degree, with Equivital being the most comfortable and least intrusive device. Wearable sensors can provide real-time monitoring, enabling proactive risk detection and mitigation. However, limitations include improving device stability and accuracy under various conditions.

The authors of [50] conducted a study at the University of Gustave Eiffel to examine the effectiveness of a tactile respiratory guidance system in reducing driving-related stress among 83 participants aged 19 to 71 in France. The study used a driving simulator with a custom-designed wristband to guide participants' breathing rhythms. Results showed that participants generally adapted to the guidance without compromising driving safety. However, individual differences, such as inattention errors and driving experience, influenced the effectiveness of the intervention. The study concluded that while tactile breathing guidance shows promise, further refinement is needed to optimize its efficacy and user experience.

The authors of [88] used EEG signals and a hybrid feature selection algorithm (mRMR-PSO-SVM) to improve mental stress state recognition in 22 healthy male participants aged 22 to 30. The research aims to identify the optimal subset of EEG features for stress detection using multi-domain feature extraction and a hybrid algorithm integrating minimum redundancy, maximum relevance (mRMR), and particle swarm optimization (PSO). The mRMR-PSO-SVM model reduced feature vector space by 70% and improved classification accuracy across multiple datasets. The study highlights its potential for stress detection applications.

4. Discussion

4.1. Interpretation of the results

Stress is a multifaceted physiological and psychological response to internal or external stressors. It activates both the autonomic nervous system and the hypothalamic–pituitary–adrenal (HPA) axis, resulting in measurable changes in cardiovascular, electrodermal, and behavioral indicators [72]. These physiological shifts are observable through increased heart rate, blood pressure, skin conductance, and respiration rate. According to the UK Health and Safety Executive, stress-related illnesses accounted for 47% of all work-related health conditions in 2023–2024 [106]. Unmanaged or prolonged stress can have serious long-term consequences, contributing to cardiovascular disease, metabolic disorders, and psychiatric conditions [44,52,69,72].

In this context, machine learning-based stress detection systems, fueled by data collected from wearable sensors, represent a promising approach for early intervention. Devices such as smartwatches, chest bands, and specialized biosensors are widely used to monitor physiological signals, including electrodermal activity (EDA), heart rate variability (HRV), skin temperature, and respiration [45,58,73,97]. These data streams undergo preprocessing, such as signal filtering and artifact correction, followed by feature extraction and classification using algorithms ranging from traditional models, like Random Forest and SVM, to deep learning architectures, including CNNs, LSTMs, and ensemble methods [28,30,51,72].

The reviewed studies demonstrate increasing diversity in population sampling and technological implementation. Older adults were studied in contexts such as chronic pain monitoring [43] and combat-related stress detection [67]. At the same time, younger and middle-aged participants were assessed in construction environments [16], vehicle automation scenarios [102], and free-living conditions using consumer-grade wearables [48,52]. Special populations included individuals with intellectual disabilities [44] and those undergoing treatment for substance use disorder [53]. Recent 2025 studies further expanded the scope. Ciccarelli et al. [26] evaluated wearable stress sensors in both laboratory and real-world settings, highlighting the challenges of generalizability. Darwish et al. [28] validated wearable biomarkers using machine learning across settings, while Singh et al. [30] applied deep learning to EDA data for early stress detection. Additionally, Schreiber et al. [37] introduced the VitaStress platform, demonstrating improved accuracy using multimodal signal integration.

Despite promising results, challenges remain in achieving reliable generalization. Controlled studies often report high accuracies, particularly when leveraging multimodal signals and sophisticated models

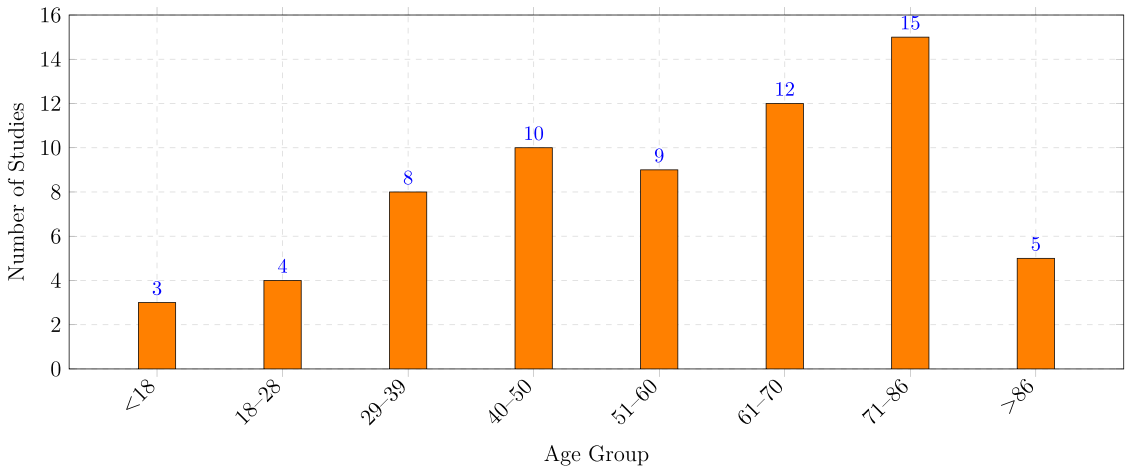


Fig. 3. Relationship between study population age intervals and the number of studies for each group.

such as CNN-LSTM or ensemble frameworks [28,30,51]. However, real-world applications face signal noise, environmental variability, and inter-individual differences. Field study performance often drops to F1 scores between 70% and 85% [45,65,72]. Techniques such as k-fold and leave-one-subject-out cross-validation are commonly used to assess robustness. However, the lack of standardized protocols, small or homogeneous samples, and inconsistent demographic reporting still limit comparability and reproducibility across studies.

In summary, current evidence, including several recent contributions from 2025, supports the feasibility and effectiveness of wearable sensor-based stress detection systems when integrated with appropriate machine learning methods. Continued research should prioritize standardized data collection, larger and more diverse samples, and robust validation in real-world settings to enable the practical deployment of these applications in healthcare, workplace monitoring, and mental wellness.

After analyzing the 66 reviewed studies, Fig. 3 shows that approximately 78% of the research focused on participants aged between 29 and 86 years. Additionally, around 25% of the studies included adolescent subjects, while 18% involved individuals over the age of 86. This distribution highlights a concentration on middle-aged and older adult populations, with comparatively fewer studies addressing stress detection in younger or elderly cohorts.

In the area of sensors, Gungormus et al. [43] played a crucial role by utilizing Electrodermal Activity (EDA) sensors to monitor stress in elderly individuals with rheumatic disease. Similarly, Al-Alim et al. [45] integrated EDA and other physiological signals into machine learning models, achieving high accuracy in free-living environments. Rescio et al. [48] focused on heart monitoring using ECG sensors in a workplace context, leveraging convolutional neural networks to detect stress levels in real-time. Nasirzadeh et al. [16] explored the use of temperature sensors for occupational health, particularly in construction workers, highlighting the relevance of thermal stress in harsh environments. More recent contributions, such as Darwish et al. [28], have combined ECG, EDA, and respiration data from wearable sensors and applied advanced algorithms, including XGBoost, to classify stress in both laboratory and real-world scenarios. Additionally, Schreiber et al. [37] introduced a multimodal platform integrating multiple vital signals, including EDA and HRV, for continuous monitoring, demonstrating improved performance over single-sensor systems. These studies underscore the growing trend toward multimodal sensing and the integration of advanced machine learning for robust stress detection across diverse contexts.

Fig. 4 illustrates the distribution of sensor categories used across the reviewed studies. Heart monitoring sensors, such as ECG and PPG, were the most prevalent, appearing in approximately 71.21% of the research. Electrodermal activity (EDA) sensors were closely followed, being utilized in 65.15% of the studies. Sensors for tracking movement and position, often through accelerometers or gyroscopes, were featured in 34.85%. Temperature sensors were used in 24.24% of the studies, while respiration monitors were used in 21.21% of the studies. Brain and neural activity sensors, such as EEG, were employed in 13.64%, and muscle activity sensors, including EMG, were the least common at 7.58%. These findings underscore the continued emphasis on cardiovascular and skin conductance signals as primary biomarkers for stress detection while also indicating a growing interest in multimodal sensor integration for more robust, context-aware systems.

When it comes to stress-related diseases, several recent studies stand out for their methodological and application advancements. Dahal et al. [57] developed a comprehensive framework for detecting stress using multimodal sensor data, achieving strong performance and demonstrating adaptability across individuals. Toshnazarov et al. [52] examined the use of consumer-grade smartwatches, illustrating the feasibility of deploying stress monitoring systems in real-life settings. Zhou et al. [75] extended this work by exploring the generalizability of physiological stress markers to anxiety disorders, contributing to the field of emotional arousal detection. Meanwhile, Mitro et al. [64] focused on high-stakes environments such as emergency evacuations, using wearable wristbands for real-time monitoring of stress-induced physiological changes.

More recent contributions from 2025 have broadened the landscape of application contexts. For instance, Darwish et al. [28] demonstrated high-accuracy stress classification in both lab-controlled and naturalistic settings using ECG and EDA signals combined with XGBoost models. Similarly, Ciccarelli et al. [26] underscored the challenges of generalizability by comparing wearable performance across structured and real-world environments. Deep learning innovations also feature in newer studies: Singh et al. [30] introduced a convolutional neural network model for real-time EDA-based stress classification, while Schreiber et al. [37] presented the VitaStress system. This multimodal platform fuses physiological signals for enhanced stress detection.

In terms of target conditions, the majority of the 2025 studies still focus on general stress detection, which represents about 72% of the reviewed work. However, there is a marked increase in attention toward mental health-related conditions: for example, Kader et al. [54] and Islam and Washington [56] focused on mental stress and personalized modeling, respectively. Specific disorders, such as depression, chronic pain, and intellectual disabilities were addressed in around 25% of studies, as seen in Gungormus et al. [43] and de Vries et al. [44]. Workplace and occupational stress contexts, including shift work, driving, and emergency response, made up about 20%, indicating ongoing interest in practical deployment scenarios [48,66,

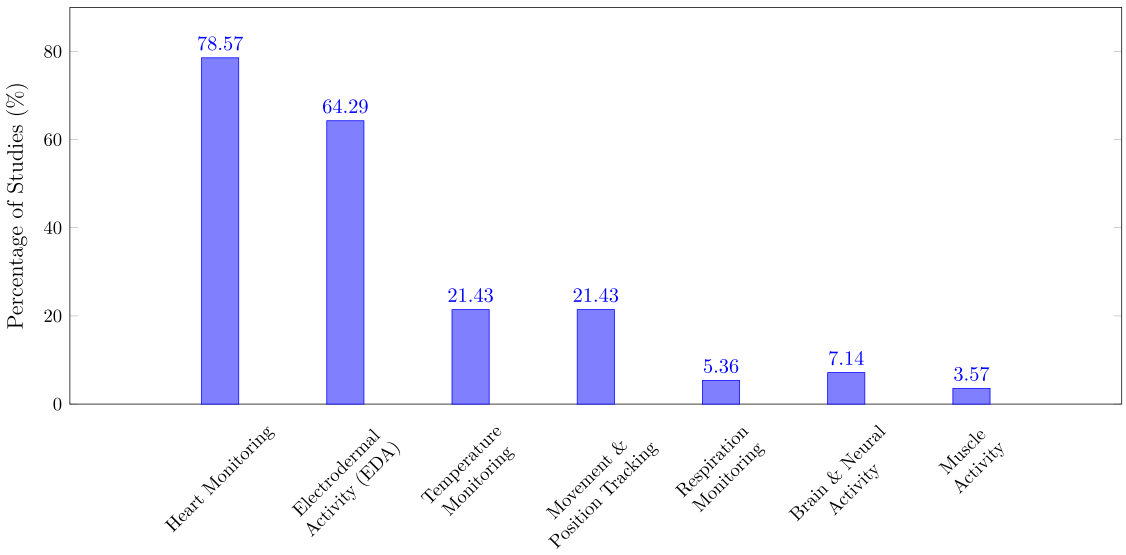


Fig. 4. Relationship between sensors and percentage of studies.

102]. These recent efforts underscore a clear trend toward multimodal integration, increased ecological validity, and specialized application contexts, marking a shift from purely experimental approaches to more clinically and operationally relevant stress-monitoring systems.

The reviewed studies revealed a consistent set of features that were most frequently used for stress detection across diverse populations and sensor modalities. Among physiological signals, heart rate variability (HRV) was the most prevalent, appearing in approximately 61% of studies. Time-domain measures, such as the standard deviation of NN intervals (SDNN), the root mean square of successive differences (RMSSD), and pNN50, were frequently used due to their responsiveness to stress-induced changes in autonomic function.

Electrodermal activity (EDA)-based features were closely followed, included in about **58%** of the studies. Skin conductance level (SCL) and skin conductance response (SCR) were the most commonly used metrics for their ability to detect short-term sympathetic arousal.

Respiration rate and amplitude were used in approximately 35% of the studies, often in combination with HRV to enhance model robustness. Skin temperature, although less prominent, appeared at roughly 27%, typically serving as a supporting indicator.

Contextual and behavioral features have become increasingly important in recent studies, with 45% of research incorporating variables such as ambient temperature, motion activity, phone usage, and posture. For example, Toshnazarov et al. [52] and Darwish et al. [28] demonstrated that contextual information significantly enhanced prediction accuracy in real-world settings. Finally, psychological indicators, including perceived stress scores and ecological momentary assessments (EMAs), were used in approximately 11% of the studies, mainly for model validation or as outcome labels, as in Zhou et al. [75].

The distribution of feature usage is illustrated in Fig. 5, which highlights the predominance of physiological features such as HRV and EDA while also showcasing the growing role of contextual and psychological variables in modern stress detection systems.

Researchers have explored a broad spectrum of modeling strategies to enhance stress detection using wearable sensor data. Rescio et al. [48] contributed to signal quality improvement by applying robust preprocessing pipelines, including advanced filtering, temporal alignment, and segmentation tailored to ECG and EDA signals. Almadhor et al. [72] demonstrated the impact of model architecture on prediction accuracy by employing deep neural networks (DNNs) and ensemble stacking strategies that integrated multiple base learners.

Expanding the modeling frontier, Parousidou et al. [58] compared conventional algorithms, such as Random Forest, with multi-task learning models, highlighting trade-offs in generalization and adaptability. From a statistical and explainability perspective, Mitro et al. [64] used regression modeling, recursive feature elimination, and statistical validation to refine stress markers during emergency simulations.

Recent studies from 2025 continued this trend. Singh et al. [30] applied a convolutional neural network (CNN) architecture optimized for electrodermal activity signals, achieving accurate real-time classification through deep feature hierarchies. Darwish et al. [28] implemented XGBoost in both lab-controlled and naturalistic settings, reinforcing the relevance of ensemble gradient boosting in heterogeneous data. Ciccarelli et al. [26] addressed domain adaptability by evaluating model performance across lab and field environments, while Schreiber et al. [37] fused deep learning with contextual awareness in their VitaStress platform, using multimodal embeddings to enhance prediction under noisy conditions.

The review also highlights the specific methods applied across studies, as shown in Fig. 6. Statistical methods, such as ANOVA, regression analysis, and correlation testing, were used in approximately 50% of the studies, often to validate features or interpret model outputs. Feature extraction techniques, including spectral analysis, time-domain signal descriptors, and dimensionality reduction, were employed in 42% of the works. Deep neural networks (DNNs) were implemented in about 38% of the studies, reflecting their growing popularity for handling complex, multimodal data. Signal preprocessing strategies, such as filtering, normalization, and segmentation, were employed in 36% of cases to enhance signal clarity and reduce artifacts. Random Forest was used in 26% of studies for classification tasks, while Support Vector Machines (SVM) appeared in 17%. XGBoost, an ensemble gradient boosting approach, was utilized in 14% of the reviewed research, particularly in recent real-world implementations.

4.2. Comparison of the studies analyzed

Fig. 7 presents direct associations between specific sensors and extracted features used in stress detection systems across the reviewed studies. Heart rate and heart rate variability (HRV) were predominantly extracted using ECG (18 instances) and PPG sensors (13 instances), confirming their foundational role in cardiovascular monitoring. Electrodermal activity (EDA) sensors were used in 22 studies to derive skin conductance levels (SCL), a primary indicator of sympathetic nervous system arousal. Accelerometers contributed to 16 studies through motion and posture features, while temperature sensors supported 12 instances of temperature-based stress estimation. Respiration belts enabled extraction of breathing rate in 11 cases. Additionally, GPS

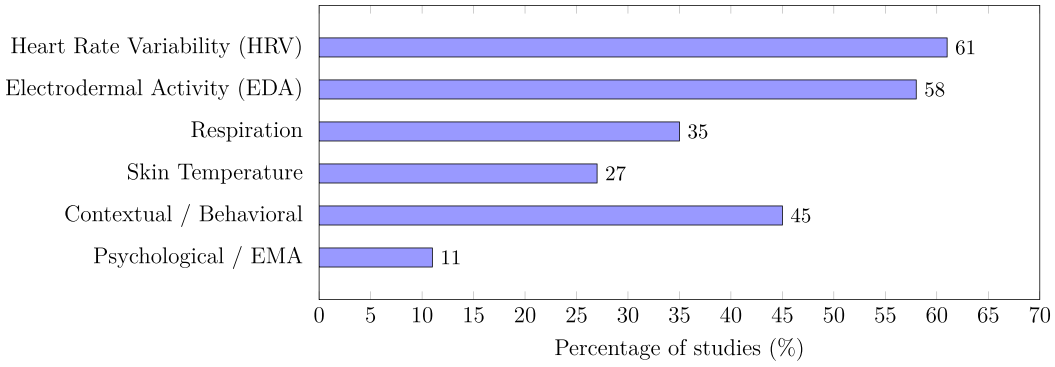


Fig. 5. Relationship between features and percentage of studies.

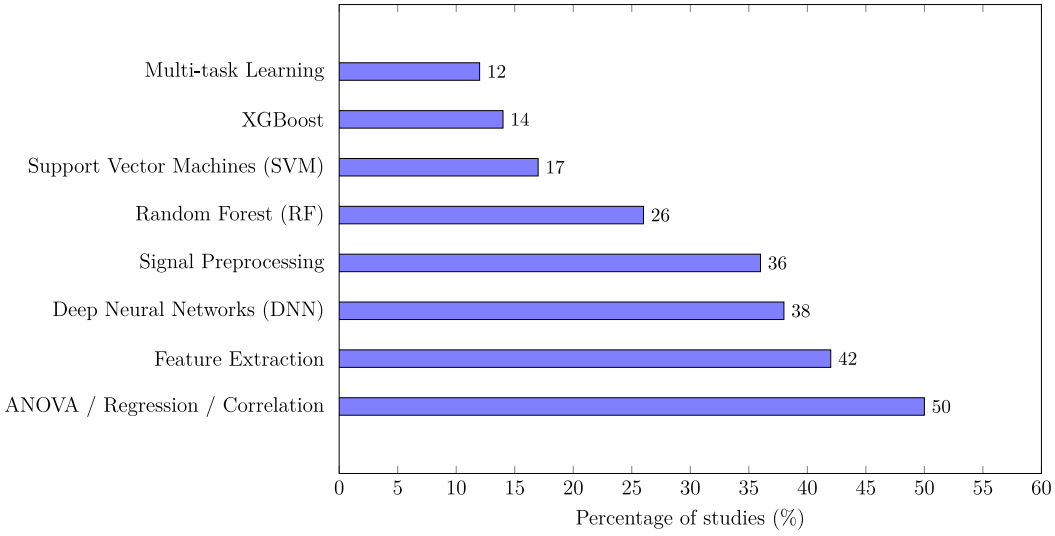


Fig. 6. Relationship between methods and percentage of studies.

modules were used in 9 studies to assess location-dependent stress contexts, microphones contributed 6 times with speech-related features, and light sensors appeared in 4 studies to provide environmental data.

Fig. 8 illustrates the interaction between various methods and sensors employed in stress detection studies. Machine learning was the most commonly used method, particularly in combination with EDA and heart-monitoring sensors (e.g., ECG and PPG), where its capability to manage continuous physiological signals proved effective. Statistical analysis and data preprocessing techniques were frequently applied across sensor types to support feature extraction and enhance signal clarity. Deep learning methods, while less frequent than traditional machine learning, were also prominently paired with EDA and heart rate data. Additionally, techniques such as cross-validation, signal filtering, and feature fusion contributed to the development of a robust model, showcasing the methodological diversity in wearable stress monitoring systems.

Fig. 9 illustrates the detailed relationship between extracted features and applied methods in wearable-based stress detection studies. Instead of grouping them into broad categories, this matrix highlights the direct use of specific machine learning and signal-processing techniques such as Random Forest, SVM, CNN, and LSTM. Heart Rate Variability (HRV) and skin conductance are the most frequently used features, appearing in studies that utilize classical methods, such as Random Forest and SVM, as well as deep learning approaches, including CNN and LSTM. Signal filtering techniques also show strong associations with HRV and skin conductance, reinforcing the importance of preprocessing in physiological data analysis. Respiration rate, movement activity, and temperature features are closely related, with frequent application in

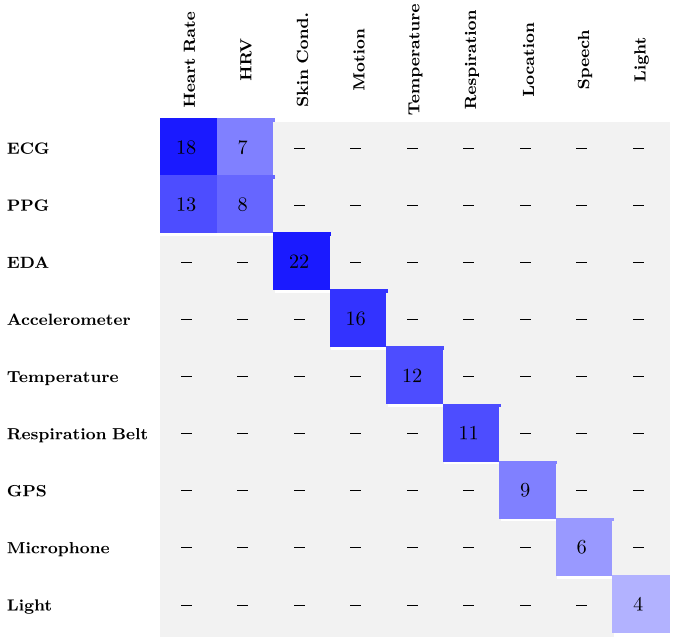


Fig. 7. Relationship between sensors and features.

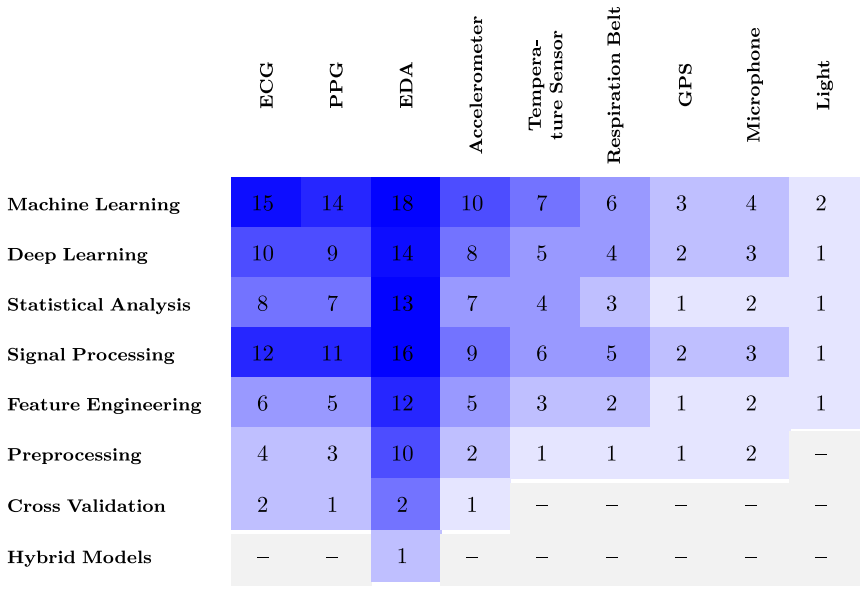


Fig. 8. Relationship between methods and sensors.

both statistical tests and XGBoost models. Contextual features, such as environmental conditions and sleep quality, appear less frequently but are often paired with fusion models or deep architectures to enhance model robustness.

Table 3 summarizes the key benefits and limitations identified across studies on stress prediction using wearable sensors and machine learning. Cross-sectional and pilot studies offer valuable early insights but often face limitations such as small sample sizes, limited generalizability, and inconsistent data quality. Many studies focus on specific populations, such as persons in high-stress occupations or those with particular health concerns, making it challenging to extrapolate the conclusions to a broader audience.

Despite these challenges, wearable technologies demonstrate tremendous potential for real-time stress monitoring through continuous data collection and individualized feedback. However, inconsistency in data collection methodologies and uncertainty about long-term effectiveness remain barriers. Addressing these challenges with more diverse sampling and consistent techniques may improve both reliability and applicability. Finally, it demonstrates significant progress in wearable-based stress detection, underscoring the need for further research to enhance consistency, scalability, and inclusivity in real-world applications.

Future research should focus on enhancing data quality, developing advanced models, and making these systems accessible for everyday use. Standardizing data formats and ensuring data accuracy are essential first steps. Future models should integrate contextual information and explore cutting-edge techniques, such as deep learning.

Testing these models in real-world settings over time is crucial to ensure they function effectively across diverse environments and populations. Researchers must also consider ethical issues, prioritize user-friendly designs, and collaborate across disciplines to create well-rounded solutions. Enhancing sensor technology and developing multifunctional devices could significantly boost the effectiveness of stress monitoring and prediction systems.

4.3. Effectiveness of the applied machine learning methods

A comprehensive evaluation of the effectiveness of machine learning (ML) methods used in stress detection requires considering multiple factors, including the models employed, the type and quality of sensor data, validation strategies and reported performance metrics. Across the reviewed studies, a wide array of ML techniques have been implemented, including classical algorithms like Support Vector Machines

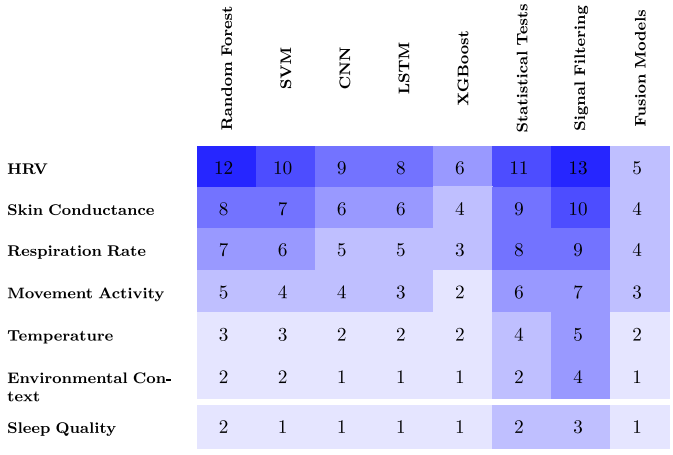


Fig. 9. Matrix showing co-occurrence between extracted features and methods in stress detection studies.

(SVM), Random Forests (RF), and k-Nearest Neighbors (kNN), as well as deep learning models such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory networks (LSTMs).

Reported performance metrics, most commonly accuracy, F1-score, and Area Under the Receiver Operating Characteristic Curve (AUC-ROC), generally range from 70% to over 95%. However, direct comparison is complicated by differences in study design, feature engineering, sensor modalities, and evaluation procedures. Studies utilizing multimodal sensor data (e.g., combining electrodermal activity, heart rate variability, and skin temperature) tended to achieve higher performance, particularly when paired with ensemble methods or deep learning architectures.

Deep learning models, particularly CNNs and LSTMs, have shown promise in capturing complex temporal patterns in physiological signals. However, their success often hinges on the availability of large, high-quality datasets, which is a limiting factor in many current studies. Conversely, traditional ML models remain competitive in smaller datasets due to their lower data demands and interpretability.

It is also notable that validation strategies varied considerably, ranging from simple holdout validation to more robust approaches, such as k-fold cross-validation or leave-one-subject-out (LOSO) validation.

Table 3
Studies benefits and limitations.

Paper	Benefits	Limitations
Ciccarelli et al. [26]	High accuracy (up to 96%) in detecting stress through wearable sensors (Empatica E4, Zephyr BioHarness); adaptable machine learning models for various stress protocols; use of multiple physiological markers (EDA, HRV, and others).	Difficulties in generalizing findings from the laboratory to real-world industrial settings; variability in experimental protocols; requires specific device configurations for different environments.
Qiao et al. [27]	High accuracy and real-time monitoring of cortisol levels in sweat using a wearable electrochemical biosensor; low detection limit (33 pg/mL); non-invasive stress monitoring.	Sensitive to biofouling in complex biological fluids (e.g., sweat); sensor performance may degrade over time due to biological contamination.
Darwish et al. [28]	High performance in a three-stage validation for real-life applicability; uses multimodal data (ECG, EDA, Respiration); high accuracy (up to 99.78%) in stress detection.	Limited to controlled environments for validation; challenges in applying findings to real-world stress detection due to variability in individual stress responses.
Singh et al. [30]	Achieved 95% accuracy and 90% F1-score in early stress detection using Electrodermal Activity (EDA) signals; open-source code for reproducibility.	Focused solely on EDA signals; may not capture the full spectrum of stress responses; relies on sensor data quality.
Tanwar et al. [36]	Achieved 96% accuracy in binary classification and 90% in 3-class classification; demonstrated the potential of autoencoders with wearable data.	Evaluation conducted in controlled settings; real-world applicability may vary; requires high-quality labeled data for training.
Schreiber et al. [37]	Achieved 89% accuracy in binary classification; 79% in ternary classification; multimodal approach for improved stress detection.	Limited to specific datasets; generalizability may be constrained; multimodal data collection increases complexity and cost.
Mai and Chung [38]	High accuracy in mental stress detection using behind-the-ear EEG; on-chip processing enhances portability and real-time performance.	Limited to specific types of mental stress (e.g., Stroop test); challenges in maintaining signal quality for long-term use; device-specific issues.
Andrić et al. [39]	Successful stress anticipation using wrist-worn biomarker data; promising results with random forest classifiers; transparency through publicly available datasets.	Focused on specific biomarkers; may not capture all physiological indicators of stress; performance may vary across individuals.
Hossain et al. [40]	Simultaneous measurement of EMG signals and sweat cortisol; on-device data processing with IoT capabilities; real-time monitoring.	Sensitive to biological contamination (sweat); limited to exercise scenarios for validation; sensor reliability issues.
Habib et al. [41]	Achieves high accuracy for detecting Major Depressive Disorder (MDD) from ECG signals using deep learning; requires only a single-channel ECG.	Limited by the size of datasets; need for real-time analysis and computational power; dependent on the quality of ECG signal data.
Fontes et al. [42]	High accuracy (95.83%) in stress detection, real-time, non-contact stress monitoring using rPPG from facial videos, computational efficiency due to hybrid deep learning models.	Limited to facial video-based rPPG analysis, may be affected by lighting conditions or facial expressions, and requires high-quality video capture for accurate results.
Gungormus et al. [43]	The research offers real-time monitoring of pain and stress, potentially improving chronic pain management in older adults.	Provides real-time monitoring of pain and stress, potentially improving chronic pain management in older adults.
de Vries et al. [44]	Improved ability to detect stress in real-time for individuals with intellectual disabilities, helping caregivers respond more effectively.	Requires further large-scale validation to confirm results.
Al-Alim et al. [45]	Real-time stress detection with high accuracy, early identification of stress to prevent chronic health issues.	Requires complex preprocessing of sensor data; performance depends on the quality of the wearable devices used.
Pickert et al. [47]	Independent of ECG; high sensitivity in resting conditions.	Lower performance under physical stress, improvements needed in real-time applications for physical activity scenarios.
Nasirzadeh et al. [16]	The study demonstrates that off-the-shelf wearable sensors can provide accurate, continuous body temperature monitoring, helping to mitigate safety risks like heat stress and physical fatigue in high-risk work environments.	The study primarily examined wearable temperature sensors from 2020–2021, focusing on accuracy and comfort, with future research exploring cost analysis and newer devices.
Rescio et al. [48]	It provides a non-intrusive, real-time monitoring system with high accuracy for stress detection in work environments.	The study was limited to a small group, and further research is needed to refine sensor placement and improve comfort over long-term use.
Nuamah [49]	The finds uncovered valuable insights into how stress affects human physiological responses and task performance	The study was limited by participant size and controlled laboratory environment
Béquet et al. [50]	The study presents a real-time stress regulation in critical driving situations	Accounts for only 34%–46% variance in stress regulation effectiveness; unaccounted factors include tactile sensitivity.
Tanwar et al. [51]	Real-time stress detection with higher accuracy compared to traditional machine learning models, offering a potential solution for wearable-based health monitoring systems.	Requires further validation on diverse datasets and a larger sample size to generalize the model across different populations.
Toshnazarov et al. [52]	Affordable stress detection using commonly available devices; no need for specialized equipment	Lower accuracy in real-world conditions due to contextual variability and sensor limitations.
Carreiro et al. [53]	Potential to provide real-time, just-in-time interventions during high-risk periods of recovery, improving access and effectiveness of SUD treatments.	The study's event detection models have limitations due to cross-validation and random split data, potentially limiting their generalizability to new users.
Kader et al. [54]	Low-cost, portable, and simple stress detection system for workplace and real-life environments.	Moderate accuracy with EEG alone and requires additional work for higher classification accuracy and extended validation.

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Table 3 (*continued*).

Paper	Benefits	Limitations
Volpes et al. [55]	A wearable biomedical device monitors physiological parameters like PPG, ECG, GSR, and motion signals from fingers, offering valuable insights into cardiovascular variability and autonomic nervous system activity.	Early-stage development, further validation required
Islam and Washington [56]	The study introduces a self-supervised pre-training method for personalizing deep learning models for stress prediction, thereby enhancing precision and personalization and improving stress management effectiveness while reducing annotation efforts.	The study's limitations include the use of a single dataset and modality, which necessitate testing across diverse biosignal datasets and health outcomes, as well as examining model explainability and applicability.
Dahal et al. [57]	The study presents a global stress detection framework that combines eight HRV features with a random forest model, achieving 99% classification accuracy in individual testing using benchmark datasets.	Limited generalization due to the small number of subjects in datasets; lack of diversity in stress causes; controlled lab environments not mimicking real-world settings
Parousidou et al. [58]	Significant improvement in stress detection accuracy through personalization.	Limitations include difficulty in applying lab results to real-world settings; lack of finer categorizations and small dataset size; binary classification of stress detection overlooks differences between high and low-stress levels
Whiston et al. [61]	Real-time, non-intrusive stress monitoring; insights into the bi-directional relationship between stress and residual symptoms in depression.	Small sample size, limited generalizability; potential for user compliance issues with wearable devices over extended periods.
Nan et al. [62]	Real-time, continuous cortisol monitoring; non-invasive approach; sensitive and accurate detection of cortisol in human sweat.	Limited to cortisol detection; dependence on environmental conditions for sensor stability.
Zheng et al. [63]	CAFS reduces the number of selected features while maintaining stress detection performance, making it efficient for real-time wearable applications.	Tested only on publicly available datasets; real-world applicability in diverse environments is not evaluated
Mitro et al. [64]	Provides real-time monitoring of stress and vital signs in emergencies with high accuracy.	Slight reduction in accuracy when used in real-world environments compared to controlled laboratory settings.
Campanella et al. [65]	Provides insight into the reliability of combining physiological sensors with self-reports for stress detection.	Requires more extensive validation for applicability in uncontrolled, real-world environments.
Costantini et al. [66]	Non-invasive, reliable for HRV measurements in no-motion conditions	Small sample size, poor reliability of EDA metrics, sensitivity to motion artifacts
Bocharov et al. [67]	Real-time, long-term monitoring of functional state, helping in decision-making regarding personnel combat readiness	Limited technological capacity for widespread use in the Defense Forces, and a focus on senior officers only
Stržinar et al. [68]	Improved stress detection with fewer data points; suitable for real-time wearable applications.	Only tested on the WESAD dataset and not yet applied in broader real-world scenarios
Vos et al. [69]	Improved stress prediction accuracy and generalization across different datasets.	Limited real-world validation; Primary based on synthesized data.
Gaitán-Padilla et al. [71]	The study presented Low-cost, non-invasive monitoring of physiological stress signals, good performance in stress classification	Small sample size, further testing needed for robust validation
Almadhor et al. [72]	Protects privacy by using a federated learning approach that does not require raw data transfer; Improved accuracy in stress detection compared to traditional methods.	The study focuses on detecting stress from wrist-based electrodermal activity, which might limit the accuracy in identifying eustress (positive stress) and distress (negative stress). Additionally, FL models face challenges in communication efficiency and data cleaning due to the decentralized nature of data storage.
Almadhor et al. [73]	Improved accuracy for stress detection compared to previous methods; applicable to wearable healthcare technologies.	Need for more data to improve the applicability of AI methodologies in healthcare
Zamani et al. [74]	High accuracy in predicting sleep quality; scalable for e-health applications.	Some limitations related to parameter tuning and the complexity of the AEA algorithm.
Zhou et al. [75]	Revealed that anxiety and stress detection models could be generalized to other high-arousal emotions, which might be used to build more robust stress detection systems.	Differences in signal quality, stimuli, and emotion annotation across datasets challenge the generalization of stress and anxiety detection models.
Momeni et al. [78]	CAFS reduces the number of selected features while maintaining stress detection performance, making it efficient for real-time wearable applications.	Its real-world applicability in diverse environments is not evaluated, as it has been tested only on publicly available datasets.
Rominger and Schwerdtfeger [79]	Offers a real-time stress detection method suitable for high-stress occupations like firefighting.	The study needs further validation in diverse real-world scenarios.
Ritsert et al. [80]	Objective, continuous monitoring of emotional states, which could enable real-time interventions and future psychological treatment methods.	Small sample size and reliance on video-induced anxiety might limit the generalizability of the results.
Chalabianloo et al. [81]	Demonstrates the potential of wearable devices in real-time stress and activity monitoring using machine learning.	Small sample size and controlled environment may limit generalizability to broader populations.
Cummings et al. [82]	Improved detection of stress during pregnancy, enabling personalized interventions to address mental health and maternal stress.	Lower adherence rates among participants with higher baseline negative affect; technical issues and skin irritation from bio-sensors reduced wearability compliance.
Ehrhart et al. [83]	Effective in balancing datasets and improving classifier performance, generation of high-quality labeled data	Generative Adversarial Network (GAN) training requires significant effort and hyperparameter tuning, risk of mode collapse still exists in small datasets

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Table 3 (*continued*).

Paper	Benefits	Limitations
Zhang et al. [84]	Offers insights into how specific urban environments (green and blue spaces) can promote or mitigate stress.	The impact of noise from nearby traffic in green and water areas was not isolated, which may affect the accuracy of stressor identification.
Zhong et al. [85]	Provides a non-invasive, real-time stress detection system for industries such as firefighting and piloting.	Challenges in accurately classifying stress due to physiological and psychological individual differences.
Thierfelder et al. [86]	Personalized detection of stress and anxiety in real-life settings, enhancing telemedicine for OCD treatment.	Limited sample size; more data are required for broader conclusions.
Iqbal et al. [87]	Provided insights into the sensitivity of different biophysiological stress indicators, contributing to the development of connected health technologies.	Lack of standardized methodology for stress detection across different individuals and environments; only healthy participants were considered.
Hag et al. [88]	Reduced feature space, improved classification accuracy, faster computation.	Limited testing on only four datasets raises concerns about potential overfitting.
Hossain et al. [90]	Enhances the usability of wearable ECG devices for long-term monitoring, particularly for atrial fibrillation detection.	Further research is needed to assess performance in various real-world scenarios, such as different noise environments.
Aristizabal et al. [92]	Provides insight into the reliability of combining physiological sensors with self-reports for stress detection.	Requires more extensive validation for applicability in uncontrolled, real-world environments.
Föll et al. [93]	Provides a comprehensive toolkit for reproducibility in wearable data research; allows easy integration and application of machine learning models.	Parameters require manual tuning for unsupported wearables; the toolkit's performance is dependent on data quality.
Dzieżyc et al. [94]	Simplifies the classification process by eliminating feature engineering, reduces dimensionality issues and increases the chances of uncovering new patterns.	The effectiveness of models varies depending on the type and intensity of the physiological stimuli.
Bolliger et al. [95]	Combines self-reported data and physiological monitoring for comprehensive stress tracking.	High demand on participants. Particularly, answering EMAs and wearing the Empatica wristband requires time, effort, and adherence to the protocol from participants and ongoing involvement from the researchers during data collection to provide support.
Delmastro et al. [96]	Helps in maintaining low-stress levels during cognitive training while improving cognitive functions in frail older adults.	The system only provides binary stress detection (stress/non-stress) and requires further enhancement to quantify stress levels more accurately.
Akbulut et al. [97]	The wearable sensor provided accurate data comparable to hospital-grade devices. The use of the mobile app enabled real-time stress tracking, which could aid in better managing psychosocial stress.	The study was conducted exclusively in a hospital setting and included only 30 participants. Future studies could include a larger population to improve the generalizability of the results.
Can et al. [98]	Significantly increases the performance of stress detection systems; enables application in real-world, ambulatory environments using off-the-shelf devices.	Limited battery life of smartwatches; data quality depends on the wearer's movements and proper device attachment.
Zubair and Yoon [99]	Improved system performance with ultra-short recordings; potential to enhance low-cost wearable devices for healthcare monitoring.	Sensor accuracy and reliability may vary with different stressors, resulting in limitations in real-world applicability due to PPG sensor constraints.
Mozgovoy [100]	Enables more accurate remote stress monitoring when user activities are not directly observed.	Further research is needed to cross-validate stress detection algorithms, and the possible variability in stress responses due to different activities was not accounted for.
Rodrigues et al. [101]	The study presented continuous stress monitoring for improved health management.	Limited information available; likely conceptual stage.
Betti et al. [102]	Demonstrates the effectiveness of using physiological sensors for real-time stress detection in working environments, providing a non-invasive method correlated with biological markers.	Small sample size; EDA data loss due to equipment malfunction; issues with reproducibility of sensor results during physical activities.
Xu et al. [103]	Accounts for intersubject differences, improving the reliability of stress evaluation in real-world applications.	Small dataset size and limited number of clusters may affect generalizability.
Seoane et al. [104]	Allows for real-time monitoring of stress and ANS responses in a non-invasive manner, potentially aiding military personnel in high-stress scenarios. The integration of textile electrodes makes the system lightweight and suitable for field use.	Not all EEG indicators were recorded in the first phase, and further validation of the garment's effectiveness in different stress-inducing tasks is necessary. Data on the best positioning of sensors (such as GSR placement) is still being analyzed.

Studies employing subject-independent evaluations generally reported lower but more realistic performance, indicating that many reported accuracies may be inflated due to overfitting in subject-dependent setups.

To support this evaluation, Table 4 summarizes the main model types used across the reviewed studies, including the number of studies, the range of accuracy, and key observations regarding their strengths and limitations.

Despite promising lab-based results, models often perform substantially worse under real-world conditions. For example, Toshnazarov et al. [52] report an F1-score drop from 0.84 (lab) to 0.71 (field) using smartwatch data, while Parousidou et al. [58] found a decline from 99% (lab) to 66% in real-world, mobile contexts. These drops reflect the effects of motion artifacts, sensor misplacement, user non-compliance, and signal noise—challenges that are rarely simulated during training. It highlights the importance of including diverse, naturalistic datasets in model development and evaluation.

One example of applied ML stress detection is in aviation. Rodrigues et al. [101] analyzed ECG-derived HRV in high-stakes tasks such as take-off and landing simulations. While ML models achieved high performance in detecting acute stress under simulated conditions, signal stability, and wearability became issues during real cockpit

Table 4
Summary of machine learning model effectiveness across reviewed studies.

Model type	No. of studies	Accuracy range	Comments
Random Forest (RF)	15	85%–99%	Strong performance with EDA and HRV signals; effective in multimodal fusion.
Support Vector Machine (SVM)	12	72%–95%	Performs well in small datasets; sensitive to feature scaling and kernel choice.
k-Nearest Neighbors (kNN)	6	70%–90%	Simplicity and interpretability; lower performance in high-dimensional data.
Convolutional Neural Networks (CNNs)	8	90%–98%	Captures spatial patterns in multimodal or raw signal input; needs large datasets.
Long Short-Term Memory (LSTM)	5	88%–97%	Effective for temporal sequence modeling; data-intensive and prone to overfitting.
XGBoost	4	87%–99%	High accuracy and robustness with tabular physiological features.

operations, where motion, harness friction, and noise affected sensor reliability. It emphasizes the need for robust preprocessing, wearable design optimization, and adaptive model calibration for operational use.

Privacy-preserving approaches such as federated learning are still rare despite their importance for long-term, real-world deployments. Furthermore, few studies address regulatory compliance (*e.g.,* GDPR, HIPAA, CE/FDA standards), which is essential for clinical translation. Most studies also draw from homogeneous populations, typically young, healthy adults, limiting generalizability to elderly, clinical, or high-stress occupational groups. Broader, inclusive research is essential for equitable and scalable stress detection technologies.

In summary, while many ML methods demonstrate high reported effectiveness, the lack of standardization in evaluation and data collection protocols makes it difficult to draw definitive conclusions. Future research should prioritize the use of standardized benchmarks, public datasets, and rigorous cross-validation to provide a clearer picture of comparative model performance and its real-world applicability.

4.4. Comparison with other studies

This review integrates wearable sensor data and machine learning techniques for predicting stress. It emphasizes key physiological features such as heart rate variability (HRV), electrodermal activity (EDA), and photoplethysmography (PPG) while exploring ML models such as Random Forest (RF), Support Vector Machines (SVM), and deep neural networks (DNN).

Reviews like [7,17] provide a comprehensive overview of stress detection technologies, informing domain-specific applications such as pregnancy and academic settings. These studies highlight more specific domains compared to the present review.

Authors such as Smirthy et al. [107] and Bolpagni et al. [108] focus on advanced ML techniques and biosignal analysis, which are foundational to all reviewed applications. These advancements address challenges like algorithm accuracy and multimodal data integration, supporting use cases like those explored in [109].

The focus on usability and real-time solutions in [110] aligns with discussions on practical implementation across various settings, as seen in [111,112]. Finally, this review explores the intersection between the potential of wearable sensor data and machine learning (ML) techniques for predicting stress episodes.

Additionally, the work by [113] provides valuable insights into the integration of sensor data and machine learning techniques for stress prediction. This study examines how wearable devices that capture physiological features, such as HRV and EDA, can be combined with machine learning algorithms to predict stress. The authors focus on the use of both traditional sensors and emerging technologies, such as photoplethysmography (PPG), which is highly relevant to the present review's discussion of sensor technologies for stress detection. Specifically, their findings support the idea that combining multiple physiological sensors, such as EDA and HRV, enhances predictive accuracy. The study also examines the challenges of collecting data from diverse sensors and integrating them into machine learning models for real-time stress monitoring. It aligns with the review's emphasis on the need for multimodal data fusion and robust ML models that can handle real-time physiological data. It bridges the gap between theoretical research on sensor technologies and practical applications, demonstrating the effectiveness of real-time stress prediction systems. Moreover, the study's findings regarding the use of various sensor modalities (such as HRV, EDA, and PPG) underscore the growing importance of sensor diversity in enhancing stress detection accuracy, a key aspect consistently emphasized throughout this review.

4.5. Limitations

The methodology's inclusion criteria, presented in Section 2.3, which prefer research using wearable sensors and machine learning (ML) approaches for stress prediction, naturally influence the findings of this systematic review. Although this emphasis guarantees applicability, it could unintentionally restrict the scope, possibly leaving out research using different methodologies or data sources that might offer insightful information.

The significant heterogeneity of the primary studies is a significant limitation of this review. Differences in study design, sample size, participant demographics, stress-induction protocols, and data-collection environments introduce substantial variability. Moreover, most experiments were conducted with younger adults in controlled laboratory settings, constraining the generalizability of the findings to other age groups, occupational contexts, and real-world conditions. Comparability is further hindered by the wide range of wearable devices and datasets used to train machine-learning models, as well as by inconsistent preprocessing pipelines, such as signal denoising, feature extraction and selection, and validation strategies, which can significantly influence the reported performance metrics.

Publication bias may be introduced by focusing on research published in English or indexed in particular databases. The review's results may be skewed toward research with noteworthy or positive outcomes due to the underrepresentation of studies with null or inconclusive results, as well as those conducted in languages other than English.

Although the review's time frame (January 2010–June 2025) encompasses wearable technology developments, it may overlook new developments or recently released works that fall outside this timeframe. Similarly, because stress predictors and behaviors can differ significantly between cultures and situations, the geographic and cultural contexts of the included studies might not accurately represent worldwide variation.

Although care was taken to extract all relevant data from full texts, we acknowledge that in a few cases the level of methodological detail in some studies was limited. To address this, such studies were

either excluded or noted in our analysis as lacking full methodological transparency.

The review's dependence on machine learning techniques presents methodological issues. The variety of machine learning models (such as random forests and deep learning architectures) and their parameter settings make direct comparisons between research more difficult. Standardizing efficacy interpretations is also challenging because studies use a variety of measures. Additionally, much research focuses on identifying acute stress rather than long-term stress patterns, which are crucial for understanding chronic stress and its effects.

In addition, while many studies reported high classification accuracies, sometimes exceeding 95%, such metrics must be interpreted with caution. These results may reflect overfitting to specific datasets or experimental conditions, especially when models are not validated on external datasets or in real-world settings. The lack of standardized benchmarks and limited external validation across studies raises concerns about the reproducibility and generalizability of these machine learning models. Furthermore, insufficient reporting of hyperparameter configurations, data splits, and open-source code availability hinders replication and independent verification. Addressing these issues is crucial to avoid inflated performance claims and to ensure that proposed stress prediction systems are robust, reliable, and clinically useful.

Furthermore, there may be obstacles to the practical application of the findings. The practical application of stress prediction systems is restricted by the high cost of specialized wearable sensors, which are less available to the general public. The inconsistent treatment of data privacy, user consent, and confidence in ML models across studies highlights a gap in user-centric research. Also, the dynamic, noisy, and multifaceted nature of stress in real-world contexts needs to be considered by the controlled settings of many studies.

Another notable limitation of this review is its predominant focus on studies conducted in controlled laboratory environments, which may not adequately capture the complexities of real-world deployment. Several factors, including motion artifacts, variable environmental conditions, user compliance, device misplacement, battery life constraints, and long-term wearability, pose significant challenges in field settings. These elements can degrade signal quality and reduce model performance when transitioning from lab to real-world use. Furthermore, user acceptance and consistent engagement with wearable devices over prolonged periods remain underexplored yet are critical for sustained monitoring and practical adoption in everyday life. Future studies should prioritize in-the-wild evaluations to bridge the gap between theoretical efficacy and real-world feasibility.

Privacy-preserving techniques, such as federated learning and edge computing, are emerging as important enablers of scalable stress monitoring systems that respect user data sovereignty. However, the adoption of these approaches remains limited in the literature reviewed. Additionally, regulatory barriers, such as the need for compliance with GDPR, HIPAA, or FDA medical device standards, pose significant hurdles to the clinical translation of wearable-based stress prediction tools. These aspects are seldom addressed explicitly, highlighting a gap between research prototypes and deployable, legally compliant solutions.

The study concludes by recognizing the necessity of more uniform protocols in data collecting, sensor integration, and ML model evaluation to increase replicability and comparability in subsequent studies. To overcome these constraints, multidisciplinary cooperation and wider integration of various populations, tools, and approaches will be necessary.

4.6. Final remarks

This systematic review explores the use of machine learning in predicting stress episodes using data from wearable sensors. It highlights the importance of physiological indicators, such as heart rate variability, electrodermal activity, and sleep patterns, in identifying stress episodes. The review uses advanced machine learning algorithms to understand real-time stress responses. The findings highlight the transformative potential of wearable devices in proactive health management. However, challenges like small sample sizes and data variability must be addressed. The review recommends standardization and interdisciplinary approaches to improve the effectiveness and accessibility of wearable-assisted stress prediction systems. It also presents a roadmap for future research in wearable technology and machine learning.

The main conclusions, which answer the research questions (Section 2.1), were based on the analysis of 66 studies. Regarding RQ1, "What are the most commonly used sensors to detect stress episodes?", it is verified that electrodermal activity (EDA) sensors, such as GSR, and heart monitoring devices, including ECG, PPG, HRV, and pulse oximeters, are the most widely used sensors for identifying stress episodes. Additionally, common are temperature sensors, respiration monitoring devices, and movement trackers such as gyroscopes and accelerometers. EEG sensors are frequently used to monitor brain activity. However, wearable biosensors and specialized sensors, such as laser-induced graphene (LIG), can improve accuracy. Furthermore, imaging tools that aid stress detection include thermal infrared sensors and RGB cameras. These multimodal systems offer thorough behavioral and physiological data for precise stress monitoring and prediction.

Concerning RQ2, "What features are most frequently extracted from wearable devices for the identification of stress episodes?", physiological metrics like heart rate (HR) and heart rate variability (HRV), electrodermal activity (EDA) parameters like skin conductance level (SCL) and response (SCR), respiratory rate and patterns, and skin temperature variations are the most commonly extracted features from wearable devices for stress identification. Along with sleep parameters such as onset delay, efficiency, and cycles, accelerometer movement variables, including activity magnitude and posture, are also frequently examined. Time-domain metrics (like mean and standard deviation) and frequency-domain information (like spectral power) are examples of signal processing characteristics. To better understand stress responses, contextual information, such as ambient circumstances and GPS-based activity tracking, is combined with psychological indicators, including subjective stress levels. These elements enable comprehensive, data-driven stress monitoring.

Finally, based on RQ3, "How effective are the methods applied in these studies?", the machine learning approaches used for stress detection demonstrate high effectiveness under controlled conditions. Models, such as Random Forest (RF) and Support Vector Machines (SVM), frequently report accuracies exceeding 95%, while deep learning architectures, particularly CNN-LSTM hybrids and Gated Recurrent Units (GRUs), perform well in complex, multimodal scenarios. Robust preprocessing techniques, including noise filtering and signal normalization, contribute significantly to data quality and model precision. Although lab-based studies often achieve high accuracy, real-world applications encounter greater variability due to environmental noise and individual physiological differences, resulting in lower but still respectable F1-scores, ranging from 70% to 85%. Validation strategies such as k-fold cross-validation and leave-one-subject-out (LOSO) contribute to assessing model robustness. Nevertheless, challenges, such as small sample sizes and inconsistent data collection protocols, remain barriers to broader applicability, underscoring the need for further research and standardization to ensure generalizability in practical deployments.

The study highlights the need for future studies to focus on larger sample sizes and more diverse populations, thereby improving the generalizability of the results. It promotes longitudinal research to provide solid proof of the efficacy of interventions. It recommends repeating studies in various contexts and demographics to confirm and bolster the body of evidence. Recommendations to provide comprehensive and effective interventions include integrating patient-centered research, embracing interdisciplinary approaches, and prioritizing the

affordability and accessibility of technologies. For the profession to advance, existing study constraints must be addressed, including the improvement of study designs and the expansion of sample sizes. The report outlines its contributions beyond data synthesis and provides a research path for wearable sensor-based health devices and stress detection.

5. Conclusion

With an emphasis on their applicability in healthcare and mental health management, this paper systematically evaluated 66 studies on wearable sensors and machine learning applications for stress prediction. It highlights how physiological monitoring, including electrodermal activity and heart rate variability, can be used as diagnostic tools to guide individualized stress management. The evaluation highlights the importance of integrating cutting-edge technologies with healthcare policies to enhance real-time monitoring and forecasting capabilities. In contrast to other studies that focus on more general applications of wearable technologies for health outcomes, a structured technique was employed to gather and analyze the data. The review focuses only on wearable-assisted stress detection's diagnostic and prognostic capabilities, especially for groups dealing with mental health issues or chronic stress.

The study identifies significant gaps in the existing literature, including small sample sizes, a need for greater diversity in the populations examined, and challenges in extrapolating results to practical settings. These restrictions make it more difficult to establish strong causation and reduce the findings' applicability to larger or more diverse groups.

The paper emphasizes that these gaps must be addressed in future research by focusing on larger, more diverse samples, conducting longitudinal studies, and improving the affordability and usability of wearable technology. It promotes patient-centered research and interdisciplinary approaches to create more specialized interventions. The study highlights how technology can enhance healthcare outcomes and the potential of personalized stress-reduction plans tailored to wearable sensor data.

CRediT authorship contribution statement

António Oseas Pataca: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Eftim Zdravevski:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Paulo Jorge Coelho:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Nuno M. Garcia:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Margot Deryuck:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Carlos Albuquerque:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ivan Miguel Pires:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Ethics statement

Not applicable.

Declaration of competing interest

The authors declare no conflicts of interest.

Acknowledgments

This work was supported by FCT - Fundação para a Ciência e Tecnologia, I.P. by project reference UID/50008/2023 IT.

This work was funded by FCT Pluriannual Funding UID/308: Instituto de Engenharia de Sistemas e Computadores de Coimbra - INESC Coimbra.

This work was also funded by National Funds through the FCT - Foundation for Science and Technology, I.P., within the scope of the project UIDB/00742/2020 and DOI identified https://doi.org/10.54499/UIDB/00742/2020.

This work was also supported by National Funds through the FCT - Fundação para a Ciência e Tecnologia, I.P., within the scope of the project UIDB/00645/2020 and DOI identifier https://doi.org/10.54499/UIDB/00645/2020.

This research is also supported by Portuguese National Distributed Computing Infrastructure (INCD) and FCT - Fundação para a Ciência e Tecnologia, I.P., through the FCT Advanced Computing Projects with the project 2023.10865.CPCA.A1 and DOI identifier https://doi.org/10.54499/2023.10865.CPCA.A1.

References

[1] Jessilyn Dunn, Ryan Runge, Michael Snyder, Wearables and the medical revolution, Pers. Med. 15 (5) (2018) 429–448.

[2] Muhammad Usman Tariq, Advanced wearable medical devices and their role in transformative remote health monitoring, in: Transformative Approaches To Patient Literacy and Healthcare Innovation, IGI Global, 2024, pp. 308–326.

[3] Ehizogie Paul Adeghe, Chioma Anthonia Okolo, Olumuyiwa Tolulope Ojeyinka, A review of wearable technology in healthcare: Monitoring patient health and enhancing outcomes, OARJ Multidiscip. Stud. 7 (01) (2024) 142–148.

[4] Lin Lu, Jiayao Zhang, Yi Xie, Fei Gao, Song Xu, Xinghuo Wu, Zhewei Ye, et al., Wearable health devices in health care: narrative systematic review, JMIR MHealth UHealth 8 (11) (2020) e18907.

[5] Marie Chan, Daniel Estève, Jean-Yves Fourniols, Christophe Escriba, Eric Campo, Smart wearable systems: Current status and future challenges, Artif. Intell. Med. 56 (3) (2012) 137–156.

[6] Mark A. Griffin, Sharon Clarke, Stress and Well-Being at Work, American Psychological Association, 2011.

[7] Georgios Taskasaplidis, Dimitris A Fotiadis, Panagiotis Bamidis, Review of stress detection methods using wearable sensors, IEEE Access (2024).

[8] Blake Anthony Hickey, Taryn Chalmers, Phillip Newton, Chin-Teng Lin, David Sibbritt, Craig S McLachlan, Roderick Clifton-Bligh, John Morley, Sara Lal, Smart devices and wearable technologies to detect and monitor mental health conditions and stress: A systematic review, Sensors 21 (10) (2021) 3461.

[9] Chioma Virginia Anikwe, Henry Friday Nweke, Anayo Chukwu Ikegwu, Chukwunonso Adolphus Egwuonwu, Fergus Uchenna Onu, Uzoma Rita Alo, Ying Wah Teh, Mobile and wearable sensors for data-driven health monitoring system: State-of-the-art and future prospect, Expert Syst. Appl. 202 (2022) 117362.

[10] Richard J. McNally, Richard A. Bryant, Anke Ehlers, Does early psychological intervention promote recovery from posttraumatic stress? Psychol. Sci. Public Interes. 4 (2) (2003) 45–79.

[11] Iris Cohen, Willem-Paul Brinkman, Mark A. Neerincx, Modelling environmental and cognitive factors to predict performance in a stressful training scenario on a naval ship simulator, Cogn. Technol. Work. 17 (2015) 503–519.

[12] Aishwarya Goyal, Shailendra Singh, Dharam Vir, Dwarka Pershad, Automation of stress recognition using subjective or objective measures, Psychol. Stud. 61 (2016) 348–364.

[13] Pedro Elkind Velmovitsky, Use of Smart Technology Tools for Supporting Public Health Surveillance: From Development of a Mobile Health Platform to Application in Stress Prediction, University of Waterloo, 2023.

[14] Stanford University, A wearable device measures cortisol in sweat, 2018, Retrieved from https://news.stanford.edu/stories/2018/07/wearable-device-measures-cortisol-sweat.

[15] C. Wang, et al., High-precision flexible sweat self-collection sensor for mental stress evaluation, Npj Flex. Electron. 8 (47) (2024) http://dx.doi.org/10.1038/s41528-024-00333-z.

[16] Farnad Nasirzadeh, Chandan Karmakar, Ahsan Habib, Kevin Benny Neelangal, Mostafa Mir, Sang Hyun Lee, Tony Arnel, Continuous monitoring of body temperature for objective detection of health and safety risks in construction sites: An analysis of the accuracy and comfort of off-the-shelf wearable sensors, Heliyon (ISSN: 24058440) 10 (2024) http://dx.doi.org/10.1016/j.heliyon.2024.e26947.

[17] Shruti Gedam, Sanchita Paul, A review on mental stress detection using wearable sensors and machine learning techniques, IEEE Access 9 (2021) 84045–84066.

[18] Alexios-Fotios A. Mentis, Donghoon Lee, Panos Roussos, Applications of artificial intelligence- machine learning for detection of stress: a critical overview, Mol. Psychiatry 29 (6) (2024) 1882–1894.

[19] Rajdeep Kumar Nath, Himanshu Thapliyal, Allison Caban-Holt, Machine learning based stress monitoring in older adults using wearable sensors and cortisol as stress biomarker, J. Signal Process. Syst. (2022) 1–13.

[20] Darius Nahavandi, Roohallah Alizadehsani, Abbas Khosravi, U Rajendra Acharya, Application of artificial intelligence in wearable devices: Opportunities and challenges, Comput. Methods Programs Biomed. 213 (2022) 106541.

[21] Jigna Jadav, Uttam Chauhan, Analysis of personalized stress recognition in the office environment, Reliab.: Theory Appl. 19 (2 (78)) (2024) 377–389.

[22] Emilio Ferrara, Large language models for wearable sensor-based human activity recognition, health monitoring, and behavioral modeling: A survey of early trends, datasets, and challenges, Sensors 24 (15) (2024) 5045.

[23] Ali Tazarv, Sina Labbaf, Stephanie M Reich, Nikil Dutt, Amir M Rahmani, Marco Levorato, Personalized stress monitoring using wearable sensors in everyday settings, in: 2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society, EMBC, IEEE, 2021, pp. 7332–7335.

[24] Peter Tugwell, David Tovey, PRISMA 2020, 134, 2021, pp. A5–A6.

[25] Eftim Zdravevski, Petre Lameski, Vladimir Trajkovik, Ivan Chorbev, Rossitza Goleva, Nuno Pombo, Nuno M Garcia, Automation in systematic, scoping and rapid reviews by an NLP toolkit: a case study in enhanced living environments, Enhanc. Living Environ.: Algorithms Archit. Platforms Syst. (2019) 1–18.

[26] Marianna Ciccarelli, Asya Pesaresi, Alessandra Papetti, Michele Germani, Wearable devices for stress detection: insights on exploring generalizability in controlled laboratory settings, Procedia Comput. Sci. 253 (2025) 2014–2023.

[27] Xiujuan Qiao, Zhenying Xu, Zhen Wei, Yiting Hou, Fengxian Gao, Xijuan Yu, Xiliang Luo, A wearable electrochemical biosensor based on antifouling and conducting polyaniline hydrogel for cortisol detection in sweat, Chin. Chem. Lett. (2025) 110884.

[28] Basil A Darwish, Shafiq Ul Rehman, Ibrahim Sadek, Nancy M Salem, Ghada Kareem, Lamees N Mahmoud, From lab to real-life: A three-stage validation of wearable technology for stress monitoring, MethodsX (2025) 103205.

[29] Philip Schmidt, Attila Reiss, Robert Duerichen, Claus Marberger, Kristof Van Laerhoven, Introducing WESAD, a multimodal dataset for wearable stress and affect detection, in: Proceedings of the 2018 ACM International Conference on Multimodal Interaction, ICMI, 2018, http://dx.doi.org/10.24432/C57K5T.

[30] Ashutosh Singh, Amit Kumar, Khushdeep Singh, Santosh Kumar, A deep learning approach for early stress detection using electrodermal activity through wearable devices, in: International Conference on Pattern Recognition, Springer, 2025, pp. 49–66.

[31] Vishal Yadav, et al., VerBIO: A multimodal bio-behavioral dataset of individuals' affective responses during public speaking tasks, 2020, URL https://hubbs.engr.tamu.edu/resources/verbio-dataset/, (Accessed 23 June 2025).

[32] Chen Chen, Roozbeh Jafari, Nasser Kehtarnavaz, UTD-MHAD: A multimodal dataset for human action recognition utilizing a depth camera and a wearable inertial sensor, 2015, URL https://www.utdallas.edu/~kehtar/UTD-MHAD.html, (Accessed 23 June 2025).

[33] S.J. Koldijk, M. Sappelli, S. Verberne, M.A. Neerincx, W. Kraaij, The SWELL knowledge work dataset for stress and user modeling research, in: Proceedings of the 16th ACM International Conference on Multimodal Interaction, ACM, New York, NY, USA, 2014, pp. 291–298, http://dx.doi.org/10.1145/2663204.2663263, URL https://resolver.tno.nl/uuid:283de7e3-bea8-4ead-bcc0-06f0047c7446.

[34] Seyedmajid Hosseini, Raju Gottumukkala, Satya Katragadda, Ravi Teja Bhupatiraju, Ziad Ashkar, Christoph W. Borst, Kenneth Cochran, A multimodal sensor dataset for continuous stress detection of nurses in a hospital, Scientific Data (ISSN: 2052-4463) 9 (1) (2022) 255, http://dx.doi.org/10.1038/s41597-022-01361-y, https://doi.org/10.1038/s41597-022-01361-y.

[35] Jennifer Healey, Rosalind W. Picard, Stress recognition in automobile drivers, 2005, URL https://archive.physionet.org/pn3/drivedb/, (Accessed 23 June 2025).

[36] Ritu Tanwar, Ghanapriya Singh, Pankaj Kumar Pal, Stress-wed: Stress recognition autoencoder using wearables data, in: International Conference on Artificial Intelligence: Towards Sustainable Intelligence, Springer, 2024, pp. 77–88.

[37] Paul Schreiber, Beyza Cinar, Lennart Mackert, Maria Maleshkova, VitaStress–multimodal vital signs for stress detection, in: International Conference on Human-Computer Interaction, Springer, 2025, pp. 205–215.

[38] Ngoc-Dau Mai, Wan-Young Chung, On-chip mental stress detection: Integrating a wearable behind-the-ear EEG device with embedded tiny neural network, IEEE J. Biomed. Heal. Inform. (2024).

[39] Marina Andrić, Mauro Dragoni, Francesco Ricci, Anticipating stress: Harnessing biomarker signals from a wrist-worn device for early prediction, in: International Conference on Artificial Intelligence in Medicine, Springer, 2024, pp. 397–408.

[40] Nafize Ishtiaque Hossain, Tanzila Noushin, Shawana Tabassum, StressFit: A hybrid wearable physicochemical sensor suite for simultaneously measuring electromyogram and sweat cortisol, Sci. Rep. 14 (1) (2024) 29667.

[41] Ahsan Habib, Shruthi Narayanan Vaniya, Ahsan Khandoker, Chandan Karmakar, MddBranchNet: A deep learning model for detecting major depressive disorder using ECG signal, IEEE J. Biomed. Heal. Inform. 28 (7) (2024) 3798–3809.

[42] Laura Fontes, Pedro Machado, Doratha Vinkemeier, Salisu Yahaya, Jordan J Bird, Isibor Kennedy Ihianle, Enhancing stress detection: a comprehensive approach through rppg analysis and deep learning techniques, Sensors 24 (4) (2024) 1096.

[43] Dogukan Baran Gungormus, Francisco M. Garcia-Moreno, Maria Bermudez-Edo, Laura Sánchez-Bermejo, José Luis Garrido, María José Rodríguez-Fórtiz, José Manuel Pérez-Mármol, A semi-automatic mhealth system using wearable devices for identifying pain-related parameters in elderly individuals, Int. J. Med. Inform. (ISSN: 18728243) 184 (2024) http://dx.doi.org/10.1016/j.ijmedinf.2024.105371.

[44] Stefan de Vries, Fransje van Oost, Hanneke Smaling, Nanda de Knegt, Pierre Cluitmans, Reon Smits, Erwin Meinders, Real-time stress detection based on artificial intelligence for people with an intellectual disability, Assist. Technol. : Off. J. RESNA (ISSN: 19493614) 36 (2024) 232–240, http://dx.doi.org/10.1080/10400435.2023.2261045.

[45] Mohamed Abd Al-Alim, Roaa Mubarak, Nancy M. Salem, Ibrahim Sadek, A machine-learning approach for stress detection using wearable sensors in free-living environments, Comput. Biol. Med. (ISSN: 18790534) 179 (2024) http://dx.doi.org/10.1016/j.compbiomed.2024.108918.

[46] Qiriro, Swell dataset, 2019, URL https://www.kaggle.com/datasets/qiriro/swell-heart-rate-variability-hrv.

[47] Marie Cathrine Pickert, Tabea Tharra, Ulf Kulau, Christoph Richter, Philip Zettner, Michael Marschollek, Klaus Hendrik Wolf, DR.BEAT: Rule-based algorithm for SCG analysis without ECG reference, Stud. Health Technol. Inform. (ISSN: 18798365) 316 (2024) 492–496, http://dx.doi.org/10.3233/SHTI240456.

[48] Gabriele Rescio, Andrea Manni, Marianna Ciccarelli, Alessandra Papetti, Andrea Caroppo, Alessandro Leone, A deep learning-based platform for workers' stress detection using minimally intrusive multisensory devices, Sensors (ISSN: 14248220) 24 (2024) http://dx.doi.org/10.3390/s24030947.

[49] Joseph Nuamah, Effect of recurrent task-induced acute stress on task performance, vagally mediated heart rate variability, and task-evoked pupil response, Int. J. Psychophysiol. (ISSN: 18727697) 198 (2024) http://dx.doi.org/10.1016/j.ijpsycho.2024.112325.

[50] A.J. Béquet, C. Jallais, J. Quick, D. Ndiaye, A.R. Hidalgo-Muñoz, Road to serenity: Individual variations in the efficacy of unobtrusive respiratory guidance for driving stress regulation, Appl. Ergon. (ISSN: 18729126) 120 (2024) http://dx.doi.org/10.1016/j.apergo.2024.104334.

[51] Ritu Tanwar, Orchid Chetia Phukan, Ghanapriya Singh, Pankaj Kumar Pal, Sanju Tiwari, Attention based hybrid deep learning model for wearable based stress recognition, Eng. Appl. Artif. Intell. (ISSN: 09521976) 127 (2024) http://dx.doi.org/10.1016/j.engappai.2023.107391.

[52] Kobiljon Toshnazarov, Uichin Lee, Byung Hyung Kim, Varun Mishra, Lismer Andres Caceres Najarro, Youngtae Noh, SOSW: Stress sensing with off-the-shelf smartwatches in the wild, IEEE Internet Things J. (ISSN: 23274662) 11 (2024) 21527–21545, http://dx.doi.org/10.1109/JIOT.2024.3375299.

[53] Stephanie Carreiro, Pravitha Ramanand, Melissa Taylor, Rebecca Leach, Joshua Stapp, Sloke Sherestha, David Smelson, Premananda Indic, Evaluation of a digital tool for detecting stress and craving in sud recovery: An observational trial of accuracy and engagement, Drug Alcohol Depend. (ISSN: 18790046) 261 (2024) http://dx.doi.org/10.1016/j.drugalcdep.2024.111353.

[54] Lamis Abdul Kader, Fares Al-Shargie, Usman Tariq, Hasan Al-Nashash, One-channel wearable mental stress state monitoring system, Sensors (ISSN: 14248220) 24 (2024) http://dx.doi.org/10.3390/s24165373.

[55] Gabriele Volpes, Simone Valenti, Giuseppe Genova, Chiara Barà, Antonino Parisi, Luca Faes, Alessandro Busacca, Riccardo Pernice, Wearable ring-shaped biomedical device for physiological monitoring through finger-based acquisition of electrocardiographic, photoplethysmographic, and galvanic skin response signals: Design and preliminary measurements, Biosensors (ISSN: 20796374) 14 (2024) http://dx.doi.org/10.3390/bios14040205.

[56] Tanvir Islam, Peter Washington, Individualized stress mobile sensing using self-supervised pre-training, Appl. Sci. (Switzerland) (ISSN: 20763417) 13 (2023) http://dx.doi.org/10.3390/app132112035.

[57] Kamana Dahal, Brian Bogue-Jimenez, Ana Doblas, Global stress detection framework combining a reduced set of hrv features and random forest model, Sensors (ISSN: 14248220) 23 (2023) http://dx.doi.org/10.3390/s23115220.

[58] Vasiliki Parousidou, Sofia Yfantidou, Christina Karagianni, Athena Vakali, Stress beats: A continuum of learning methods for personalized stress detection, in: Proceedings - 2023 22nd IEEE/WIC International Conference on Web Intelligence and Intelligent Agent Technology, WI-IAT 2023, Institute of Electrical and Electronics Engineers Inc., ISBN: 9798350309188, 2023, pp. 40–47, http://dx.doi.org/10.1109/WI-IAT59888.2023.00012.

[59] Ramesh Kumar Sah, Michael McDonell, Patricia Pendry, Sara Parent, Hassan Ghasemzadeh, Michael J Cleveland, Adarp: A multi modal dataset for stress and alcohol relapse quantification in real life setting, in: 2022 IEEE-EMBS International Conference on Wearable and Implantable Body Sensor Networks, BSN, IEEE, 2022, pp. 1–4.

[60] Sofia Yfantidou, Christina Karagianni, Stefanos Efstathiou, Athena Vakali, Joao Palotti, Dimitrios Panteleimon Giakatos, Thomas Marchioro, Andrei Kazlouski, Elena Ferrari, Šarūnas Girdzijauskas, LifeSnaps, a 4-month multi-modal dataset capturing unobtrusive snapshots of our lives in the wild, Sci. Data 9 (1) (2022) 663.

[61] Aoife Whiston, Eric R. Igou, Dònal G. Fortune, Maria Semkovska, Longitudinal interactions between residual symptoms and physiological stress in the remitted symptom network structure of depression, Acta Psychol. (ISSN: 18736297) 241 (2023) http://dx.doi.org/10.1016/j.actpsy.2023.104078.

[62] Minghui Nan, Bobby Aditya Darmawan, Gwangjun Go, Shirong Zheng, Junhyeok Lee, Seokjae Kim, Taeksu Lee, Eunpyo Choi, Jong Oh Park, Doyeon Bang, Wearable localized surface plasmon resonance-based biosensor with highly sensitive and direct detection of cortisol in human sweat, Biosensors (ISSN: 20796374) 13 (2023) http://dx.doi.org/10.3390/bios13020184.

[63] Hehui Zheng, Han Wang, Kunran Yi, Jian Lin, An Chen, Lingming Chen, Zebiao Zou, Maolin Liu, Yuchen Ji, Lingzhi Dong, Zhenpei Lin, Wearable LIG flexible stress sensor based on spider web bionic structure, Coatings (ISSN: 20796412) 13 (2023) http://dx.doi.org/10.3390/coatings13010155.

[64] Nikos Mitro, Katerina Argyri, Lampros Pavlopoulos, Dimitrios Kosyvas, Lazaros Karagiannidis, Margarita Kostovasili, Fay Misichroni, Eleftherios Ouzounoglou, Angelos Amditis, AI-enabled smart wristband providing real-time vital signs and stress monitoring, Sensors (ISSN: 14248220) 23 (2023) http://dx.doi.org/10.3390/s23052821.

[65] Sara Campanella, Ayham Altaleb, Alberto Belli, Paola Pierleoni, Lorenzo Palma, A method for stress detection using empatica E4 bracelet and machine-learning techniques, Sensors (ISSN: 14248220) 23 (2023) http://dx.doi.org/10.3390/s23073565.

[66] Simone Costantini, Mattia Chiappini, Giorgia Malerba, Carla Dei, Anna Falivene, Sara Arlati, Vera Colombo, Emilia Biffi, Fabio Alexander Storm, Wrist-worn sensor validation for heart rate variability and electrodermal activity detection in a stressful driving environment, Sensors (Basel, Switzerland) (ISSN: 14248220) 23 (2023) http://dx.doi.org/10.3390/s23208423.

[67] Mykhailo Bocharov, Vasyl Stasiuk, Vasyl Osyodlo, Tetyana Ryzhenko, Vlad Malanin, Dmytro Chumachenko, Illya Chaikovsky, Assessment of the activities physiological cost of the defense forces officers in Ukraine using miniature ECG device, Front. Cardiovasc. Med. (ISSN: 2297055X) 10 (2023) http://dx.doi.org/10.3389/fcvm.2023.1239128.

[68] Žiga Stržinar, Araceli Sanchis, Agapito Ledezma, Oscar Sipele, Boštjan Pregelj, Igor Škrjanc, Stress detection using frequency spectrum analysis of wrist-measured electrodermal activity, Sensors (ISSN: 14248220) 23 (2023) http://dx.doi.org/10.3390/s23020963.

[69] Gideon Vos, Kelly Trinh, Zoltan Sarnyai, Mostafa Rahimi Azghadi, Ensemble machine learning model trained on a new synthesized dataset generalizes well for stress prediction using wearable devices, J. Biomed. Inform. (ISSN: 15320464) 148 (2023) http://dx.doi.org/10.1016/j.jbi.2023.104556.

[70] J. Lesley Crow, Barbara C. Harmeling, Development of a consensus and evidence-based standardised clinical assessment and record form for neurological inpatients: the neuro dataset, Physiotherapy 88 (1) (2002) 33–46.

[71] María Gaitán-Padilla, Marcela Múnera, Carlos A. Cifuentes, Maxwell E. Monteiro, Maria J. Pontes, Camilo A.R. Diaz, Stress classification using a low-cost optical fiber physiological sensor: A preliminary study, in: 20th SBMO/IEEE MTT-S International Microwave and Optoelectronics Conference, IMOC 2023, Institute of Electrical and Electronics Engineers Inc., ISBN: 9798350320671, 2023, pp. 310–312, http://dx.doi.org/10.1109/IMOC57131.2023.10379658.

[72] Ahmad Almadhor, Gabriel Avelino Sampedro, Mideth Abisado, Sidra Abbas, Ye Jin Kim, Muhammad Attique Khan, Jamel Baili, Jae Hyuk Cha, Wrist-based electrodermal activity monitoring for stress detection using federated learning, Sensors (ISSN: 14248220) 23 (2023) http://dx.doi.org/10.3390/s23083984.

[73] Ahmad Almadhor, Gabriel Avelino Sampedro, Mideth Abisado, Sidra Abbas, Efficient feature-selection-based stacking model for stress detection based on chest electrodermal activity, Sensors (ISSN: 14248220) 23 (2023) http://dx.doi.org/10.3390/s23156664.

[74] Abu Sarwar Zamani, Aisha Hassan Abdalla Hashim, Md Mobin Akhtar, Faizan Samdani, Ahmad Talha Siddiqui, Adel Alluhayb, Manar Ahmed Hamza, Naved Ahmad, The prediction of sleep quality using wearable-assisted smart health monitoring systems based on statistical data, J. King Saud Univ. - Sci. (ISSN: 10183647) 35 (2023) http://dx.doi.org/10.1016/j.jksus.2023.102927.

[75] Emily Zhou, Mohammad Soleymani, Maja J. Mataric, Investigating the generalizability of physiological characteristics of anxiety, in: Proceedings - 2023 2023 IEEE International Conference on Bioinformatics and Biomedicine, BIBM 2023, Institute of Electrical and Electronics Engineers Inc., ISBN: 9798350337488, 2023, pp. 4848–4855, http://dx.doi.org/10.1109/BIBM58861.2023.10385292.

[76] Hashini Senaratne, Levin Kuhlmann, Kirsten Ellis, Glenn Melvin, Sharon Oviatt, A multimodal dataset and evaluation for feature estimators of temporal phases of anxiety, in: Proceedings of the 2021 International Conference on Multimodal Interaction, 2021, pp. 52–61.

[77] Karan Sharma, Claudio Castellini, Egon L Van Den Broek, Alin Albu-Schaeffer, Friedhelm Schwenker, A dataset of continuous affect annotations and physiological signals for emotion analysis, Sci. Data 6 (1) (2019) 196.

[78] Niloofar Momeni, Adriana Arza Valdés, João Rodrigues, Carmen Sandi, David Atienza, CAFS: Cost-aware features selection method for multimodal stress monitoring on wearable devices, IEEE Trans. Biomed. Eng. (ISSN: 15582531) 69 (2022) 1072–1084, http://dx.doi.org/10.1109/TBME.2021.3113593.

[79] Christian Rominger, Andreas R. Schwerdtfeger, Feelings from the heart part II: Simulation and validation of static and dynamic HRV decrease-trigger algorithms to detect stress in firefighters, Sensors (ISSN: 14248220) 22 (2022) http://dx.doi.org/10.3390/s22082925.

[80] Florian Ritsert, Mohamed Elgendi, Valeria Galli, Carlo Menon, Heart and breathing rate variations as biomarkers for anxiety detection, Bioengineering (ISSN: 23065354) 9 (2022) http://dx.doi.org/10.3390/bioengineering9110711.

[81] Niaz Chalabianloo, Yekta Said Can, Muhammad Umair, Corina Sas, Cem Ersoy, Application level performance evaluation of wearable devices for stress classification with explainable AI, Pervasive Mob. Comput. (ISSN: 15741192) 87 (2022) http://dx.doi.org/10.1016/j.pmcj.2022.101703.

[82] P. Cummings, A. Petitclerc, J. Moskowitz, D. Tandon, Y. Zhang, L.A. MacNeill, N. Alshurafa, S. Krogh-Jespersen, J.L. Hamil, A. Nili, J. Berken, W. Grobman, A. Rangarajan, L. Wakschlag, Feasibility of passive ECG bio-sensing and EMA emotion reporting technologies and acceptability of just-in-time content in a well-being intervention, considerations for scalability and improved uptake, Affect. Sci. (ISSN: 2662205X) 3 (2022) 849–861, http://dx.doi.org/10.1007/s42761-022-00147-0.

[83] Maximilian Ehrhart, Bernd Resch, Clemens Havas, David Niederseer, A conditional GAN for generating time series data for stress detection in wearable physiological sensor data, Sensors (ISSN: 14248220) 22 (2022) http://dx.doi.org/10.3390/s22165969.

[84] Zhaoxi Zhang, Prince M. Amegbor, Torben Sigsgaard, Clive E. Sabel, Assessing the association between urban features and human physiological stress response using wearable sensors in different urban contexts, Heal. Place (ISSN: 18732054) 78 (2022) http://dx.doi.org/10.1016/j.healthplace.2022.102924.

[85] Jun Zhong, Yongfeng Liu, Xiankai Cheng, Liming Cai, Weidong Cui, Dong Hai, Gated recurrent unit network for psychological stress classification using electrocardiograms from wearable devices, Sensors (ISSN: 14248220) 22 (2022) http://dx.doi.org/10.3390/s22228664.

[86] Annika Thierfelder, Jonas Primbs, Bjorn Severitt, Carolin S. Hohnecker, Jan Kuhnhausen, Annika K. Alt, Anja Pascher, Ursula Worz, Helene Passon, Jens Seemann, Christian Ernst, Heinrich Lautenbacher, Martin Holderried, Enkelejda Kasneci, Martin A. Giese, Andreas Bulling, Michael Menth, Gottfried M. Barth, Winfried Ilg, Karsten Hollmann, Tobias J. Renner, Multimodal sensor-based identification of stress and compulsive actions in children with obsessive-compulsive disorder for telemedical treatment, in: Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBS, 2022-July, Institute of Electrical and Electronics Engineers Inc., ISBN: 9781728127828, 2022, pp. 2976–2982, http://dx.doi.org/10.1109/EMBC48229.2022.9871899, ISSN: 1557170X.

[87] Talha Iqbal, Pau Redon-Lurbe, Andrew J. Simpkin, Adnan Elahi, Sandra Ganly, William Wijns, Atif Shahzad, A sensitivity analysis of biophysiological responses of stress for wearable sensors in connected health, IEEE Access (ISSN: 21693536) 9 (2021) 93567–93579, http://dx.doi.org/10.1109/ACCESS.2021.3082423.

[88] Ala Hag, Dini Handayani, Maryam Altalhi, Thulasyammal Pillai, Teddy Mantoro, Mun Hou Kit, Fares Al-Shargie, Enhancing EEG-based mental stress state recognition using an improved hybrid feature selection algorithm, Sensors (ISSN: 14248220) 21 (2021) http://dx.doi.org/10.3390/s21248370.

[89] Sander Koelstra, Christian Muhl, Mohammad Soleymani, Jong-Seok Lee, Ashkan Yazdani, Touradj Ebrahimi, Thierry Pun, Anton Nijholt, Ioannis Patras, Deap: A database for emotion analysis; using physiological signals, IEEE Trans. Affect. Comput. 3 (1) (2011) 18–31.

[90] Md Billal Hossain, Syed Khairul Bashar, Jesus Lazaro, Natasa Reljin, Yeonsik Noh, Ki H. Chon, A robust ECG denoising technique using variable frequency complex demodulation, Comput. Methods Programs Biomed. (ISSN: 18727565) 200 (2021) http://dx.doi.org/10.1016/j.cmpb.2020.105856.

[91] George B. Moody, Roger G. Mark, The impact of the MIT-BIH arrhythmia database, IEEE Eng. Med. Biol. Mag. 20 (3) (2001) 45–50.

[92] Sara Aristizabal, Kunjoon Byun, Nadia Wood, Aidan F. Mullan, Paige M. Porter, Carolina Campanella, Anja Jamrozik, Ivan Z. Nenadic, Brent A. Bauer, The feasibility of wearable and self-report stress detection measures in a semi-controlled lab environment, IEEE Access (ISSN: 21693536) 9 (2021) 102053–102068, http://dx.doi.org/10.1109/ACCESS.2021.3097038.

[93] Simon Föll, Martin Maritsch, Federica Spinola, Varun Mishra, Filipe Barata, Tobias Kowatsch, Elgar Fleisch, Felix Wortmann, FLIRT: A feature generation toolkit for wearable data, Comput. Methods Programs Biomed. (ISSN: 18727565) 212 (2021) http://dx.doi.org/10.1016/j.cmpb.2021.106461.

[94] Maciej Dzieżyc, Martin Gjoreski, Przemysł aw Kazienko, Stanisł aw Saganowski, Matjaž Gams, Can we ditch feature engineering? End-to-end deep learning for affect recognition from physiological sensor data, Sensors (Switzerland) (ISSN: 14248220) 20 (2020) 1–21, http://dx.doi.org/10.3390/s20226535.

[95] Larissa Bolliger, Junoš Lukan, Mitja Luštrek, Dirk De Bacquer, Els Clays, Protocol of the STRess at work (STRAW) project: How to disentangle day-to-day occupational stress among academics based on EMA, physiological data,

and smartphone sensor and usage data, Int. J. Environ. Res. Public Heal. 17 (23) (2020) 8835.

[96] Franca Delmastro, Flavio Di Martino, Cristina Dolciotti, Cognitive training and stress detection in MCI frail older people through wearable sensors and machine learning, IEEE Access (ISSN: 21693536) 8 (2020) 65573–65590, http://dx.doi.org/10.1109/ACCESS.2020.2985301.

[97] Fatma Patlar Akbulut, Baris Ikitimur, Aydin Akan, Wearable sensor-based evaluation of psychosocial stress in patients with metabolic syndrome, Artif. Intell. Med. (ISSN: 18732860) 104 (2020) http://dx.doi.org/10.1016/j.artmed.2020.101824.

[98] Yekta Said Can, Niaz Chalabianloo, Deniz Ekiz, Javier Fernandez-Alvarez, Giuseppe Riva, Cem Ersoy, Personal stress-level clustering and decision-level smoothing to enhance the performance of ambulatory stress detection with smartwatches, IEEE Access (ISSN: 21693536) 8 (2020) 38146–38163, http://dx.doi.org/10.1109/ACCESS.2020.2975351.

[99] Muhammad Zubair, Changwoo Yoon, Multilevel mental stress detection using ultra-short pulse rate variability series, Biomed. Signal Process. Control. (ISSN: 17468108) 57 (2020) http://dx.doi.org/10.1016/j.bspc.2019.101736.

[100] Vadym Mozgovoy, Stress pattern recognition through wearable biosensors in the workplace: Experimental longitudinal study on the role of motion intensity, in: Proceedings - 6th Swiss Conference on Data Science, SDS 2019, Institute of Electrical and Electronics Engineers Inc., ISBN: 9781728131054, 2019, pp. 37–45, http://dx.doi.org/10.1109/SDS.2019.00-10.

[101] Susana Rodrigues, Joana S. Paiva, Duarte Dias, Gonçalo Pimentel, Mariana Kaiseler, João Paulo S. Cunha, Wearable biomonitoring platform for the assessment of stress and its impact on cognitive performance of firefighters: An experimental study, Clin. Pr. Epidemiology Ment. Heal. (ISSN: 1745-0179) 14 (2018) 250–262, http://dx.doi.org/10.2174/1745017901814010250.

[102] Stefano Betti, Raffaele Molino Lova, Erika Rovini, Giorgia Acerbi, Luca Santarelli, Manuela Cabiati, Silvia Del Ry, Filippo Cavallo, Evaluation of an integrated system of wearable physiological sensors for stress monitoring in working environments by using biological markers, in: IEEE Transactions on Biomedical Engineering, vol. 65, IEEE Computer Society, (ISSN: 15582531) 2018, pp. 1748–1758, http://dx.doi.org/10.1109/TBME.2017.2764507.

[103] Qianli Xu, Tin Lay Nwe, Cuntai Guan, Cluster-based analysis for personalized stress evaluation using physiological signals, IEEE J. Biomed. Heal. Inform. (ISSN: 21682208) 19 (2015) 275–281, http://dx.doi.org/10.1109/JBHI.2014.2311044.

[104] Fernando Seoane, Javier Ferreira, Lorena Alvarez, Ruben Buendia, David Ayllón, Cosme Llerena, Roberto Gil-Pita, Sensorized garments and textrode-enabled measurement instrumentation for ambulatory assessment of the autonomic nervous system response in the ATREC project, Sensors (Switzerland) (ISSN: 14248220) 13 (2013) 8997–9015, http://dx.doi.org/10.3390/s130708997.

[105] Saskia Koldijk, Maya Sappelli, Suzan Verberne, Mark A Neerincx, Wessel Kraaij, The swell knowledge work dataset for stress and user modeling research, in: Proceedings of the 16th International Conference on Multimodal Interaction, 2014, pp. 291–298.

[106] Health and Safety Executive, Work-related ill health and occupational disease in great britain, 2024, URL https://www.hse.gov.uk/statistics/causdis/.

[107] M. Smirthy, M. Dhanushree, G.R. Ashisha, Investigation of machine learning techniques and sensing devices for mental stress detection, in: 2023 4th International Conference on Signal Processing and Communication, ICSPC, IEEE, 2023, pp. 287–291.

[108] Marco Bolpagni, Susanna Pardini, Marco Dianti, Silvia Gabrielli, Personalized stress detection using biosignals from wearables: A scoping review, Sensors 24 (10) (2024) 3221.

[109] C Bambang Dwi Kuncoro, Adhan Efendi, Maria Mahardini Sakanti, Wearable sensor for psychological stress monitoring of pregnant woman-state of the art, Measurement (2023) 113556.

[110] Ryan Cannady, Catherine Warner, Aaron Yoder, Jeffrey Miller, Kaitlyn Crosby, Donald Elswick, Kristina W Kintziger, The implications of real-time and wearable technology use for occupational heat stress: A scoping review, Saf. Sci. 177 (2024) 106600.

[111] Laura P Jiménez-Mijangos, Jorge Rodríguez-Arce, Rigoberto Martínez-Méndez, José Javier Reyes-Lagos, Advances and challenges in the detection of academic stress and anxiety in the classroom: A literature review and recommendations, Educ. Inf. Technol. 28 (4) (2023) 3637–3666.

[112] Shruti Gedam, Sanchita Paul, Automatic stress detection using wearable sensors and machine learning: A review, in: 2020 11th International Conference on Computing, Communication and Networking Technologies, ICCCNT, IEEE, 2020, pp. 1–7.

[113] Elena Smets, Emmanuel Rios Velazquez, Giuseppina Schiavone, Imen Chakroun, Ellie D'Hondt, Walter De Raedt, Jan Cornelis, Olivier Janssens, Sofie Van Hoecke, Stephan Claes, et al., Large-scale wearable data reveal digital phenotypes for daily-life stress detection, NPJ Digit. Med. 1 (1) (2018) 67.